PEARSON EDEXCEL INTERNATIONAL A LEVEL CHEMISTRY · WCH14

Unit 4: Rates, Equilibria and Further Organic Chemistry

Revision notes, spec point by spec point (Topics 11 to 15)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

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MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

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Topic 11 – Kinetics

SPEC 11.1 Key terms in kinetics

WHAT THIS SPEC POINT ASKS

Know the meaning of: rate of reaction, rate equation (rate = $k[A]^m[B]^n$, with m and n equal to 0, 1 or 2), order with respect to a substance, overall order, rate constant, half-life, rate-determining step, activation energy, and homogeneous and heterogeneous catalysts.

11.1 The nine definitions

Term	What it means
Rate of reaction	The **change in concentration** of a reactant or product **per unit time**. Units: **mol dm^{-3} s^{-1}**. On a concentration–time graph it is the **gradient** (of a tangent) at that moment.
Rate equation	rate = **k**$[A]^m[B]^n$. It links the rate to the concentrations of the species that affect it. It is found **by experiment**, never from the balanced equation.
Order with respect to a substance	The **power** that substance's concentration is raised to in the rate equation (m for A, n for B).
Overall order	The **sum** of the individual orders, m + n.
Rate constant, k	The constant of proportionality in the rate equation. It is fixed **at a given temperature**: change the temperature (or add a catalyst) and k changes; change a concentration and it doesn't. A bigger k means a faster reaction.
Half-life, t½	The **time taken for the concentration of a reactant to fall to half** of its starting value.
Rate-determining step (RDS)	The **slowest step** in a multi-step mechanism. It limits the rate of the whole reaction.
Activation energy, Ea	The **minimum energy** that colliding particles must have for the collision to lead to a reaction.
Homogeneous catalyst	A catalyst in the **same phase** (state) as the reactants, e.g. H^+(aq) in the iodination of propanone (11.6).
Heterogeneous catalyst	A catalyst in a **different phase** from the reactants, usually a solid with gaseous or liquid reactants, e.g. iron in the Haber process (11.11).

A catalyst speeds up a reaction by giving an **alternative route with a lower activation energy**, and it is **regenerated** at the end.

11.1 What the orders tell you

Order in A	Rate equation term	[A] × 2	[A] × 3	[A] × 10
0	$[A]^0$ = 1, so A drops out	rate × 1 (no change)	× 1	× 1
1	[A]	rate × 2	× 3	× 10
2	$[A]^2$	rate × 4	× 9	× 100

- A **zero-order** species is not in the rate equation at all, but it still reacts. It just doesn't take part in (or before) the slow step.
- The rate equation can include a **catalyst** (such as H^+) that isn't in the stoichiometric equation. It never includes a species that only appears after the slow step.

11.1 Units of k

Rearrange the rate equation, then cancel units: k = rate ÷ ($[A]^m[B]^n$).

Overall order	Units of k
0	mol dm^{-3} s^{-1} (the same as the rate)
1	s^{-1}
2	dm^3 mol^{-1} s^{-1}
3	dm^6 mol^{-2} s^{-1}
4	dm^9 mol^{-3} s^{-1}

Pattern: each extra order adds dm^3 mol^{-1} to the units. Learn the method, not just the table.

WORKED EXAMPLE: UNITS OF K

rate = $k[NO]^2[O_2]$, so k = rate ÷ ($[NO]^2[O_2]$).
Units = mol dm^{-3} s^{-1} ÷ (mol dm^{-3})^3 = mol dm^{-3} s^{-1} ÷ (mol^3 dm^{-9})
= mol^{-2} dm^6 s^{-1}, written **dm^6 mol^{-2} s^{-1}** (any order of the units is fine).

MARK-SCHEME LANGUAGE

- Rate equation: "rate = $k[NO]^2[O_2]$". **k must be there**, and concentrations go in **square brackets**. Round brackets are not credited.
- Units of k: "dm^6 mol^{-2} s^{-1}" (third order); "dm^3 mol^{-1} s^{-1}" (second); "s^{-1}" (first). Any order of the units is allowed.
- Effect of temperature on k: "as T increases, k increases"; "a greater proportion of collisions have energy ≥ the activation energy". "More successful collisions" on its own is ignored.
- Homogeneous catalyst: "same state/phase as the reactants"; catalyst: "speeds up the reaction and is regenerated" (WCH14 Jan 2022). If both words are in bold in the question, explain both.

Often asked as: MCQ on units of k or on the effect of changing a concentration; write a rate equation from given orders; calculate k with units; name a catalyst type.

BEYOND THE SPEC, BUT EXAMINED

- Rate equations with a **fourth overall order**, e.g. rate = $k[BrO_3^-][Br^-][H^+]^2$, so k has units dm^9 mol^{-3} s^{-1} (WCH14 Jun 2021, Jan 2022).
- **Zero order overall**: rate = k, so k has the same units as the rate (WCH14 Oct 2021).
- The rate equation can contain a species that isn't in the balanced equation as written, e.g. CN^- rather than HCN (WCH14 Oct 2024).

COMMON MISTAKE

Mixing up the **rate constant k** with the **equilibrium constant K**. Le Chatelier and ΔStotal = R ln K tell you nothing about k. Explain changes in k with particle energies and Ea (examiner report, Jan 2020).
Leaving k out of the rate equation, or quoting orders from the balanced equation's coefficients.
Answering "explain why the rate increases with temperature" by writing out the Arrhenius equation. Describe what the particles do.

SPEC 11.2

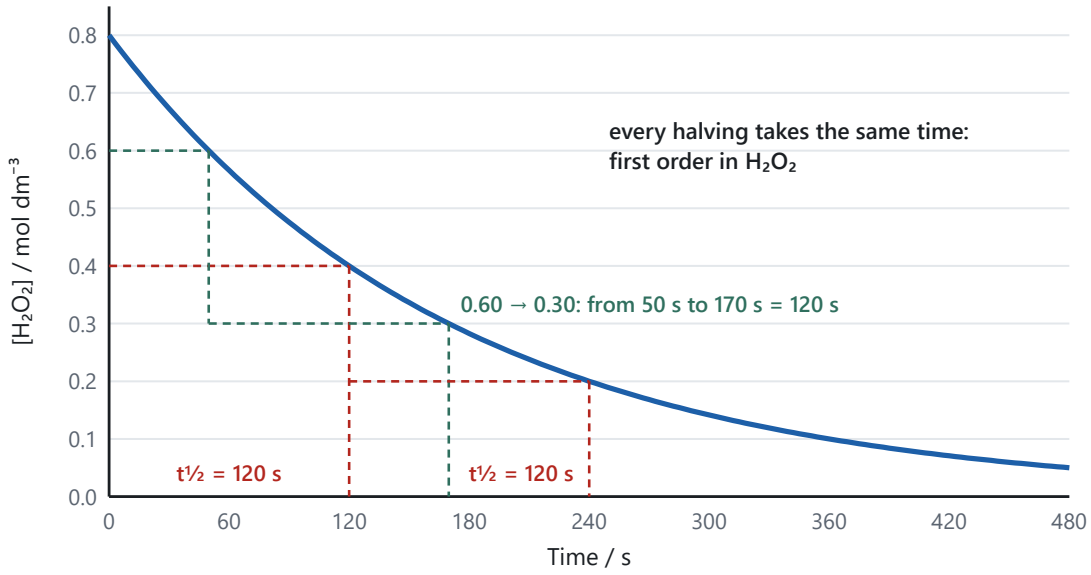
Half-life from a graph: constant half-life means first order

WHAT THIS SPEC POINT ASKS

Find the half-life of a reaction from a suitable graph, and recognise that a constant half-life means the reaction is first order.

11.2 Reading half-lives

1. On the concentration–time curve, pick a starting concentration (say 0.80 mol dm^{-3}) and read the time.
2. Find when the concentration has fallen to **half** that value (0.40) and read the time. The difference is one half-life.
3. Repeat from another concentration. It **doesn't have to be consecutive**: 0.60 → 0.30 works just as well as 0.40 → 0.20.
4. Draw the construction lines on the graph. They earn the method mark.



A first-order reaction: 0.80 → 0.40, 0.40 → 0.20 and 0.60 → 0.30 all take 120 s.

11.2 Half-life and order

Order	Half-life as the reaction goes on
First	**constant** (it doesn't depend on concentration)
Zero	gets **shorter**
Second	gets **longer**

Only first order has a constant half-life, so "the half-life is constant" is the evidence the mark scheme wants for first order. Radioactive decay is the classic first-order process.

WORKED EXAMPLE: HALF-LIFE TO K

From the graph above, t½ = 120 s (three half-lives all 120 s, so first order).
For a first-order reaction, k = ln 2 ÷ t½ = 0.693 ÷ 120 = **5.78×10^{-3} s^{-1}**.
Initial rate = k[H_2O_2] = $5.78 \times 10^{-3} \times 0.80$ = **4.6×10^{-3} mol dm^{-3} s^{-1}**.
If t½ is read in minutes (say 132 min), convert first: 132 × 60 = 7920 s, so k = 0.693 ÷ 7920 = 8.75×10^{-5} s^{-1}.

MARK-SCHEME LANGUAGE

- "working for at least one half-life shown on graph". Lines on the graph are required.
- "two half-lives … within the range", then "(first order as) the half-life is constant". Stating the value alone does not score the conclusion.
- k = ln 2 ÷ t½ = 0.693 ÷ 120 = 5.78×10^{-3} s^{-1} (IGNORE SF except 1 SF).

Often asked as: show two half-lives on a printed graph and deduce the order; read a half-life (MCQ); turn a half-life into k.

BEYOND THE SPEC, BUT EXAMINED

- **k = ln 2 ÷ t½** for a first-order reaction (WCH14 Oct 2021 Q17, Oct 2022 Q16).
- The half-life is the time to reach **half the starting concentration**, not half-way to the final concentration. MCQ distractors use the second (WCH14 Jan 2025 Q15).

COMMON MISTAKE

Showing only one half-life, or not writing "constant". Assuming the second half-life must start where the first ended: any halving works (examiner report, Jun 2024).
Leaving k per minute when the question wants s^{-1}.

SPEC 11.3

Choosing a technique to follow the rate

WHAT THIS SPEC POINT ASKS

Pick, and justify, a suitable method to get rate data for a given reaction: titration, colorimetry, mass change, volume of gas, or another suitable technique.

11.3 The principle

Find a property that **changes as the reaction goes** and is **proportional to the concentration** of one reactant or product. Then measure it against time. The justification must name **what changes in this reaction**.

Technique	Use it when...	How it works	Watch out for
Titration (sample and quench)	an acid, alkali or iodine is used up or made	remove samples at timed intervals, **quench** them (stop the reaction), then titrate	quenching is needed; the reaction itself mustn't make the species you titrate
Colorimetry	a reactant or product is **coloured** (I_2, Br_2, MnO_4^-, NO_2, $Cr_2O_7^{2-}$)	absorbance ∝ concentration of the coloured species; use a **calibration curve** and a filter of the **complementary colour**	useless if everything is colourless; a precipitate makes the mixture cloudy
Mass change	a **dense gas** escapes (CO_2 from a carbonate + acid)	flask on a balance; mass falls as gas leaves (loose cotton-wool plug stops spray)	H_2 is too light: the mass change is too small to measure accurately
Volume of gas	any gas is given off	gas syringe (or inverted burette / measuring cylinder over water)	gases that dissolve in water (NH_3, HCl, some CO_2) need a syringe
Other	ions or H^+ change, a precipitate forms	**pH** probe; **electrical conductivity** (number or charge of ions changes); turbidity / "disappearing cross" for a precipitate	conductivity needs a change in total ions

11.3 Quenching

To stop a sample reacting before you titrate it:

- **dilute** it with a large volume of ice-cold water (slows it a lot), or
- **remove the catalyst**: add sodium hydrogencarbonate to neutralise an acid catalyst, or filter off a solid catalyst.

WORKED EXAMPLE: WHICH METHOD WON'T WORK?

$Br_2(aq) + HCOOH(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$
- Colorimetry works: orange-brown bromine is used up.
- Gas volume and mass loss work: CO_2 is given off.
- Conductivity works: ions are produced.
- **Quench and titrate with acid does not**: H^+ is a product, so titrating with acid tells you nothing useful.

MARK-SCHEME LANGUAGE

- Colorimetry: "the reactants are colourless but the product NO_2 is reddish-brown" (WCH14 Oct 2023). Name the coloured species.
- Manganate(VII) titration of H_2O_2: "the concentration (of H_2O_2 is) proportional to the volume of manganate(VII)"; "5:2 ratio" allowed (WCH14 Jun 2025).
- Solid catalyst: quench by "separating the reaction mixture from the catalyst" (WCH14 Jun 2025).

Often asked as: MCQ "which technique is (not) suitable"; justify colorimetry for a named reaction; how to stop a sample reacting.

BEYOND THE SPEC, BUT EXAMINED

- Colours you need from AS: NO_2 is **brown** (thermal decomposition of Group 2 nitrates), $Br_2(aq)$ orange/yellow, $I_2(aq)$ yellow-brown, MnO_4^- purple (WCH14 Oct 2023 Q15, Jan 2025 Q20).
- Quench an acid-catalysed reaction with **sodium hydrogencarbonate** (WCH14 Oct 2025 Q11).

COMMON MISTAKE

Writing "there must be a colour change" without saying **which** species is coloured. The commonest slip: saying NO is brown and NO_2 colourless (it's the other way round).
Describing a thiosulfate titration when the question is about a clock reaction.

SPEC **11.4**

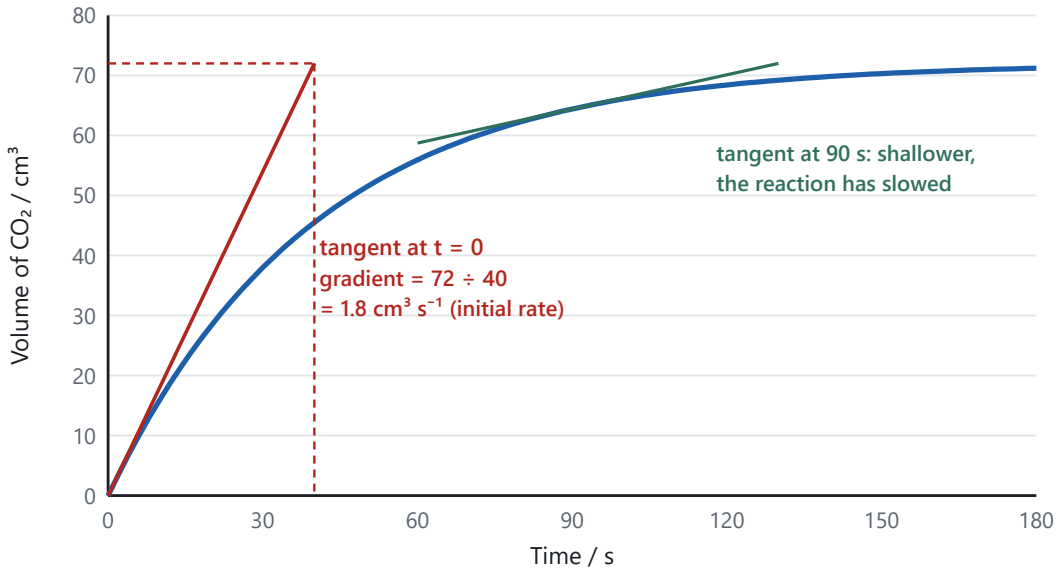
Initial-rate and continuous-monitoring methods

WHAT THIS SPEC POINT ASKS

Understand two ways of getting rate data: (i) the initial-rate method, with separate experiments at different starting concentrations of one reagent (a clock reaction is an acceptable approximation); (ii) continuous monitoring, giving a concentration–time or volume–time graph.

11.4 Initial-rate method

- Run **several separate experiments**. In each one change the starting concentration of **one** reagent and keep all the others (and the temperature) the same.
- Measure the rate at the very start, t = 0, when the concentrations are exactly the ones you made up.
- Make up each mixture to the **same total volume** with water, so that changing one volume doesn't dilute the others differently.
- From a concentration–time (or volume–time) curve, the initial rate is the **gradient of the tangent drawn at t = 0**.



The initial rate is the gradient of the tangent at t = 0. Later tangents are shallower because the reactants are being used up.

11.4 Clock reactions

A clock reaction gives one time, t: the time for a sudden colour change. That time measures the initial rate because:

- A **small, fixed amount** of a product has to form before the colour appears.
- So little reactant has been used by then that the concentrations have **hardly changed**: the rate is still the initial rate.
- **Rate ∝ 1/t** (shorter time, faster reaction).

In the iodine clock (Core Practical 9b, 11.12): iodine is made slowly; a small amount of **thiosulfate** turns it straight back into iodide. When all the thiosulfate is used up, free iodine meets the **starch**, and the mixture suddenly turns **blue-black**.

11.4 Continuous monitoring

- **One** experiment. Measure the concentration (or a property proportional to it) at **many times** as the reaction goes.
- Plot concentration (or volume, or absorbance) against time. Then:
  - read **half-lives** (11.2), or
  - draw **tangents** at several points to get rates at different concentrations, then plot rate against concentration (11.5).
- Examples: gas syringe readings every 10 s; a colorimeter logging absorbance; sampling and titrating (11.6).

WORKED EXAMPLE: RATE FROM A CLOCK TIME

5.0 cm^3 of 0.010 mol dm^{-3} $Na_2S_2O_3$ is in a total volume of 50.0 cm^3. The blue-black colour appears after 40 s. Find the rate of formation of iodine.

Moles of $S_2O_3^{2-}$ = 0.0050 × 0.010 = 5.0×10^{-5} mol

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$, so moles of I_2 formed before the colour = $5.0 \times 10^{-5} \div 2 = 2.5 \times 10^{-5}$ mol

Concentration of I_2 = $2.5 \times 10^{-5} \div 0.0500 = 5.0 \times 10^{-4}$ mol dm^{-3}

Rate = $5.0 \times 10^{-4} \div 40$ = **1.3×10^{-5} mol dm^{-3} s^{-1}**

MARK-SCHEME LANGUAGE

- Clock chemistry: "sodium thiosulfate reacts with the iodine formed"; "when (all) the thiosulfate is used up the iodine reacts with the starch giving a blue-black colour"; "the reciprocal of the time taken … is a measure of the rate".
- Why it's an initial rate: "the thiosulfate is used up before the concentrations of the other reagents change significantly".
- Colour change: "**colourless to blue-black**" (blue or black allowed). "Clear" is not a colour.
- Initial rate from a graph: "tangent drawn at t = 0" and its gradient, in mol dm^{-3} s^{-1}.

Often asked as: explain how a clock reaction measures an initial rate; state the colour change; calculate a rate from a clock time; draw a tangent at t = 0.

BEYOND THE SPEC, BUT EXAMINED

- The iodine clock used in WCH14 is often **hydrogen peroxide + iodide + acid**, not propanone + iodine (WCH14 Jan 2023 Q17, Jan 2024 Q16).
- Moles of I_2 = ½ × moles of thiosulfate; divide by the **mixture volume** and by the **time** (WCH14 Jan 2024 Q16).

COMMON MISTAKE

Dividing moles of iodine by volume but forgetting to divide by time. Using a single point instead of a tangent. Giving the rate per minute when it should be per second.

Saying starch reacts with iodide ions. Starch detects **iodine**, I_2.

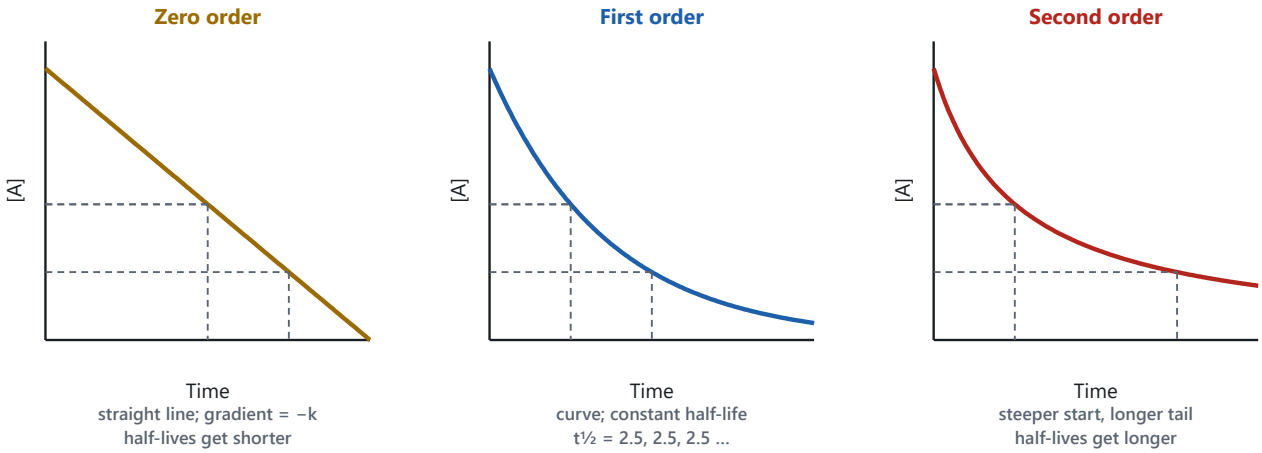
SPEC 11.5

Deducing orders from graphs and initial-rate data

WHAT THIS SPEC POINT ASKS

Deduce the order (0, 1 or 2) with respect to a substance from (i) a concentration–time graph, (ii) a rate–concentration graph, (iii) initial-rate data.

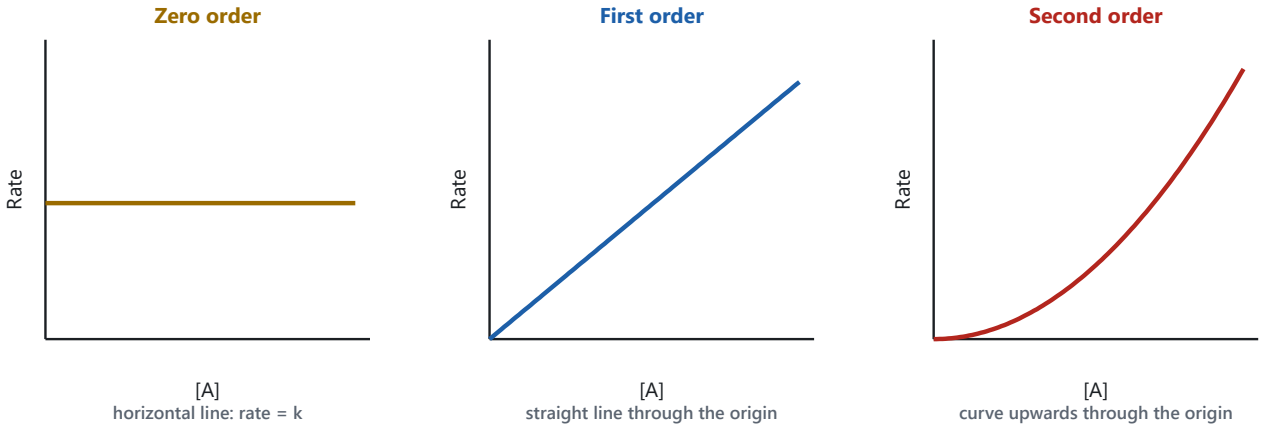
11.5 (i) Concentration–time graphs



Zero order: straight line. First order: curve with a constant half-life. Second order: falls fast, then a long tail; half-lives get longer.

- **Zero order**: a **straight line** going down. The gradient (the rate) is constant and equals −k.
- **First order**: a curve with a **constant half-life**.
- **Second order**: a curve that starts steeper and levels off more slowly; **half-lives increase**.

11.5 (ii) Rate–concentration graphs



Rate against concentration: horizontal (zero), straight line through the origin (first), upward curve through the origin (second).

- **Zero order**: **horizontal** line. Changing [A] doesn't change the rate.
- **First order**: **straight line through the origin**. Rate ∝ [A]; the gradient is k.
- **Second order**: **upward curve** through the origin. To prove it, plot rate against $[A]^2$: you get a straight line through the origin.
- You get the points by drawing tangents on one concentration–time curve, or from separate initial-rate experiments.

EXAM TIP

Two graphs are straight lines and easy to mix up: **zero order on a concentration–time graph** and **first order on a rate–concentration graph**. Always read the axis labels first.

11.5 (iii) Initial-rate tables

1. Find two experiments where **only one** concentration changes.
2. Work out the factor the concentration changes by and the factor the rate changes by.
3. Same factor → first order; factor squared → second order; no change → zero order.
4. Repeat for each reactant. If no pair changes only one concentration, allow for the order you already know (see below).
5. Write the rate equation, then substitute one experiment's values to find k, with units.

WORKED EXAMPLE: ORDERS, RATE EQUATION AND K

$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$

- Exp 1: $[BrO_3^-]$ 0.10, $[Br^-]$ 0.10, $[H^+]$ 0.10; initial rate 1.2×10^{-3} mol dm^{-3} s^{-1}
- Exp 2: $[BrO_3^-]$ 0.20, $[Br^-]$ 0.10, $[H^+]$ 0.10; rate 2.4×10^{-3}
- Exp 3: $[BrO_3^-]$ 0.20, $[Br^-]$ 0.20, $[H^+]$ 0.10; rate 4.8×10^{-3}
- Exp 4: $[BrO_3^-]$ 0.10, $[Br^-]$ 0.10, $[H^+]$ 0.20; rate 4.8×10^{-3}

BrO_3^-: Exp 1 → 2, $[BrO_3^-]$ × 2 with the others constant; rate × 2. **First order.**
Br^-: Exp 2 → 3, $[Br^-]$ × 2 with the others constant; rate × 2. **First order.**
(Exp 1 → 3 would also work: both double and the rate goes × 4. BrO_3^- accounts for × 2, so Br^- must account for the other × 2.)
H^+: Exp 1 → 4, $[H^+]$ × 2; rate × 4 = 2^2. **Second order.**
rate = $k[BrO_3^-][Br^-][H^+]^2$ (fourth order overall)
From Exp 1: $k = 1.2 \times 10^{-3} \div (0.10 \times 0.10 \times 0.10^2)$ = **12 dm^9 mol^{-3} s^{-1}**

MARK-SCHEME LANGUAGE

- Name the experiments and say what's held constant: "experiment 2 compared to 1, the concentration of O_2 doubles (while NO remains constant) and the rate doubles … so first order"; "multiplying NO by four … results in an increase in rate of 16 times … second order". Calculations or annotations on the table are accepted as the explanation.
- Graphs you plot: "labelled axes with units", "points cover over half the graph paper", points plotted to ±½ square, then the line.
- First order, rate against concentration: "straight line … beginning at the origin". Concentration–time data: a "smooth curve of best fit", not a ruler.
- Not first order: "graph is not a straight line (through the origin)".

Often asked as: deduce each order from a table with justification; write the rate equation and calculate k; identify or sketch the graph for a given order; plot rate against concentration and conclude.

BEYOND THE SPEC, BUT EXAMINED

- A **ten-fold** change in concentration: rate × 10 is first order, × 100 second order (WCH14 Jan 2024 Q16).
- The y-axis of a clock-reaction graph may be 1/t or "rate" rather than concentration (WCH14 Jan 2025 Q18).

COMMON MISTAKE

Quoting orders without justifying them: that loses most of the marks (examiner reports, Jan 2024, Jan 2025).

Drawing a **rate–time** graph when a concentration–time graph is asked for. Joining concentration–time points with straight segments, or ruling a straight line through curved data.

SPEC 11.6 The acid-catalysed iodination of propanone

WHAT THIS SPEC POINT ASKS

Know how to get the data that give the orders with respect to propanone, iodine and hydrogen ions; use the orders to say which species are in the rate-determining step; and suggest a mechanism.

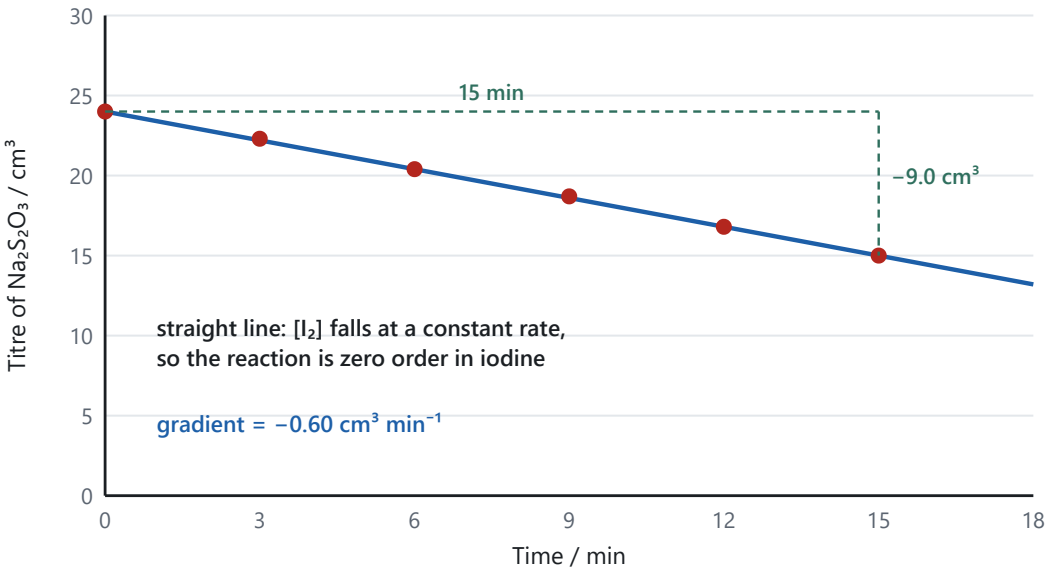
11.6 The reaction

$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$ (catalysed by H^+)

- The brown iodine colour fades as iodine is used up.
- H^+ is a **catalyst**: it is used in the mechanism and given back. (It is also a product, but with acid in large excess from the start its concentration hardly changes.)

11.6 Getting the orders (Core Practical 9a, see 11.12)

- **Iodine**: use **propanone and acid in large excess**, so their concentrations stay almost constant and only $[I_2]$ changes. Follow $[I_2]$ by sampling, quenching with $NaHCO_3$ and titrating with thiosulfate (or with a colorimeter). The plot of $[I_2]$ (or titre) against time is a **straight line**, so the rate doesn't depend on $[I_2]$: **zero order in iodine**.
- **Propanone**: repeat with double the propanone concentration (same acid and iodine). The gradient doubles: **first order**.
- **H^+**: repeat with double the acid concentration. The gradient doubles: **first order**.



Titre against time is a straight line: iodine is used at a constant rate, so the reaction is zero order in I_2.

rate = $k[CH_3COCH_3][H^+]$

11.6 What the rate equation says about the mechanism

- **Propanone and H^+** are in the rate equation, so they take part **in or before the rate-determining step**.
- **Iodine** isn't in the rate equation, so it reacts in a **fast step after** the RDS. Changing $[I_2]$ can't speed up a step it isn't part of.
- Overall order 2 → the RDS involves one propanone and one H^+ (directly, or via an intermediate formed from them in a fast step before it).

11.6 A mechanism that fits

1. **Fast, reversible**: a lone pair on the carbonyl oxygen accepts H^+.
   $CH_3COCH_3 + H^+ \rightleftharpoons CH_3C(OH^+)CH_3$
2. **Slow (RDS)**: a C–H bond on a CH_3 group breaks; H^+ is lost and the **enol** forms (C=C with an –OH on it).
   $CH_3C(OH^+)CH_3 \rightarrow CH_2=C(OH)CH_3 + H^+$
3. **Fast**: the electron-rich C=C of the enol attacks I_2; the O–H loses its H^+ and the C=O re-forms.
   $CH_2=C(OH)CH_3 + I_2 \rightarrow CH_3COCH_2I + H^+ + I^-$

Check it: the steps add up to the overall equation (H^+ cancels out as a catalyst); propanone and H^+ appear up to the slow step; I_2 appears only after it. Step 2 is slow because it breaks a strong C–H bond, so it has a high activation energy.

WORKED EXAMPLE: FROM TITRE GRADIENT TO RATE

10.0 cm^3 samples are titrated with 0.0100 mol dm^{-3} $Na_2S_2O_3$. The titre falls by 0.60 cm^3 every minute (graph above).
Thiosulfate per minute = $0.60 \times 10^{-3} \times 0.0100 = 6.0 \times 10^{-6}$ mol
I_2 per minute = $6.0 \times 10^{-6} \div 2 = 3.0 \times 10^{-6}$ mol (in 10.0 cm^3)
Change in $[I_2]$ per minute = $3.0 \times 10^{-6} \div 0.0100 = 3.0 \times 10^{-4}$ mol dm^{-3} min^{-1}
Rate = $3.0 \times 10^{-4} \div 60 =$ **5.0×10^{-6} mol dm^{-3} s^{-1}**

MARK-SCHEME LANGUAGE

- Large excess: "so only changes in iodine concentration affect the rate" / "the concentrations of propanone and acid remain (almost) constant".
- Zero order in iodine: "$[I_2]$ (or volume of thiosulfate) against time gives a straight line", or "rate against $[I_2]$ is a horizontal line".
- Quench: $HCO_3^-(aq) + H^+(aq) \rightarrow CO_2(g) + H_2O(l)$, with state symbols.
- Mechanism: the step with iodine isn't in the rate equation; the steps with propanone and H^+ are. The slow step "involves breaking a (strong) bond / has a high activation energy".

Often asked as: why the excess; which graph shows zero order; quench equation; which steps of a given mechanism fit the rate equation, and which is the RDS.

BEYOND THE SPEC, BUT EXAMINED

- Why **not** quench with NaOH: base also **catalyses** the reaction (so it wouldn't stop), and in alkali the iodoform reaction / multiple substitution happens (WCH14 Oct 2025 Q11).
- Showing the acid's anion (Cl^-) plays no part: change $[Cl^-]$ with NaCl, or use a different strong acid, and show the rate doesn't change (WCH14 Oct 2025 Q11).
- Mechanism MCQ: protonation of the C=O oxygen is fast; loss of H^+ to form the enol is the slow step (WCH14 Jan 2021 Q7).

COMMON MISTAKE

Concluding that iodine "doesn't react" because it is zero order. It reacts, but after the slow step.
Learning a mechanism by rote and not checking it against the rate equation and the overall equation.

SPEC 11.7 The rate-determining step from a rate equation, and back again

WHAT THIS SPEC POINT ASKS

Deduce the rate-determining step from a rate equation, and the rate equation from a given rate-determining step.

11.7 The rules

- The rate equation shows the species (and how many of each) that take part **in the rate-determining step or in fast steps before it**.
- **Order = number of particles of that species** that go into the RDS (directly or via an intermediate made from them before the RDS).
- Species that only react **after** the RDS are **zero order** and not in the rate equation.
- An **intermediate** is never written in the rate equation. Replace it with the species that made it.

11.7 Rate equation → slow step

Rate equation	What the slow step contains
rate = k[A]	one A only (A breaks up, e.g. a C–Br bond ionises)
rate = k[A][B]	one A and one B
rate = k[A]^2	two A
rate = k[A]^2[B]	two A and one B. A three-particle collision is unlikely, so expect 2A ⇌ A_2 (fast) then A_2 + B (slow)

11.7 Slow step → rate equation

Write the reactants of the slow step. If one is an intermediate, swap it for whatever formed it earlier.

WORKED EXAMPLE: IS THE MECHANISM CONSISTENT?

$2H_2O_2 \rightarrow 2H_2O + O_2$ is found to be **first order** in H_2O_2.
Mechanism X: slow step $H_2O_2 + H_2O_2 \rightarrow$ … (two H_2O_2). This would give rate = k$[H_2O_2]^2$, so it **isn't consistent**.
Mechanism Y: slow step $H_2O_2 \rightarrow H_2O + O$ (one H_2O_2), then fast $O + H_2O_2 \rightarrow H_2O + O_2$. This gives rate = k$[H_2O_2]$ and the steps add up to the overall equation, so it **is consistent**.

MARK-SCHEME LANGUAGE

- "the slow step … involves two molecules of H_2O_2 so the mechanism is not consistent with being a first-order reaction" (WCH14 Jun 2025).
- Two checks for "justify": "the steps add up to the overall equation" **and** "the rate equation matches the species in the slow step". Both are needed (WCH14 Oct 2023: only 14% gave both).

Often asked as: MCQ which step fits the rate equation; complete a two-step mechanism; justify whether a proposed mechanism agrees.

BEYOND THE SPEC, BUT EXAMINED

- An **intermediate** (e.g. N_2O_2) can appear in the slow step; the rate equation then shows the species that formed it (WCH14 Jan 2026 Q13).

COMMON MISTAKE

Checking only one thing (just the overall equation, or just the rate equation) in a "justify" question. Giving a stock answer without looking at the steps you've been given.

SPEC 11.8 Deducing a mechanism from the rate equation and the overall equation

WHAT THIS SPEC POINT ASKS

Use the rate equation and the balanced (stoichiometric) equation together to suggest a reaction mechanism.

11.8 Method

1. From the rate equation, write the **slow step**: it contains the species (and numbers) in the rate equation.
2. Use **fast steps** before or after it to account for everything else in the overall equation.
3. Check that the steps **add up to the overall equation** (intermediates and catalysts cancel).
4. Check that each step involves **no more than two particles** colliding.

11.8 Why most reactions have several steps

A balanced equation like $BrO_3^- + 5Br^- + 6H^+$ has 12 reactant particles. They can't all collide at once: a collision of even **three** particles at the same moment, in the right orientation and with enough energy, is very unlikely. So the reaction must happen in a series of steps involving one or two particles each.

WORKED EXAMPLE: THREE MECHANISMS

$2NO_2(g) + F_2(g) \rightarrow 2NO_2F(g)$, rate = $k[NO_2][F_2]$
Slow: $NO_2 + F_2 \rightarrow NO_2F + F$
Fast: $NO_2 + F \rightarrow NO_2F$
Sum: $2NO_2 + F_2 \rightarrow 2NO_2F$ ✓ (F is an intermediate)

$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$, rate = $k[NO_2]^2$ (zero order in CO)
Slow: $NO_2 + NO_2 \rightarrow NO_3 + NO$
Fast: $NO_3 + CO \rightarrow NO_2 + CO_2$
Sum: $NO_2 + CO \rightarrow NO + CO_2$ ✓ (CO only reacts after the slow step)

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$, rate = $k[NO]^2[O_2]$
Fast, reversible: $2NO \rightleftharpoons N_2O_2$
Slow: $N_2O_2 + O_2 \rightarrow 2NO_2$
Sum ✓. The slow step contains N_2O_2 (made from **2NO**) and O_2, which matches $[NO]^2[O_2]$ without needing a three-particle collision.

WORKED EXAMPLE: HYDROGEN PEROXIDE AND IODIDE

$H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$, rate = $k[H_2O_2][I^-]$ (zero order in H^+)
Slow: $H_2O_2 + I^- \rightarrow IO^- + H_2O$
Fast: $IO^- + H^+ \rightarrow HIO$
Fast: $HIO + I^- + H^+ \rightarrow I_2 + H_2O$
Sum: $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$ ✓

MARK-SCHEME LANGUAGE

- Must be multi-step: "there are only 4 particles in the rate equation and 12 in the equation for the reaction", or "collisions with more than two particles are unlikely" (WCH14 Jan 2022).
- "a three-particle collision is unlikely" (hard / difficult / impossible are allowed) (WCH14 Oct 2023).
- 2NO + O_2: "two moles of NO" in step 1 forming N_2O_2, and "O_2 in the rate-determining step": $N_2O_2 + O_2 \rightarrow 2NO_2$. A reversible arrow in step 1 is allowed (WCH14 Jan 2026).

Often asked as: explain why the reaction can't be one step; propose steps that fit a rate equation; say which step is slow.

COMMON MISTAKE

Linking "number of particles" questions to SN1/SN2 when the reaction has nothing to do with halogenoalkanes (examiner report, Jan 2022).
Pasting the stock "five particles can't collide at once" into a question about something else (e.g. the activation energy of a solid reaction).

SPEC 11.9 Rate equations as evidence for SN1 and SN2

WHAT THIS SPEC POINT ASKS

Understand how the rate equations for the hydrolysis of primary and tertiary halogenoalkanes give evidence for the SN2 and SN1 mechanisms.

11.9 The evidence

	Primary, e.g. 1-bromobutane	Tertiary, e.g. 2-bromo-2-methylpropane
Rate equation (with OH^-)	rate = k[RBr][OH^-]	rate = k[RBr]
Overall order	2	1
What's in the RDS	the halogenoalkane **and** OH^-	the halogenoalkane **only**
Mechanism	**SN2**: one step	**SN1**: two steps, first one slow
Energy profile	one hump (transition state)	two humps (carbocation intermediate)

S = substitution, N = nucleophilic, 1 or 2 = the number of species in the rate-determining step.

11.9 SN2 (primary): one step

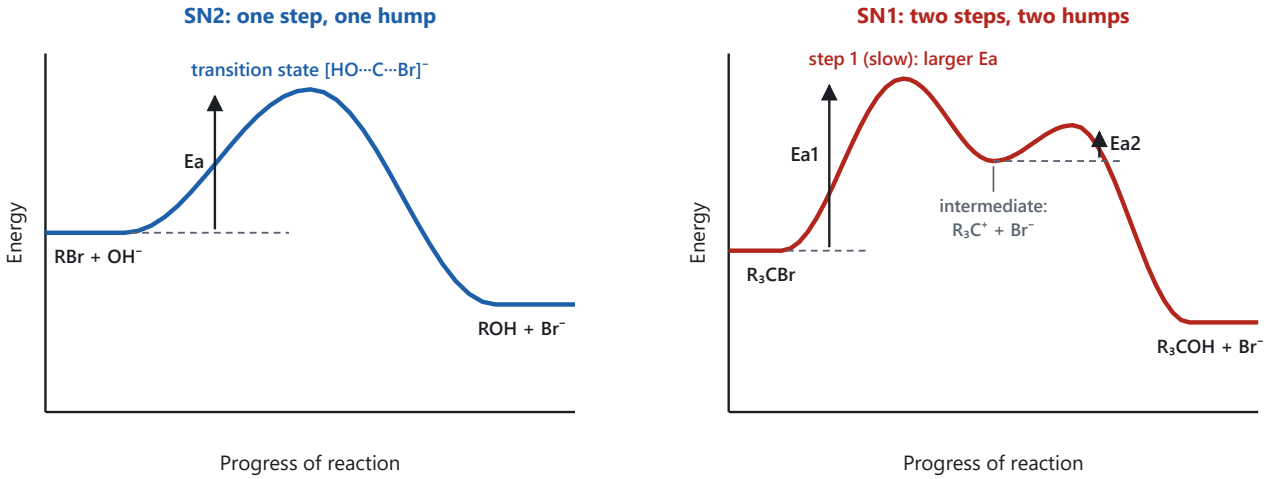
- The C–Br bond is polar: C is **δ+**, Br is **δ−**.
- A **lone pair on the oxygen of OH^-** attacks the δ+ carbon, from the side **opposite** the bromine.
- At the same time the **C–Br bond breaks** and both electrons go to Br, which leaves as Br^-.

- Halfway, there is a **transition state** [HO···C···Br]⁻: five groups round the carbon, two partial bonds, an overall **negative charge** (written outside square brackets).
- Both the halogenoalkane and OH^- are in this one slow step, so the reaction is first order in each.

11.9 SN1 (tertiary): two steps

1. **Slow**: the C–Br bond breaks **heterolytically** on its own. Both electrons go to Br; a **tertiary carbocation** $(CH_3)_3C^+$ and Br^- form. Only the halogenoalkane is involved, so rate = k[RBr].
2. **Fast**: a lone pair on OH^- (or on a water molecule) attacks the positive carbon to give the alcohol $(CH_3)_3COH$.

OH^- only reacts in the fast second step, so its concentration doesn't affect the rate.



SN2 has one transition state. SN1 has an intermediate carbocation between two transition states; the first activation energy is the larger, so step 1 is the slow step.

11.9 Why primary and tertiary go different ways

- **Tertiary → SN1**: three alkyl groups **push electrons** towards the positive carbon, so the **tertiary carbocation is stabilised**. The three bulky groups also block an OH^- attacking from the back (**steric hindrance**), so SN2 is difficult.
- **Primary → SN2**: a primary carbocation would be very unstable, but the carbon is easy to reach (little steric hindrance).
- **Secondary** halogenoalkanes can react by both.

WORKED EXAMPLE: DATA TO MECHANISM

Hydrolysis of 2-bromobutane with OH^-: doubling [2-bromobutane] doubles the rate; doubling $[OH^-]$ doubles the rate.
rate = $k[C_4H_9Br][OH^-]$, second order overall. Both species are in the slow step, so the mechanism is **SN2**: one step through a transition state.
If instead doubling $[OH^-]$ had **no effect**, rate = $k[C_4H_9Br]$, and the mechanism would be **SN1**: slow ionisation to a carbocation, then fast attack by OH^-.

MARK-SCHEME LANGUAGE

- "rate = k[RX][OH^-]" for primary, "rate = k[RX]" for tertiary; "primary SN2, tertiary SN1".
- Primary: the slow step is the halogenoalkane reacting with OH^- "via a transition state". DO NOT AWARD "intermediate" for transition state.
- Tertiary: the slow step forms "a carbocation (in a step that only involves the halogenoalkane)", then "the hydroxide ion attacks the carbocation in a (fast) second step".
- SN2 diagram: lone pair on O of OH^-; dipole on C–Br; arrow from lone pair to the δ+ carbon; arrow from the C–Br bond to Br; transition state in square brackets with the negative charge outside. A mechanism that doesn't match the given rate equation scores at most 2.
- SN1 profile: "two activation energies imply two steps / an intermediate"; "the first activation energy is larger so the first step is slower".

Often asked as: 6-mark answer comparing a primary and a tertiary bromoalkane (WCH14 Oct 2021 Q18, Jan 2025 Q18); deduce SN1 or SN2 from orders and draw it; MCQ; annotate an energy profile.

BEYOND THE SPEC, BUT EXAMINED

- **Steric hindrance** and **carbocation stability** as the reasons (WCH14 Oct 2025 Q6, Jan 2026 Q3).
- Optical activity: SN2 on one enantiomer gives **inversion** (one optically active product); SN1 goes through a **planar** carbocation that is attacked equally from both sides, giving a **racemic mixture** (WCH14 Oct 2024 Q19, Oct 2025 Q6). See 15.5.

COMMON MISTAKE

Drawing SN2 when the rate equation has no OH^- in it. Drawing SN2 without the transition state, or with arrows that don't start at a lone pair or a bond.
Writing SN1 without the Br^- leaving group, or without saying "slow" then "fast".
Using the "planar carbocation, attacked from above and below" answer in an SN2 question.

SPEC 11.10

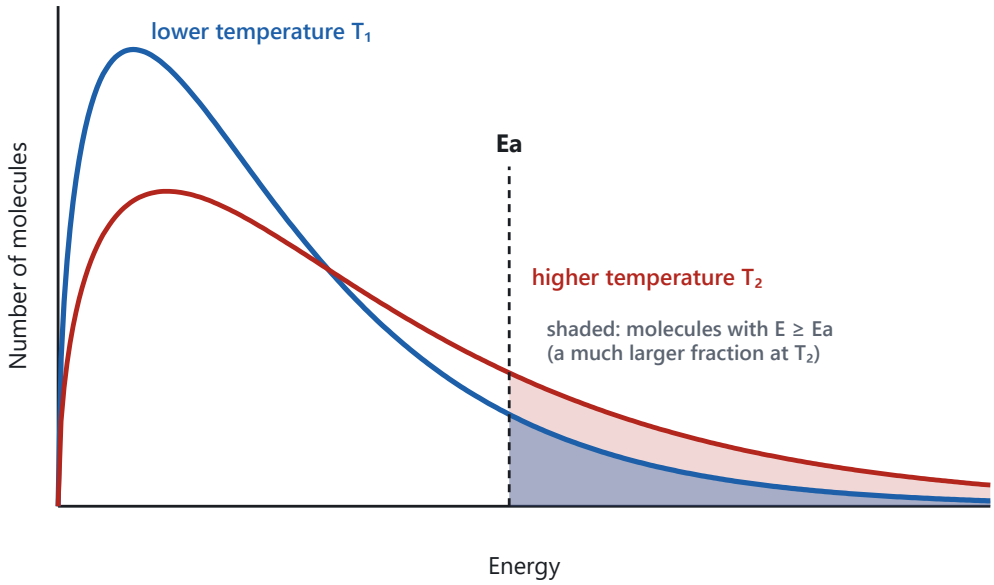
Finding the activation energy from experimental data

WHAT THIS SPEC POINT ASKS

Use calculations and graphs to find the activation energy of a reaction from experimental data. (The Arrhenius equation is given if you need it.)

11.10 Why k increases with temperature

- At a higher temperature the particles have more kinetic energy on average, so they collide more often, but that is a small effect.
- The big effect: a **much larger proportion of collisions have energy ≥ Ea** (the shaded area in the Maxwell–Boltzmann distribution grows a lot).
- So more collisions are successful per second and **k increases**. Roughly, a 10 K rise often doubles the rate.



At the higher temperature the curve flattens and shifts right; the area beyond Ea (molecules able to react) is much larger.

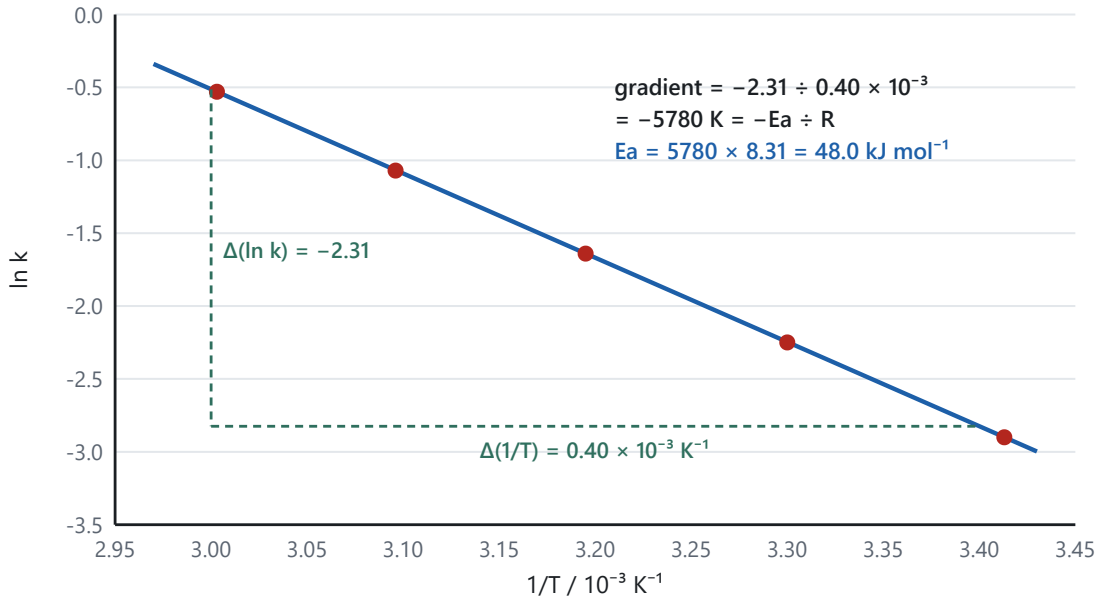
11.10 The Arrhenius equation

$k = A\, e^{-Ea/RT}$

- k = rate constant; A = the **pre-exponential (Arrhenius) constant**, linked to collision frequency and orientation; Ea in **J mol^{-1}**; R = 8.31 J K^{-1} mol^{-1}; T in **kelvin**.
- Take natural logs: **ln k = ln A − Ea/(RT)**, which is the straight-line form **ln k = (−Ea/R) × (1/T) + ln A**.

11.10 Graphical method

1. For each temperature, find k (or the rate, or 1/t for a clock reaction: at fixed concentrations all three are proportional to k).
2. Calculate **1/T** (T in K) and **ln k**.
3. Plot **ln k (y-axis) against 1/T (x-axis)**. You get a straight line with a **negative gradient**.
4. **Gradient = −Ea/R**, so **Ea = −gradient × R**. The y-intercept is ln A.



Use a large gradient triangle on the best-fit line, not two data points. 1/T values are small, so watch the power of ten.

WORKED EXAMPLE: EA FROM A GRAPH

- T = 293 K: k = 5.49 × 10^{-2} s^{-1}; 1/T = 3.413 × 10^{-3} K^{-1}; ln k = −2.90
- T = 303 K: k = 0.105 s^{-1}; 1/T = 3.300 × 10^{-3} K^{-1}; ln k = −2.25
- T = 313 K: k = 0.193 s^{-1}; 1/T = 3.195 × 10^{-3} K^{-1}; ln k = −1.64
- T = 323 K: k = 0.342 s^{-1}; 1/T = 3.096 × 10^{-3} K^{-1}; ln k = −1.07
- T = 333 K: k = 0.586 s^{-1}; 1/T = 3.003 × 10^{-3} K^{-1}; ln k = −0.53

Gradient (from the line, between 1/T = 3.00 × 10^{-3} and 3.40 × 10^{-3}) = −2.31 ÷ (0.40 × 10^{-3}) = −5780 K
Ea = −(−5780) × 8.31 = 48 000 J mol^{-1} = **+48.0 kJ mol^{-1}** (3 SF)

WORKED EXAMPLE: TWO TEMPERATURES, NO GRAPH

The rate constant doubles between 298 K and 308 K. Find Ea.
Write ln k = ln A − Ea/(RT) for each temperature and subtract (ln A cancels):
$\ln(k_2/k_1) = (Ea/R)(1/T_1 - 1/T_2)$
ln 2 = (Ea ÷ 8.31) × (1/298 − 1/308)
0.693 = (Ea ÷ 8.31) × 1.0895 × 10^{-4}
Ea = 0.693 × 8.31 ÷ (1.0895 × 10^{-4}) = 52 900 J mol^{-1} = **+52.9 kJ mol^{-1}**

WORKED EXAMPLE: ONE POINT, A GIVEN

k = 3.2 × 10^{-4} s^{-1} at 500 K and A = 1.0 × 10^{13} s^{-1}. Find Ea.
ln k = ln A − Ea/(RT), so Ea = RT(ln A − ln k) = 8.31 × 500 × (29.93 − (−8.05))
= 8.31 × 500 × 37.98 = 157 800 J mol^{-1} = **+158 kJ mol^{-1}**

MARK-SCHEME LANGUAGE

- Graph: ln k on the y-axis, **1/T / K^{-1}** on the x-axis (ln k has no unit), points covering at least half the grid both ways, plotted to ±½ square, straight best-fit line. A capital T, not t.
- Gradient: "measures the increase in ln k and the increase in 1/T"; negative sign; unit **K**.
- "Ea = −gradient × R"; positive answer; units kJ mol^{-1} or J mol^{-1}; usually 3 SF.
- Large Ea → "kinetically stable / rate is low".

Often asked as: plot ln k against 1/T from a table and calculate Ea (most papers); MCQ for the expression (−gradient × R); one-point rearrangement; two-temperature calculation; spot the anomalous point.

BEYOND THE SPEC, BUT EXAMINED

- Plotting **ln(rate)** or ln(1/t) instead of ln k: same gradient, so the same Ea. Label the axis with what you actually plotted (WCH14 Jan 2024 Q16).
- The **two-temperature** subtraction method (WCH14 Jan 2023 Q17).
- Which bond breaks first in a decomposition: the **weakest** one, from bond enthalpy data (WCH14 Oct 2022 Q16).

COMMON MISTAKE

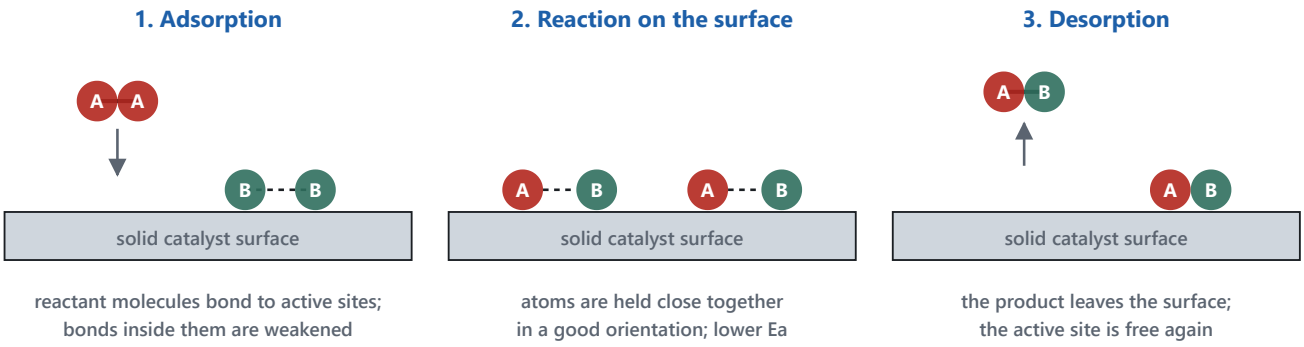
Axes the wrong way round; a scale that bunches all the points in one corner; no unit on 1/T.
Using one data pair when a graph is asked for. Reading the gradient off two data points instead of the line.
A negative Ea, J and kJ mixed up, more than 3 SF, or °C instead of K.

SPEC 11.11 Solid (heterogeneous) catalysts in industry

WHAT THIS SPEC POINT ASKS

Understand how a solid catalyst works for an industrial gas-phase reaction, by providing a surface for the reaction.

11.11 Three steps at the surface



Reactants adsorb, react on the surface, then the product desorbs and frees the active site.

1. **Adsorption**: gas molecules form weak bonds to **active sites** on the catalyst surface. This **weakens the bonds inside the reactant molecules** and holds them close together in a favourable orientation.
2. **Reaction** on the surface, by a route with a **lower activation energy**.
3. **Desorption**: the product molecules break away from the surface, leaving the site free for more reactant.

11.11 Making a good industrial catalyst

- **Large surface area**: finely divided powder, pellets, or a thin coat on a **ceramic honeycomb** (catalytic converters). More surface → more active sites → faster reaction, and less of an expensive metal is needed.
- The right **strength of adsorption**: strong enough to weaken bonds, weak enough for products to leave.
- **Poisoning**: impurities (e.g. sulfur compounds; lead in old petrol on catalytic converters) adsorb strongly and block active sites. Feed gases are purified to prevent this.

Process	Catalyst
Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$	iron
Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$	vanadium(V) oxide, V_2O_5
Hydrogenation of C=C (margarine)	nickel
Catalytic converter: CO, NO_x, hydrocarbons	platinum, palladium, rhodium

A catalyst speeds up the forward and reverse reactions **equally**: equilibrium is reached sooner, but the position and K are unchanged (13.4, 13.5).

MARK-SCHEME LANGUAGE

- "reactants adsorb onto the (palladium) catalyst surface"; "this weakens bonds in the reactants"; "products then desorb (from the catalyst surface)" (WCH14 Jun 2021 Q16).
- Catalyst and K: "no effect (on Kc) because the catalyst speeds up both the forward and backward reactions equally" (WCH14 Oct 2025).

Often asked as: explain how a solid catalyst works (3 marks); how to stop a reaction with a solid catalyst (filter it off); catalyst and equilibrium yield.

COMMON MISTAKE

Writing "absorb" for "adsorb". Adsorption is sticking to the **surface**.
Saying a catalyst "increases the yield". It only makes equilibrium arrive faster.

SPEC 11.12

Core Practicals 9a and 9b: iodine–propanone by titration, and an iodine clock

WHAT THIS SPEC POINT ASKS

Core Practical 9a: follow the rate of the iodine–propanone reaction by a titration method. Core Practical 9b: investigate a clock reaction (Harcourt–Esson iodine clock).

CORE PRACTICAL 9A: IODINE AND PROPANONE BY TITRATION

Aim: find the order with respect to iodine (and, with extra runs, propanone and H^+).

1. Mix known volumes of **propanone** and **sulfuric (or hydrochloric) acid** in a flask; both are in **large excess** over iodine.
2. Add a known volume of **iodine solution** and start the clock at the moment of mixing. Swirl.
3. Every few minutes, withdraw a **10.0 cm^3 sample** with a pipette and run it into a flask of **sodium hydrogencarbonate** solution. This **quenches** the reaction by neutralising the acid catalyst. Note the exact time.
4. Titrate the iodine left in each sample with standard **sodium thiosulfate**. Add **starch** near the end point (when the solution is pale yellow): end point **blue-black → colourless**.
5. Plot titre ($\propto [I_2]$) against time. A **straight line** shows zero order in iodine; the gradient gives the rate (worked example in 11.6).
6. Repeat with a different starting [propanone] (then a different $[H^+]$), everything else the same, and compare gradients to find those orders.

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

Control: temperature (water bath), total volume, the volume of each sample, the concentrations of the excess reagents.
Safety: propanone is highly flammable (no flames); iodine solution stains and irritates; wear eye protection.

CORE PRACTICAL 9B: AN IODINE CLOCK

Aim: find the order with respect to one reagent (e.g. iodide) by timing a colour change.
Reaction (one common version): $H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$, slow
Clock step: $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$, very fast

1. In a beaker, mix potassium iodide, sulfuric acid, a **small fixed volume** of sodium thiosulfate, starch and water.
2. Add the hydrogen peroxide, start the clock and stir.
3. Stop the clock when the mixture suddenly turns **blue-black**. Record t.
4. Repeat with different volumes of KI, making up the **same total volume** with water so only [I^-] changes. Keep the temperature constant.
5. Use **1/t as the rate**. Plot 1/t against [I^-] (or volume of KI): a straight line through the origin shows first order.

Why it works: the thiosulfate removes iodine as fast as it forms, so no blue colour appears. When the thiosulfate runs out, the next iodine formed reacts with starch. The thiosulfate is used up **before the other concentrations have changed much**, so 1/t measures the **initial rate**.
Another version uses peroxodisulfate: $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$.

11.12 Making the results reliable

- Same total volume in every run, measured with burettes or graduated pipettes.
- Thermostat the solutions before mixing; record the temperature.
- Start timing at the moment of mixing; judge the colour change against a white tile. Human reaction time matters most for short times, so choose concentrations that give times of tens of seconds at least.
- Repeat each run and take a mean.

MARK-SCHEME LANGUAGE

- Clock: "sodium thiosulfate reacts with the iodine formed"; "when the thiosulfate is used up the iodine reacts with starch giving a blue-black colour"; "1/time is a measure of rate".
- Large excess: "so the concentrations of propanone and acid remain (almost) constant".
- Colour change in the titration end point and in the clock: give both colours in the right order.

Often asked as: describe how a clock reaction measures an initial rate; calculate rates from clock times; use titration data to show zero order in iodine; explain each step of the method (WCH14 Jan 2023 Q17, Jan 2024 Q16, Oct 2025 Q11).

COMMON MISTAKE

Saying the blue-black colour appears "when the iodine is used up". It appears when the **thiosulfate** is used up.
Changing the volume of one reagent without making up the total volume with water: then every concentration changes.

SPEC 11.13 Core Practical 10: finding the activation energy

WHAT THIS SPEC POINT ASKS

Core Practical 10: find the activation energy of a reaction.

11.13 Principle

Keep **every concentration the same** and change only the **temperature**. Then the rate (or 1/t for a clock reaction) is proportional to k, so ln(1/t) against 1/T gives a line of gradient −Ea/R.

CORE PRACTICAL 10: METHOD IN BRIEF

A common version uses a clock reaction between bromide and bromate(V) ions in acid:
$5Br^- + BrO_3^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$
A small fixed amount of **phenol** removes the bromine as it forms. When the phenol is used up, bromine **bleaches methyl red** indicator: time how long the red colour takes to disappear.

1. Put the bromide/bromate mixture, phenol and methyl red in one boiling tube and the sulfuric acid in another.
2. Stand both tubes in a **water bath** until they reach the bath temperature. Record it with a thermometer in the mixture.
3. Mix, start the clock, and stop it when the red colour disappears. Record t.
4. Repeat at about five temperatures between room temperature and about 60 °C, with **identical volumes** each time.
5. Calculate 1/T (T in K) and ln(1/t). Plot ln(1/t) against 1/T; Ea = −gradient × R.

Other versions: the peroxodisulfate/iodide clock at different temperatures; magnesium ribbon in hydrochloric acid (time for the ribbon to disappear).
Safety: phenol is toxic and corrosive (gloves, eye protection); bromine fumes; hot water.

WORKED EXAMPLE: FROM CLOCK TIMES TO EA

At 20 °C, t = 125 s; at 40 °C, t = 34.6 s.
T_1 = 293 K, $1/T_1 = 3.413 \times 10^{-3}$ K^{-1}, $\ln(1/t_1) = \ln(0.00800) = -4.83$
T_2 = 313 K, $1/T_2 = 3.195 \times 10^{-3}$ K^{-1}, $\ln(1/t_2) = \ln(0.0289) = -3.54$
Gradient = $(-3.54 - (-4.83)) \div (3.195 \times 10^{-3} - 3.413 \times 10^{-3}) = 1.29 \div (-0.218 \times 10^{-3}) = -5920$ K
Ea = 5920 × 8.31 = 49 200 J mol^{-1} ≈ **+49 kJ mol^{-1}**
(In the exam you'd plot all five points and take the gradient from the line.)

11.13 Errors and improvements

- Temperature drifts while the reaction runs: use a thermostatted bath; record the temperature at the start and end and use the mean.
- Judging the end point: use a white tile; a colorimeter removes the judgement.
- Mixing time is significant at high temperatures (short times): choose concentrations that give longer times.

MARK-SCHEME LANGUAGE

- The practical procedure is rarely examined; the **data analysis** is (see 11.10): axes, scale, best-fit line, gradient with sign and unit (K), Ea = −gradient × R, sign, units, SF.

Often asked as: plot ln k or ln(rate) against 1/T from given data and find Ea (WCH14 Jan 2024 Q16, Oct 2025 Q2).

COMMON MISTAKE

Changing a concentration as well as the temperature: then the rate isn't proportional to k.
Using °C for T, or forgetting that ln(1/t) is negative when t > 1 s.

Topic 12 – Entropy and Energetics

SPEC 12.1 Enthalpy alone doesn't decide whether a reaction happens

WHAT THIS SPEC POINT ASKS

Understand that, because some endothermic reactions happen on their own at room temperature, the enthalpy change can't be the only thing that controls whether a reaction occurs.

12.1 Endothermic but spontaneous

If "downhill in enthalpy" were the whole story, no endothermic change could ever happen by itself. Plenty do:

Change	What you see	Why it can happen
Ammonium nitrate dissolving in water (cold packs)	solution gets cold	solid lattice → free ions in solution
$Ba(OH)_2{\cdot}8H_2O(s) + 2NH_4Cl(s) \rightarrow BaCl_2 + 2NH_3(g) + 10H_2O(l)$	gets so cold that water freezes the flask to a wet block	3 moles of solid → 13 moles, including a gas and a liquid
$2CH_3COOH + (NH_4)_2CO_3 \rightarrow 2CH_3COONH_4 + H_2O + CO_2(g)$	fizzes and cools	a gas is made
Ice melting above 0 °C		solid → liquid

What these have in common: the products are **more disordered** than the reactants (more particles, more freedom of movement, gases formed). The extra quantity that decides the direction of change is **entropy** (12.2).

- **Feasible** (spontaneous) means a change can happen without a continuous input of energy from outside. It says nothing about how **fast** (12.11).

MARK-SCHEME LANGUAGE

- "ΔSsystem must be positive"; "(as the compound does dissolve) ΔStotal is > 0"; "TΔSsystem larger than ΔH" also allowed (WCH14 Jun 2021 Q15).
- Endothermic reaction of solids that makes a gas: "moles increase from 4 to 7" **and** "a gas (and liquid products) are produced (from solids)". Give both reasons (WCH14 Jan 2025 Q19).

Often asked as: explain why an endothermic dissolving or reaction is feasible; MCQ on which statement explains an endothermic change.

COMMON MISTAKE

Saying ΔSsurroundings is positive for an endothermic change. Heat is taken **from** the surroundings, so ΔSsurroundings is **negative**; it's the system's entropy that makes up for it.

SPEC 12.2 Entropy: dispersal of particles and energy

WHAT THIS SPEC POINT ASKS

Understand entropy, S, as a measure of the disorder of a system: the random spreading out of molecules and of quanta of energy among them.

12.2 What entropy measures

- **Entropy (S)** measures disorder: the **number of ways the particles can be arranged and the energy can be shared out among them**.
- Energy in molecules comes in fixed packets (**quanta**). The more particles there are, and the more freely they move, the more ways there are to share the quanta. The more ways there are, the more likely that state is, and the higher its entropy.
- Units: **J K^{-1} mol^{-1}** (note: joules, not kilojoules).

12.2 Counting ways: a feel for it

Share **3 identical quanta** between particles:

Particles	Possible shares	Number of ways
2	(3,0) (2,1) (1,2) (0,3)	4
3	(3,0,0) (0,3,0) (0,0,3) (2,1,0) (2,0,1) (1,2,0) (0,2,1) (1,0,2) (0,1,2) (1,1,1)	10

Add one particle and the number of ways more than doubles. With real numbers of particles (10^{23}) the number of ways is astronomically large, so a system almost always ends up in the arrangement with the most ways: the most disordered one.

EXAM TIP

Two ideas earn the mark: **disorder** (arrangement of particles) and the **number of ways of distributing energy quanta**. Use either, but tie it to the specific change (e.g. "fewer moles of gas, so fewer ways of distributing energy").

MARK-SCHEME LANGUAGE

- "disorder / number of ways of distributing energy quanta decreases as the number of molecules (of gas) decreases". DO NOT AWARD just "ΔSsystem is negative" (WCH14 Oct 2020 Q19).
- "a measure of the number of ways that energy can be distributed or particles arranged" (examiner report, Jun 2024).
- "moles (decrease) from 5 to 3", "decrease in disorder" (WCH14 Jun 2024 Q15).

Often asked as: explain the sign of ΔSsystem in words; MCQ ordering entropies.

BEYOND THE SPEC, BUT EXAMINED

- Boltzmann's equation S = k ln W (W = number of ways) is a neat summary but isn't required. The **number-of-ways wording** is rewarded (examiner report, Jun 2024).

COMMON MISTAKE

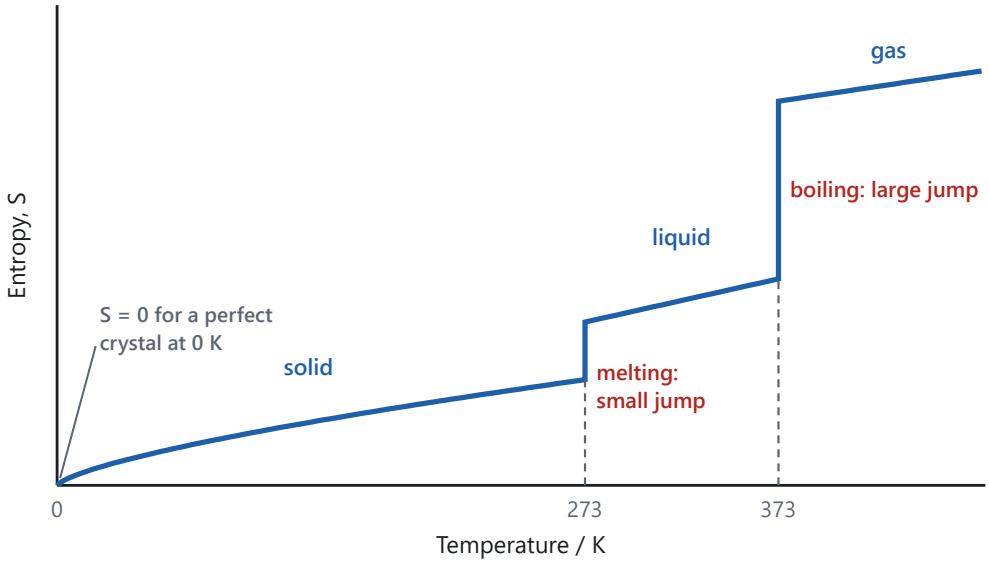
Talking only about physical states when the question is about a change in the **number of moles of gas** (examiner report, Jun 2024).

SPEC 12.3

How entropy depends on temperature and state

WHAT THIS SPEC POINT ASKS

Know that a substance's entropy increases with temperature, increases from solid → liquid → gas, and that a perfect crystal at 0 K has zero entropy.



Entropy rises steadily with temperature, jumps a little at melting and a lot at boiling.

- **0 K, perfect crystal: S = 0.** Every particle is fixed in a perfect lattice and there's no energy to share, so there is only **one** possible arrangement.
- **Heating** a substance gives its particles more energy quanta to share, so S rises gradually.
- **Melting**: particles go from fixed positions to moving past each other: a **small jump** in S.
- **Boiling**: particles become completely free and far apart: a **big jump** in S. So S(gas) ≫ S(liquid) > S(solid).

12.3 Comparing entropies without data

1. **State** first: gas ≫ liquid > solid.
2. Same state: **more atoms per molecule** (more complex) → more ways to store energy (rotating, vibrating) → higher S. Heavier particles also tend to have higher S.
3. Hard, strongly bonded solids (diamond) have very low S; softer solids higher.

Substance	$S^{\ominus}$ / J K^{-1} mol^{-1}
C(s, diamond)	2.4
C(s, graphite)	5.7
Mg(s)	32.7
NaCl(s)	72.1
$H_2O(l)$	69.9
$Br_2(l)$	152.2
$H_2O(g)$	188.7
$O_2(g)$	205.0
$CO_2(g)$	213.6
$Br_2(g)$	245.4

MARK-SCHEME LANGUAGE

- Sketch S against T: "general shape of increase from left to right" (a straight line is allowed); "two vertical stages for melting and boiling"; mark 273 K and 373 K for water (WCH14 Jun 2021 Q20).
- Ordering: larger, more complex gas molecules have more entropy than simple gases; a liquid metal such as Hg has less than most gases (WCH14 Jan 2025 Q14).

Often asked as: sketch S against T; MCQ ordering substances by entropy (e.g. Os < Hg < He < O_2).

COMMON MISTAKE

Saying elements have zero entropy, by analogy with ΔfH. Only a **perfect crystal at 0 K** has S = 0; every substance at 298 K has a positive S.

SPEC 12.4

The natural direction of change: total entropy increases

WHAT THIS SPEC POINT ASKS

Interpret the natural direction of change as the direction in which total entropy increases (ΔStotal positive), including gases spreading through a room.

- Left alone, changes go the way that **increases the total entropy** (system + surroundings). That is the second law of thermodynamics in Unit 4 language.
- **Gas spreading**: open a jar of bromine in a room. Bromine molecules spread until they fill the room evenly. They never gather back into the jar, because the spread-out state has vastly more arrangements.
  - One molecule: chance of finding it in the original half of a container = ½.
  - N molecules: chance of finding them all there = $(½)^N$. For N = 100 that is about 10^{-30}; for a mole, effectively zero.
- Heat flows from hot to cold for the same reason: the energy quanta become spread over more particles.
- Feasible means **ΔStotal > 0**. At equilibrium the total entropy of the mixture is at its maximum and nothing further changes overall (13.8).

MARK-SCHEME LANGUAGE

- "chemical reactions are feasible when ΔStotal is positive" (WCH14 Jan 2026 Q17).

Often asked as: the opening point of a feasibility answer; rarely asked on its own.

COMMON MISTAKE

Saying a reaction happens because "the system's entropy increases". The test is the **total** entropy.

SPEC 12.5

Entropy changes: changes of state, dissolving and changing numbers of moles

WHAT THIS SPEC POINT ASKS

Explain why entropy changes during (i) changes of state, (ii) dissolving a solid ionic lattice, (iii) reactions where the number of moles changes from reactants to products.

12.5 (i) Changes of state

Change	ΔSsystem	Why
melting, boiling, subliming	**positive**	particles gain freedom; boiling gives the biggest increase
freezing, condensing	**negative**	particles become more ordered

12.5 (ii) Dissolving an ionic solid

Two effects pull in opposite directions:

- The **lattice breaks up**: ions go from fixed positions to moving freely in solution → entropy **increases**.
- Water molecules become **ordered round each ion** (hydration shells) → entropy **decreases**. This effect is biggest for **small, highly charged ions** (Mg^{2+}, Al^{3+}), which hold water tightly.

So ΔSsystem for dissolving is usually positive (e.g. about +43 J K^{-1} mol^{-1} for NaCl, +109 for NH_4NO_3) but can be negative (about −115 for $MgCl_2$). Don't assume the sign.

12.5 (iii) Changes in the number of moles

- **More moles of gas** in the products → ΔSsystem **positive** and large.
- Fewer moles of gas → negative. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$: 4 moles of gas → 2, so ΔSsystem is negative (−199 J K^{-1} mol^{-1}).
- With no gases, count all moles, and look at state changes: $Ba(OH)_2{\cdot}8H_2O(s) + 2NH_4Cl(s) \rightarrow BaCl_2(s) + 2NH_3(g) + 10H_2O(l)$ goes from 3 moles of solid to 13 moles, including a gas.

EXAM TIP

Predict the sign in this order: (1) any change in moles of **gas**? That dominates. (2) Any solid → liquid or solution? (3) Then total moles.

MARK-SCHEME LANGUAGE

- "1 mole of reactant becomes 7 moles of products (and more moles have higher entropy)" OR "a solid becomes (a solid and) a liquid (and liquids have higher entropy than solids)" (WCH14 Oct 2025 Q9). "mols / molecules / particles" are all allowed.
- "one molecule (of reactant) forms two molecules (of product) in reaction 1"; "no change in the number of molecules in reaction 2" (WCH14 Oct 2022 Q17).

Often asked as: predict and explain the sign of ΔSsystem with no data; explain why solids don't react but their solutions do.

BEYOND THE SPEC, BUT EXAMINED

- Why a mixture of two ionic solids doesn't react but their solutions do: "in solids the particles are fixed in a lattice (so unable to move)", "so they cannot collide and react", "but when dissolved the particles are mobile" (WCH14 Jan 2025 Q19). This is AS collision theory applied here.

COMMON MISTAKE

Vague "it's more disordered" answers. **Count** the molecules or name the change of state.

SPEC 12.6

Total entropy change = system + surroundings

WHAT THIS SPEC POINT ASKS

Know that the total entropy change of a reaction is the sum of the entropy change of the system and that of the surroundings: ΔStotal = ΔSsystem + ΔSsurroundings.

ΔStotal = ΔSsystem + ΔSsurroundings

- **System** = the reacting chemicals. ΔSsystem comes from the entropies of products and reactants (12.7).
- **Surroundings** = everything else (the solvent, flask, air). ΔSsurroundings comes from the heat the reaction gives out or takes in (12.8).
- Only **ΔStotal** decides feasibility: positive means feasible.

Exothermic or endothermic?	ΔSsurroundings
Exothermic (ΔH < 0): heat goes **into** the surroundings	**positive**
Endothermic (ΔH > 0): heat comes **out of** the surroundings	**negative**

MARK-SCHEME LANGUAGE

- "ΔStotal = ΔSsystem + ΔSsurroundings and ΔSsurroundings = −ΔH ÷ T" (WCH14 Jan 2026 Q17).

Often asked as: the first indicative point in a feasibility 6-mark answer.

COMMON MISTAKE

Adding ΔSsystem in J K^{-1} mol^{-1} to ΔSsurroundings in kJ K^{-1} mol^{-1}. Put both in **J K^{-1} mol^{-1}** before adding.

SPEC 12.7 Calculating ΔSsystem

WHAT THIS SPEC POINT ASKS

Calculate the entropy change of the system for a reaction from the entropies of the reactants and products.

ΔSsystem = ΣS(products) − ΣS(reactants)

- **Multiply** each S value by its number of moles in the equation.
- Elements have **non-zero** entropies: include them.
- Use the value for the right **state** ($H_2O(l)$ 69.9, $H_2O(g)$ 188.7).
- Units J K^{-1} mol^{-1}, with a **sign**.

WORKED EXAMPLE: THREE REACTIONS

Haber process: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
S: N_2 191.6, H_2 130.6, NH_3 192.3 J K^{-1} mol^{-1}
ΔSsystem = (2 × 192.3) − (191.6 + 3 × 130.6) = 384.6 − 583.4 = **−198.8 J K^{-1} mol^{-1}** (4 mol gas → 2 mol gas)

Decomposing limestone: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
S: $CaCO_3$ 92.9, CaO 39.7, CO_2 213.6
ΔSsystem = (39.7 + 213.6) − 92.9 = **+160.4 J K^{-1} mol^{-1}** (a gas is made)

Burning magnesium: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
S: Mg 32.7, O_2 205.0, MgO 26.9
ΔSsystem = (2 × 26.9) − (2 × 32.7 + 205.0) = 53.8 − 270.4 = **−216.6 J K^{-1} mol^{-1}** (a gas is used up)

MARK-SCHEME LANGUAGE

- "Σ entropy of products − Σ entropy of reactants" with values substituted, e.g. (109.2 + 6 × 69.9) − 343 = +185.6 J K^{-1} mol^{-1} (WCH14 Oct 2025 Q9).
- Sign **and** units J K^{-1} mol^{-1}. "Penalise incorrect units once only."

Often asked as: in almost every paper, as the first part of a ΔStotal or feasibility-temperature calculation.

BEYOND THE SPEC, BUT EXAMINED

- Working backwards to the **S value of one substance** from a given ΔSsystem (WCH14 Jan 2022 Q22).
- Finding ΔSsystem from ΔStotal − ΔSsurroundings (WCH14 Oct 2020 Q19).

COMMON MISTAKE

Forgetting to multiply by 2 or 3. Subtracting products from reactants. Quoting 4 or more SF when the data are to 3 or 4 SF.

SPEC 12.8 ΔSsurroundings = −ΔH/T, and ΔStotal

WHAT THIS SPEC POINT ASKS

Calculate the entropy change of the surroundings using ΔSsurroundings = −ΔH/T, and hence ΔStotal.

ΔSsurroundings = −ΔH ÷ T

- ΔH must be in **J mol^{-1}** (multiply kJ by 1000). T in **kelvin** (°C + 273).
- The **minus sign**: an exothermic reaction (ΔH negative) gives heat to the surroundings, so their entropy **rises**.
- Dividing by **T**: the same amount of heat makes a **bigger difference to cold surroundings** than to hot ones, which are already highly disordered. (Like a shout in a quiet library compared with a busy street.)

WORKED EXAMPLE: ΔSTOTAL FOR THE HABER PROCESS AT 298 K

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$; ΔH = −92.2 kJ mol^{-1}; ΔSsystem = −198.8 J K^{-1} mol^{-1}
ΔSsurroundings = −ΔH ÷ T = −(−92 200) ÷ 298 = **+309.4 J K^{-1} mol^{-1}**
ΔStotal = −198.8 + 309.4 = **+110.6 J K^{-1} mol^{-1}**
Positive, so the reaction is **feasible** at 298 K (although it is extremely slow without a catalyst).

WORKED EXAMPLE: ΔSTOTAL FOR BURNING MAGNESIUM

$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$; ΔH = −1203.4 kJ mol^{-1}; ΔSsystem = −216.6 J K^{-1} mol^{-1}
ΔSsurroundings = +1 203 400 ÷ 298 = +4038 J K^{-1} mol^{-1}
ΔStotal = −216.6 + 4038 = **+3822 J K^{-1} mol^{-1}**: hugely positive, so feasible.

WORKED EXAMPLE: BACK TO ΔH

An endothermic reaction at 298 K has ΔSsurroundings = −234.9 J K^{-1} mol^{-1}. Find ΔH.
ΔH = −T × ΔSsurroundings = −298 × (−234.9) = +70 000 J mol^{-1} = **+70.0 kJ mol^{-1}**

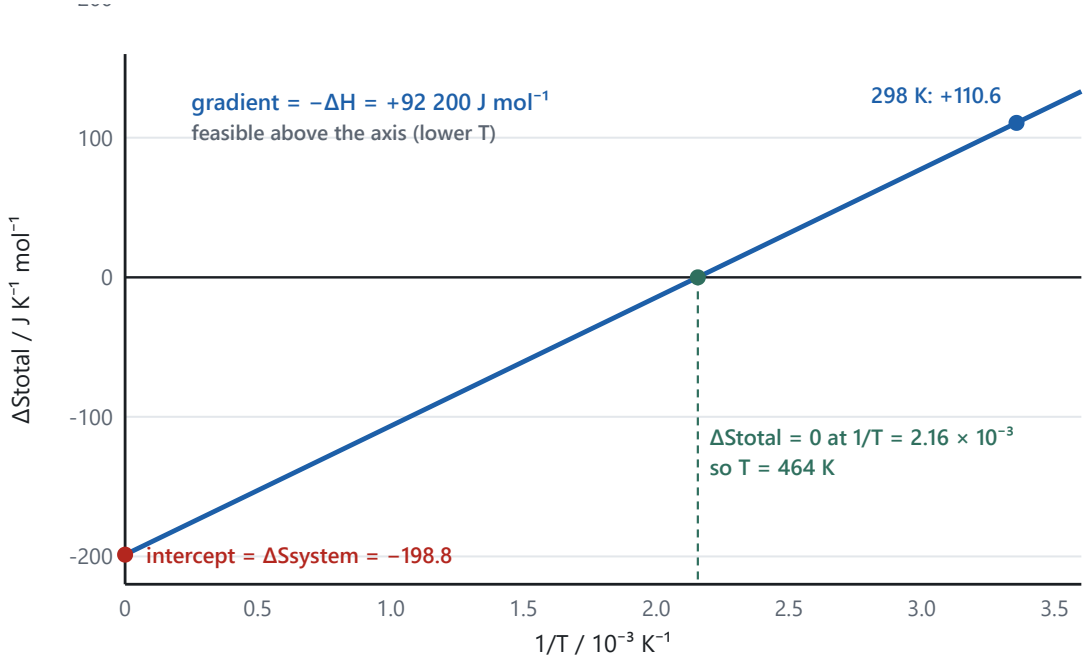
MARK-SCHEME LANGUAGE

- "= −(−631.3 × 10^3) ÷ 298 = +2118.5 J K^{-1} mol^{-1}" (WCH14 Jan 2020 Q19): the kJ → J conversion and K are both checked.
- Temperature in °C must be converted, e.g. 23.0 + 273 = 296 K (WCH14 Jun 2024 Q15).
- Sign **and** units required, e.g. −295.64 J K^{-1} mol^{-1} or −0.29564 kJ K^{-1} mol^{-1} (WCH14 Oct 2025 Q9).

Often asked as: calculate ΔSsurroundings and ΔStotal and comment on feasibility; work backwards to ΔH.

BEYOND THE SPEC, BUT EXAMINED

- **ΔStotal against 1/T graph.** ΔStotal = ΔSsystem − ΔH × (1/T), a straight line: **gradient = −ΔH**, **intercept = ΔSsystem**, and where the line crosses ΔStotal = 0 gives the temperature at which feasibility changes (WCH14 Jan 2022 Q22).
- Back-calculating ΔH = −ΔSsurroundings × T (WCH14 Jan 2025 Q19).



$N_2 + 3H_2 \rightarrow 2NH_3$: gradient = $-\Delta H = +92\,200$ J mol^{-1}, intercept = $\Delta S_{system} = -198.8$ J K^{-1} mol^{-1}. ΔS_{total} is positive (feasible) below 464 K, i.e. to the right of $1/T = 2.16 \times 10^{-3}$ K^{-1}.

COMMON MISTAKE

The top three: forgetting ×1000; dropping the minus sign (especially when ΔH is already negative: −(−) = +); using °C.

SPEC 12.9

Feasibility: balancing the entropy terms, and the effect of temperature

WHAT THIS SPEC POINT ASKS

Understand that whether a reaction is feasible depends on (i) the balance between ΔSsystem and ΔSsurroundings, so even endothermic reactions can be spontaneous, and (ii) temperature, because a higher temperature makes ΔSsurroundings smaller in magnitude. Calculate the temperature at which a reaction becomes feasible. (ΔG = ΔH − TΔSsystem may be used but isn't required.)

12.9 The four combinations

ΔSsystem hardly changes with temperature (unless something changes state), and neither does ΔH. Only the ΔSsurroundings term, −ΔH/T, changes, and **it gets smaller in size as T rises**.

ΔH	ΔSsystem	ΔSsurroundings	Feasible?
negative (exo)	positive	positive	**always**: both terms positive
negative (exo)	negative	positive	at **low T**, while −ΔH/T is bigger than the size of ΔSsystem
positive (endo)	positive	negative	at **high T**, once ΔSsystem outweighs ΔH/T
positive (endo)	negative	negative	**never**: both terms negative

12.9 Temperature at which feasibility changes

Set $\Delta S_{total} = 0$: $\Delta S_{system} - \Delta H/T = 0$, so **$T = \Delta H \div \Delta S_{system}$** (ΔH in J mol^{-1}).

WORKED EXAMPLE: DECOMPOSING CALCIUM CARBONATE

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$; $\Delta H = +178$ kJ mol^{-1}; $\Delta S_{system} = +160.4$ J K^{-1} mol^{-1}
At 298 K: $\Delta S_{surroundings} = -178\,000 \div 298 = -597.3$; $\Delta S_{total} = 160.4 - 597.3 =$ **-436.9 J K^{-1} mol^{-1}**: not feasible.
Feasible when $\Delta S_{total} \geq 0$: $T = 178\,000 \div 160.4 =$ **1110 K** (about 837 °C). Above this, $CaCO_3$ decomposes: this is why a lime kiln runs at over 900 °C.

WORKED EXAMPLE: THE HABER PROCESS

$\Delta H = -92.2$ kJ mol^{-1}, $\Delta S_{system} = -198.8$ J K^{-1} mol^{-1}: exothermic with a negative ΔS_{system}.
$T = -92\,200 \div -198.8 =$ **464 K**. The reaction is feasible **below** 464 K. The industrial plant runs at about 700 K for **rate**, so ΔS_{total} there is negative and the equilibrium yield is low: that is the compromise (13.4, 13.8).

12.9 Using ΔG instead (allowed, not required)

$\Delta G = \Delta H - T\Delta S_{system}$, and $\Delta G = -T\Delta S_{total}$. A reaction is feasible when **ΔG is negative** (ΔS_{total} positive). Setting $\Delta G = 0$ gives the same $T = \Delta H / \Delta S_{system}$.

MARK-SCHEME LANGUAGE

- Feasibility 6-mark (WCH14 Jan 2026 Q17): exothermic + positive ΔS_{system}: always feasible; exothermic + negative ΔS_{system}: feasible only if $-\Delta H/T > |\Delta S_{system}|$; endothermic + positive ΔS_{system}: feasible only if $\Delta S_{system} > \Delta H/T$; endothermic + negative: never feasible. The ΔG route is allowed throughout.
- Always feasible: "$\Delta S_{surroundings}$ and ΔS_{system} are (both) positive so ΔS_{total} will always be positive". Answers mentioning only ΔS_{total} or only $\Delta S_{surroundings}$ scored zero (WCH14 Jun 2022).
- Higher T: "$\Delta S_{surroundings}$ will decrease (become less positive); ΔS_{system} does not change (significantly); so ΔS_{total} decreases".
- $T = \Delta H \div \Delta S_{system} = 88\,100 \div 185.6 = 474.68$ K (WCH14 Oct 2025 Q9).

Often asked as: minimum temperature for feasibility (most papers); why feasible at all temperatures; why a higher temperature lowers ΔS_{total}; 6-mark on the four combinations; Haber temperature as a compromise.

BEYOND THE SPEC, BUT EXAMINED

- "Justify" a plant temperature means using the entropy/yield data, not reciting collision theory (examiner report, Jan 2020).
- A feasibility temperature that's impractically high: "the temperature required is too high" (WCH14 Oct 2023 Q16).
- If ΔH is close to zero, $\Delta S_{surroundings}$ hardly changes with T, so feasibility (and K) barely change (WCH14 Jan 2024 Q19).

COMMON MISTAKE

Writing "$\Delta S_{surroundings}$ gets bigger/smaller" without saying "more positive" or "less negative". Be precise about signs.
Bullet lists in the 6-mark question: link the points in sentences.
Mixing J and kJ in $T = \Delta H \div \Delta S_{system}$.

SPEC 12.10 A reaction can go even if one entropy term is negative

WHAT THIS SPEC POINT ASKS

Understand that a reaction can happen as long as ΔStotal is positive, even if ΔSsystem or ΔSsurroundings is negative.

- What matters is the **sum**. Compare the **sizes** of the two terms.
- **ΔSsystem negative, ΔSsurroundings positive and bigger**: burning magnesium (−216.6 + 4038 = +3822); the Haber process at 298 K (−198.8 + 309.4 = +110.6); freezing water below 0 °C.
- **ΔSsurroundings negative, ΔSsystem positive and bigger**: dissolving ammonium nitrate at 298 K:

WORKED EXAMPLE: AN ENDOTHERMIC CHANGE THAT HAPPENS

$NH_4NO_3(s) \rightarrow NH_4^+(aq) + NO_3^-(aq)$; $\Delta solH = +25.7$ kJ mol^{-1}; ΔSsystem = +108.7 J K^{-1} mol^{-1}
ΔSsurroundings = −25 700 ÷ 298 = −86.2 J K^{-1} mol^{-1}
ΔStotal = 108.7 − 86.2 = **+22.5 J K^{-1} mol^{-1}** → feasible, so ammonium nitrate dissolves even though the water gets cold.

MARK-SCHEME LANGUAGE

- The verdict is "ΔStotal is positive (so the reaction is feasible)", backed by the numbers (WCH14 Jan 2020 Q19: ΔSsystem −337.8, ΔSsurroundings +2118.5).

Often asked as: part of 12.8/12.9 calculations, ending "comment on your answer".

COMMON MISTAKE

Concluding "not feasible" because ΔSsystem is negative. Always finish with ΔStotal.
Forgetting that "feasible" doesn't mean "fast": if asked why a feasible reaction doesn't happen, talk about activation energy (12.11).

SPEC 12.11 Thermodynamic stability and kinetic stability

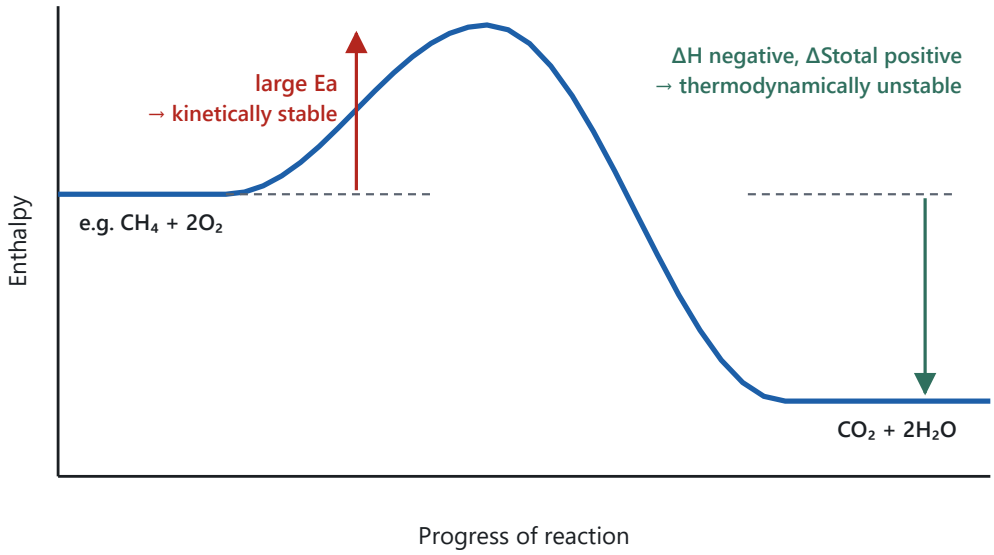
WHAT THIS SPEC POINT ASKS

Understand and tell apart thermodynamic stability and kinetic stability.

	Thermodynamic stability	Kinetic stability
Question it answers	**Can** it react? (Which way is "downhill" overall?)	**How fast** will it react?
Decided by	**ΔStotal** (or ΔG, or K)	**activation energy, Ea** (and so the rate)
Stable means	ΔStotal for the reaction is **negative**: the reaction isn't feasible	Ea is **so high** that the rate is negligible at that temperature

- A substance can be **thermodynamically unstable but kinetically stable**: the reaction is feasible (ΔStotal > 0) but too slow to notice because Ea is high.
  - **Methane in air**: combustion is extremely feasible, but nothing happens until a spark supplies the activation energy.

  - **Diamond → graphite** is feasible at room temperature, but the strong C–C bonds give a huge Ea, so diamonds last.
  - **Hydrogen peroxide** decomposes slowly; MnO_2 (a catalyst) lowers Ea and it fizzes.
  - **SiO_2** in a feasible reaction at room temperature: no reaction, because breaking strong Si–O bonds in a giant lattice needs a lot of energy.
- **Thermodynamically stable** means it won't react however long you wait (e.g. $CaCO_3$ decomposing at 298 K: $\Delta S_{total} = -437$ J K^{-1} mol^{-1}).



Feasible (products far below reactants, ΔStotal positive) but kinetically stable (large Ea): the reaction needs a spark, heat or a catalyst to start.

MARK-SCHEME LANGUAGE

- "(thermodynamically) feasible because ΔStotal is positive"; "activation energy (is) high so the reaction is very slow (at low temperatures)" (WCH14 Jun 2023 Q20).
- "a large amount of energy is needed to break the (strong Si–O) bonds" (WCH14 Jun 2024 Q15). "Kinetically stable" on its own is ignored: give the reason.
- "a lot of energy is required to break the strong C–C bonds in cyclopropane" (WCH14 Jan 2020 Q18).
- Only call a reaction "kinetically stable / too slow" if your ΔStotal is **positive**. If ΔStotal is negative the reaction simply isn't feasible (WCH14 Oct 2025 Q9).

Often asked as: why a feasible reaction doesn't happen at room temperature; MCQ "kinetically stable and thermodynamically unstable".

COMMON MISTAKE

Explaining "no reaction" with collision frequency of solids, or with intermolecular forces. The point is the **energy needed to break strong bonds** (Ea) (examiner report, Jun 2024).
Commenting only on ΔStotal when the question also wants the rate (examiner report, Jun 2023).

SPEC 12.12

Definitions: atomisation, electron affinity, lattice energy

WHAT THIS SPEC POINT ASKS

Define (i) the standard enthalpy change of atomisation, ΔatH; (ii) electron affinity; (iii) lattice energy, as the exothermic change when one mole of an ionic solid forms from its gaseous ions.

12.12 The definitions, each with an equation

Term	Definition	Example equation	Sign
Standard enthalpy change of atomisation, $\Delta atH^{\ominus}$	enthalpy change when **one mole of gaseous atoms** is formed from the element in its **standard state**, under standard conditions	$Na(s) \rightarrow Na(g)$, $+107\ kJ\ mol^{-1}$ $\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$, $+122\ kJ\ mol^{-1}$ $\frac{1}{2}Br_2(l) \rightarrow Br(g)$ $\frac{1}{2}I_2(s) \rightarrow I(g)$	always **positive**
First electron affinity	enthalpy change when **each atom in one mole of gaseous atoms gains one electron**, forming one mole of gaseous 1− ions	$Cl(g) + e^- \rightarrow Cl^-(g)$, $-349\ kJ\ mol^{-1}$ $O(g) + e^- \rightarrow O^-(g)$, $-141\ kJ\ mol^{-1}$	usually **negative**
Second electron affinity	enthalpy change when each ion in one mole of gaseous 1− ions gains one electron, forming one mole of gaseous 2− ions	$O^-(g) + e^- \rightarrow O^{2-}(g)$, $+798\ kJ\ mol^{-1}$	**positive**
Lattice energy (LE)	enthalpy change when **one mole of an ionic solid is formed from its gaseous ions** (under standard conditions)	$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$, $-787\ kJ\ mol^{-1}$ $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s)$	always **negative**

You also need, from Units 1 and 2: **first ionisation energy** ($Na(g) \rightarrow Na^+(g) + e^-$, always positive), **second ionisation energy** ($Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$) and **enthalpy change of formation** ($Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$).

12.12 Why the signs are what they are

- **Atomisation**: bonds (metallic or covalent) are broken: endothermic.
- **First EA**: the incoming electron is attracted by the nucleus: energy released.
- **Second EA**: the electron is added to an ion that is **already negative**, so it is **repelled**. Energy must be put in: endothermic.
- **Lattice energy**: oppositely charged ions attract and come together: energy released. The more negative it is, the stronger the ionic bonding. It can't be measured directly, so it's found from a Born–Haber cycle (12.13).

MARK-SCHEME LANGUAGE

- Atomisation MCQ: "one mole of gaseous atoms", and "the standard state of iodine is solid", so $\frac{1}{2}I_2(s) \rightarrow I(g)$ (WCH14 Jan 2025 Q6); bromine: $\frac{1}{2}Br_2(l) \rightarrow Br(g)$ (WCH14 Jan 2021).
- Second EA: $O^-(g) + e^- \rightarrow O^{2-}(g)$. The first and second together are not one step (WCH14 Jan 2025 Q7).
- Why second EA is endothermic: "the electron is being added to a negative ion and so there is repulsion (so energy is required)" (WCH14 Jun 2022 Q21).

Often asked as: pick the equation that represents a defined quantity (MCQ); explain the sign of the second electron affinity.

COMMON MISTAKE

Writing $Cl_2(g) \rightarrow 2Cl(g)$ for the atomisation of chlorine: that's two moles of atoms. Use $\frac{1}{2}Cl_2$.
Saying an electron affinity is "larger" or "higher" when you mean more or less exothermic.
Confusing electron affinity (gaining electrons) with ionisation energy (losing them).

SPEC 12.13

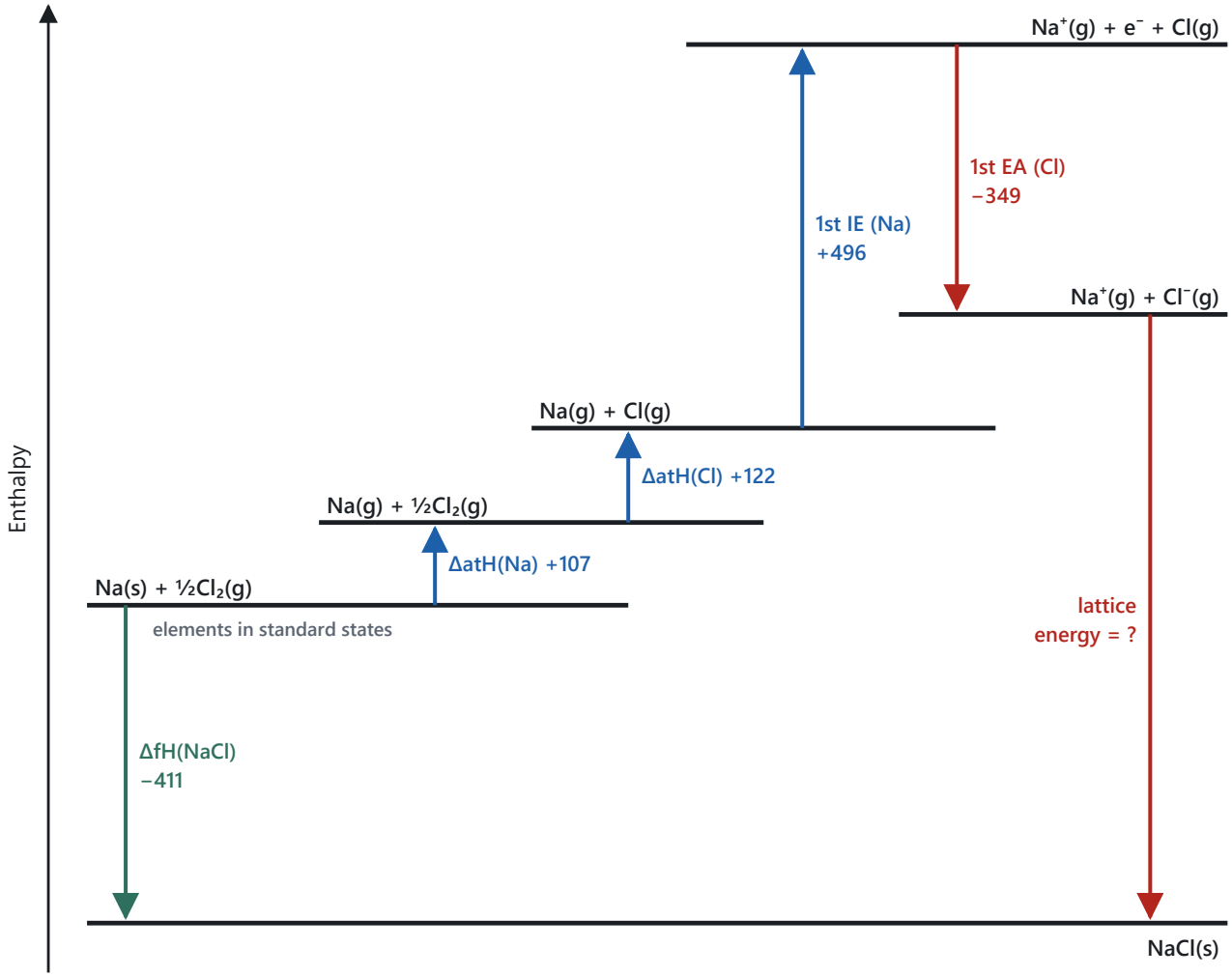
Born–Haber cycles

WHAT THIS SPEC POINT ASKS

Construct Born–Haber cycles and use them to calculate an unknown enthalpy change.

12.13 Building the cycle as an energy-level diagram

1. Draw a line for the **elements in their standard states**, about a third of the way up. Energy increases up the page.
2. **Upward** (endothermic) steps: atomise the metal, atomise the non-metal, ionise the metal (first, then second, ionisation energy).
3. **Downward** steps for exothermic first electron affinities; an **upward** step for a second electron affinity.
4. You now have **gaseous ions**. The **lattice energy** arrow goes **down** from here to the ionic solid.
5. The **enthalpy of formation** links the elements line to the ionic solid (usually down).
6. Every line has species with **state symbols**, and **one** arrowhead per arrow.



LE = −411 − (107 + 122 + 496 − 349) = −787 kJ mol^{-1}

Born–Haber cycle for NaCl. Going round either way from the elements to NaCl(s) gives the same total (Hess's law).

12.13 The calculation

By Hess's law, the direct route (ΔfH) equals the long way round:

ΔfH = ΔatH(metal) + ΔatH(non-metal) + ionisation energies + electron affinities + LE

Rearrange for whichever value is missing.

WORKED EXAMPLE 1: LATTICE ENERGY OF NACL

ΔfH(NaCl) = −411; ΔatH(Na) = +107; IE_1(Na) = +496; ΔatH(Cl) = +122; EA_1(Cl) = −349 (all kJ mol^{-1})

LE = ΔfH − (ΔatH(Na) + IE_1 + ΔatH(Cl) + EA_1)

= −411 − (107 + 496 + 122 − 349) = −411 − 376 = **−787 kJ mol^{-1}**

WORKED EXAMPLE 2: LATTICE ENERGY OF MGCL$_2$ (A 2:1 COMPOUND)

ΔfH($MgCl_2$) = −641; ΔatH(Mg) = +148; IE_1 + IE_2 of Mg = +738 + 1451; ΔatH(Cl) = +122; EA_1(Cl) = −349

Two chloride ions, so **double** the chlorine atomisation and electron affinity:

Sum of steps to Mg^{2+}(g) + 2Cl^-(g) = 148 + 738 + 1451 + 2(122) + 2(−349) = +1883

LE = −641 − 1883 = **−2524 kJ mol^{-1}**

WORKED EXAMPLE 3: SECOND ELECTRON AFFINITY OF OXYGEN, FROM MGO

$\Delta fH(MgO) = -602$; $\Delta atH(Mg) = +148$; $IE_1 + IE_2 = +738 + 1451$; $\Delta atH(O) = +249$; $EA_1(O) = -141$; $LE(MgO) = -3845$

$\Delta fH = 148 + 738 + 1451 + 249 + (-141) + EA_2 + (-3845)$

$-602 = -1400 + EA_2$

EA_2 = **+798 kJ mol^{-1}**. Positive, as expected: the electron is repelled by O^-.

MARK-SCHEME LANGUAGE

- Cycle marks: "arrows upwards for first and second ionisation energies … downward arrows for electron affinity and lattice enthalpy … correct species including state symbols on horizontal lines" (WCH14 Jan 2020 Q19).
- The unlabelled arrow from elements to compound is "enthalpy (change) of formation" (WCH14 Jan 2026 Q14).
- Calculations: correct answer with working gets full marks; one slip usually costs 1 mark.

Often asked as: complete a cycle with species and arrows; calculate LE, ΔfH, an electron affinity or a second ionisation energy from a cycle.

BEYOND THE SPEC, BUT EXAMINED

- **Why $MgCl_3$ doesn't exist**: the **third ionisation energy** of Mg (removing an electron from an inner shell) is enormous and "would not be compensated for by the lattice energy", so ΔfH would be very positive (WCH14 Jan 2024 Q17).
- Any value in the cycle can be the unknown, including IE_2 of a Group 2 metal (WCH14 Jun 2022 Q21) or ΔfH (WCH14 Jan 2026 Q14: −552 kJ mol^{-1}, which must be negative).

COMMON MISTAKE

Not doubling the atomisation and electron affinity of chlorine for $MgCl_2$ (the commonest error in several examiner reports).

Arrowheads on both ends; missing state symbols; arrows in the wrong direction.

A positive ΔfH for a stable ionic compound: recheck your signs.

SPEC 12.14 Experimental and theoretical lattice energies: evidence for covalency

WHAT THIS SPEC POINT ASKS

Understand that comparing the experimental lattice energy (from a Born–Haber cycle) with the theoretical value (from electrostatic calculations) shows how much covalent bonding a compound has.

12.14 Two ways to get a lattice energy

- **Experimental**: from a Born–Haber cycle, using measured enthalpy changes. This is the **real** value.
- **Theoretical**: calculated from a **100% ionic model**: the ions are **perfect spheres** (point charges) with **no distortion**, touching, held only by electrostatic attraction.

12.14 What the comparison tells you

- **Values close** → the ionic model is good → the compound is **almost purely ionic**.

- **Experimental more exothermic** than theoretical → there is **extra** bonding the model leaves out: **partial covalent character** caused by polarisation (12.15). The bigger the gap, the more covalent.

Compound	Experimental LE / kJ mol^{-1}	Theoretical LE / kJ mol^{-1}	Difference	Verdict
NaCl	−780	−770	1%	almost purely ionic
AgCl	−905	−833	9%	some covalency
AgBr	−891	−816	9%	some covalency
AgI	−889	−778	14%	most covalent

(Typical data-book values; use whatever the question gives you.) For MgO, the two values in one paper agreed to within 4 kJ mol^{-1} out of about 3800: O^{2-} is small and hard to polarise, so MgO is essentially ionic (WCH14 Oct 2025 Q3).

MARK-SCHEME LANGUAGE

- Model assumptions: "the bonding is 100% ionic / the ions are in contact / perfect spheres / point charges" (any two) (WCH14 Oct 2024 Q20).
- "the experimental value is more exothermic … the chloride ion is polarised (by the magnesium ion) … some covalent character".
- "calcium chloride is almost completely ionic"; "calcium iodide has partially covalent character"; "the iodide ion is larger, so more (easily) polarised" (WCH14 Jun 2023 Q17).
- MgO v MgS: MgO "very nearly purely ionic because the oxide ion is small and very difficult to polarise"; MgS "the sulfide ion is large and more easily polarised" (WCH14 Jan 2026 Q14). Talk about **ions**, not atoms.
- Use "**more exothermic / less exothermic**", not "bigger / higher" for negative numbers.

Often asked as: 6-mark comparison of lattice energies (WCH14 Jun 2021 Q18: NaCl and MgI_2; Jan 2024 Q17: NaF, MgF_2, $MgCl_2$); 4-mark comparing two calcium or magnesium halides.

BEYOND THE SPEC, BUT EXAMINED

- Near-identical experimental and theoretical values are **evidence of 100% ionic bonding**, not experimental error (WCH14 Oct 2025 Q3; Jan 2020 examiner report).

COMMON MISTAKE

Blaming the difference on "experimental error".
Writing about atoms, atomic radius, molecules, or "the compound being polarised". It's the **anion** that is polarised, by the **cation**.
Confusing **polarising power** (a cation property) with **polarisability** (an anion property).

SPEC 12.15

Polarisation: how cations make ionic bonds partly covalent

WHAT THIS SPEC POINT ASKS

Understand that polarisation of anions by cations gives some covalent character to an ionic bond, based on evidence from Born–Haber cycles.

12.15 What happens

- A cation attracts the electron cloud of a neighbouring anion. The anion's cloud is **distorted** towards the cation.
- Some electron density now sits **between** the two nuclei, so the bond is partly **shared**: partial covalent character.
- This extra attraction makes the real (experimental) lattice energy **more exothermic** than the purely ionic model predicts.

12.15 Which ions do it most

	Increases with…	Examples (most → least)
Polarising power of the cation	**high charge density**: high charge, small radius	Al^{3+} > Mg^{2+} > Na^{+}; Li^{+} > Na^{+} > K^{+}
Polarisability of the anion	**large radius** (electrons far from the nucleus, well shielded) and high charge	I^{-} > Br^{-} > Cl^{-} > F^{-}; S^{2-} > O^{2-}

The most covalent ionic compounds pair a small, highly charged cation with a large anion: AlI_3, MgI_2, LiI. The least covalent pair a large, low-charge cation with a small anion: CsF, NaF.

WORKED EXAMPLE: RANKING COVALENT CHARACTER

Which has more covalent character, NaF or $MgCl_2$?
Mg^{2+} has a higher charge and smaller radius than Na^{+}, so a **higher charge density** and more **polarising power**.
Cl^{-} is larger than F^{-}, so it is **more polarisable**.
Both effects point the same way: **$MgCl_2$** has more covalent character, and its experimental lattice energy will be further from (more exothermic than) the theoretical value.

MARK-SCHEME LANGUAGE

- "the calcium ion is polarising" and "the iodide ion … is polarised / distorted" (WCH14 Jan 2020 Q19).
- "the magnesium ion is more polarising than the sodium ion"; "the chloride ion is larger / more polarisable than the fluoride ion" (WCH14 Jan 2024 Q17).

Often asked as: inside the 12.14 questions.

COMMON MISTAKE

Saying "the chlorine is polarised". It's the **chloride ion**.

SPEC 12.16 Definitions: enthalpy change of solution and of hydration

WHAT THIS SPEC POINT ASKS

Define the enthalpy change of solution, ΔsolH, and the enthalpy change of hydration of an ion, ΔhydH.

Term	Definition	Example	Sign
Enthalpy change of solution, ΔsolH	enthalpy change when **one mole of an ionic solid dissolves** in enough water to form an **infinitely dilute solution**	$NaCl(s) + aq \rightarrow Na^+(aq) + Cl^-(aq)$	can be positive or negative
Enthalpy change of hydration, ΔhydH	enthalpy change when **one mole of gaseous ions** dissolves in enough water to form an **infinitely dilute solution**	$Na^+(g) + aq \rightarrow Na^+(aq)$ $Cl^-(g) + aq \rightarrow Cl^-(aq)$	always **negative**

- **Infinitely dilute**: so much water that adding more gives **no further heat change**. Then the ions don't interact with each other, only with water.
- Hydration is exothermic because of **ion–dipole attractions**: the δ− oxygen of water points towards a cation; the δ+ hydrogens point towards an anion. New attractions form; no bonds break.

MARK-SCHEME LANGUAGE

- Hydration: "enthalpy change when 1 mol of gaseous ions" is "completely hydrated / forms an infinitely dilute solution / dissolves until there is no further heat change" (WCH14 Jun 2025 Q19). DO NOT AWARD "heat needed" or "dissolved in 1 mol of water". IGNORE "standard conditions".
- Hydration enthalpy of an ion is always negative (WCH14 Oct 2022 Q4).

Often asked as: define ΔhydH; MCQ on which equation shows hydration or solution.

COMMON MISTAKE

Defining hydration for the solid compound, or for 1 mol of water. It is for **gaseous ions**.

SPEC 12.17

Energy cycles for the enthalpy change of solution

WHAT THIS SPEC POINT ASKS

Use energy cycles and energy-level diagrams to calculate the enthalpy change of solution of an ionic compound from hydration enthalpies and the lattice energy.

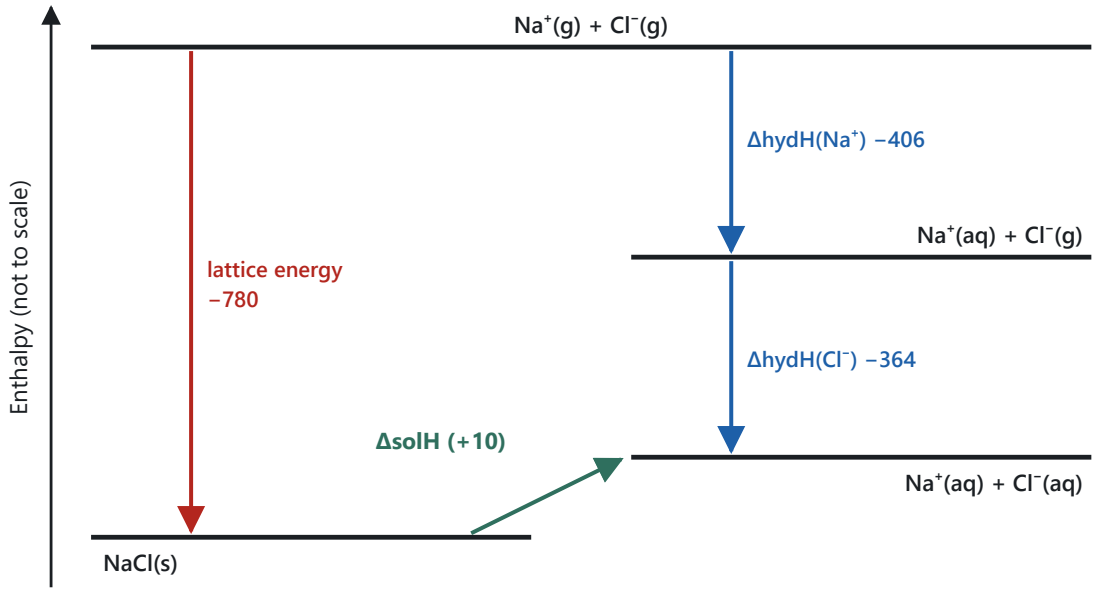
12.17 The cycle

Two routes from the solid to the solution:

- Direct: **ΔsolH**.
- Indirect: break the lattice into gaseous ions (**−LE**, endothermic, because LE is defined as the exothermic formation), then hydrate each ion (**ΣΔhydH**).

$\Delta_{sol}H = \Sigma\Delta_{hyd}H - LE$

For $MgCl_2$ this is $\Delta_{sol}H = \Delta_{hyd}H(Mg^{2+}) + \mathbf{2} \times \Delta_{hyd}H(Cl^-) - LE(MgCl_2)$.



ΔsolH = ΔhydH(Na^+) + ΔhydH(Cl^-) − LE = −770 − (−780) = +10 kJ mol^{-1}

NaCl: the lattice energy and the total hydration enthalpy are both large and nearly cancel, leaving a small ΔsolH.

WORKED EXAMPLE 1: ΔSOLH OF NACL

LE(NaCl) = −780; ΔhydH(Na^+) = −406; ΔhydH(Cl^-) = −364 kJ mol^{-1}

ΔsolH = (−406 + −364) − (−780) = −770 + 780 = **+10 kJ mol^{-1}** (slightly endothermic)

WORKED EXAMPLE 2: A MISSING HYDRATION ENTHALPY

For $MgCl_2$: ΔsolH = −155; LE = −2526; ΔhydH(Cl^-) = −364. Find ΔhydH(Mg^{2+}).

ΔhydH(Mg^{2+}) = ΔsolH + LE − 2 × ΔhydH(Cl^-)

= −155 + (−2526) − 2(−364) = −155 − 2526 + 728 = **−1953 kJ mol^{-1}**

WORKED EXAMPLE 3: FROM ΔSOLH TO A TEMPERATURE CHANGE

2.00 g of anhydrous $MgCl_2$ (Mr 95.3) dissolves in 50.0 g of water. ΔsolH = −155 kJ mol^{-1}; c = 4.18 J g^{-1} K^{-1}.

Moles = 2.00 ÷ 95.3 = 0.02099 mol

Heat released = 155 000 × 0.02099 = 3253 J

ΔT = 3253 ÷ (50.0 × 4.18) = **+15.6 °C** (temperature rises because ΔsolH is negative)

MARK-SCHEME LANGUAGE

- Expression: "ΔsolH = ΔhydH(M^{n+}) + ΔhydH(X^{n-}) − lattice energy"; signs reversed scores nothing (WCH14 Jun 2022 Q16).
- Cycle: "enthalpy of solution arrow up from the solid" (when endothermic); "lattice energy … gaseous ions, arrow down"; "hydration arrows down to both aqueous ions" (WCH14 Jan 2025 Q19).
- $CuCl_2$: ΣΔhydH = −2100 + 2(−378) = −2856; ΔsolH = −2856 − (−2811) = **−45 kJ mol^{-1}**; +45 scores only 1 (WCH14 Jun 2025 Q19).

Often asked as: complete a solution cycle; calculate ΔsolH or a missing hydration enthalpy; link to a calorimetry experiment.

BEYOND THE SPEC, BUT EXAMINED

- **Calorimetry of dissolving** (q = mcΔT with the mass of **water**, moles from the correct formula mass, sign from the temperature change): WCH14 Oct 2024 Q20, Jun 2025 Q19.
- Hydrated salts: use the Mr of the hydrate (e.g. $CuSO_4 \cdot 5H_2O$) when that's what was weighed (WCH14 Jun 2025 Q19).

COMMON MISTAKE

Not doubling ΔhydH(Cl^-) for $MgCl_2$ or $CuCl_2$. Getting the sign of LE wrong in the expression.
In calorimetry, using the mass of the salt (or salt + water) as m.

SPEC 12.18 How ionic charge and radius affect lattice and hydration enthalpies

WHAT THIS SPEC POINT ASKS

Understand the effect of ionic charge and ionic radius on the hydration enthalpies of ions and the lattice energy of ionic compounds.

12.18 The rule

Both lattice energy and hydration enthalpy come from **electrostatic attraction**, so both become **more exothermic** when:

- the ions have **higher charges** (stronger attraction), and
- the ions are **smaller** (charges closer together: closer to each other in the lattice, or closer to the water dipoles).

In one phrase: higher **charge density** → more exothermic.

12.18 Lattice energy

Compare	Lattice energy	Reason
NaCl v KCl v CsCl	about −780, −711, −661 kJ mol^{-1}	cation larger → ions further apart → weaker attraction → less exothermic
NaF v NaCl v NaI	about −918, −780, −705	anion larger → less exothermic
NaCl v MgO	about −780 v −3800	2+ and 2− charges (and smaller ions) → much more exothermic

12.18 Hydration enthalpy

Cation	ΔhydH / kJ mol^{-1}	Cation	ΔhydH	Anion	ΔhydH
Li^+	−519	Mg^{2+}	−1920	F^-	−506
Na^+	−406	Ca^{2+}	−1650	Cl^-	−364
K^+	−322	Sr^{2+}	−1480	Br^-	−335
Cs^+	−276	Ba^{2+}	−1360	I^-	−293

- Down a group the ions get bigger → hydration less exothermic.
- 2+ ions are hydrated about four times more strongly than 1+ ions of similar size.

EXAM TIP

Compare like with like: change **one** thing (charge **or** radius) and keep the other ion the same. Then say *why*: "smaller ion, so its charge is closer to the δ− oxygen of water, so stronger ion–dipole attraction, so more energy released".

MARK-SCHEME LANGUAGE

- Lattice energy "becomes less exothermic as the radius of the metal ion increases (and the charge remains the same)"; "weaker forces of attraction between ions" (WCH14 Oct 2020 Q16).
- Cu^+ v Cu^{2+}: "Cu^+ less highly charged … weaker (electrostatic) forces of attraction with water (so less energy is released when hydrated)" (WCH14 Jun 2025 Q19). DO NOT AWARD atoms or "intermolecular forces".
- Ammonium halides: "the bromide ion is larger than the chloride ion"; its hydration enthalpy is "less exothermic"; the lattice energy of ammonium chloride is "more exothermic"; then link both to ΔsolH (WCH14 Jun 2022 Q16).

Often asked as: MCQ on charge density; explain a trend in lattice energy or hydration enthalpy; explain the difference in ΔsolH between two salts.

COMMON MISTAKE

Atomic radius instead of **ionic** radius; "halogens" instead of "halides".
"Weaker" or "larger" without saying what is weaker or larger.

SPEC 12.19

Predicting solubility from entropy and enthalpy

WHAT THIS SPEC POINT ASKS

Use entropy changes and enthalpy changes of solution to predict whether an ionic compound dissolves, and discuss the solubility trends from Unit 2.

12.19 Is dissolving feasible?

A salt dissolves if **ΔStotal for dissolving is positive**:

ΔStotal = ΔSsystem + ΔSsurroundings = ΔSsystem − ΔsolH/T

- **Exothermic ΔsolH**: ΔSsurroundings is positive, so dissolving is usually feasible.
- **Small positive ΔsolH** (NaCl, KCl, NH_4NO_3): feasible as long as **ΔSsystem > ΔsolH/T**.
- **Large positive ΔsolH**: ΔSsurroundings is very negative, so the salt is insoluble (or only sparingly soluble).
- ΔSsystem itself is the balance between the lattice breaking up (+) and water being ordered round the ions (−) (12.5).

WORKED EXAMPLE: HOW POSITIVE MUST ΔSSYSTEM BE?

A salt has ΔsolH = +25.7 kJ mol^{-1}. It dissolves at 298 K. What can you say about ΔSsystem?
ΔSsurroundings = −25 700 ÷ 298 = −86.2 J K^{-1} mol^{-1}
For ΔStotal > 0, **ΔSsystem must be more positive than +86.2 J K^{-1} mol^{-1}**. (For NH_4NO_3 it is +108.7, so it dissolves.)

12.19 Unit 2 trends: Group 2 hydroxides and sulfates

Both lattice energy and hydration enthalpy get **less exothermic** as the cation gets bigger down Group 2. What matters is **which falls faster**, because ΔsolH = ΣΔhydH − LE.

	Group 2 sulfates (large anion)	Group 2 hydroxides (small anion)
Lattice energy down the group	changes **only a little**: the inter-ion distance is dominated by the big SO_4^{2-}	becomes much less exothermic: OH^- is small, so the cation's size matters a lot
Hydration enthalpy of M^{2+}	becomes **much** less exothermic	becomes less exothermic, but by less than the lattice energy
ΔsolH down the group	becomes **less exothermic / more endothermic**	becomes **more exothermic / less endothermic**
Solubility down the group	**decreases** ($BaSO_4$ is insoluble)	**increases** ($Mg(OH)_2$ sparingly soluble, $Ba(OH)_2$ fairly soluble)

Entropy changes play a part as well, so this enthalpy argument is a simplification, but it is the one you are expected to give.

MARK-SCHEME LANGUAGE

- ΔSsystem for a salt that dissolves: "ΔSsystem must be positive"; "ΔSsystem > (ΔH/T value)"; "ΔStotal > 0 as it dissolves" (WCH14 Jun 2021 Q15).
- Least exothermic lattice energy in a 100% ionic model: the compound with **singly charged, larger ions** (WCH14 Jan 2026 Q7).

Often asked as: decide whether a salt dissolves from ΔsolH and ΔSsystem; MCQ on lattice energies; link hydration and lattice changes to ΔsolH.

BEYOND THE SPEC, BUT EXAMINED

- NaCl dissolving with ΔsolH = +1.0 kJ mol^{-1} is feasible, so its ΔSsystem must be positive. The correct MCQ answer is "the change has a positive entropy change of the system", not "a positive ΔSsurroundings" (WCH14 Oct 2025 Q4).

COMMON MISTAKE

Using ΔsolH alone to decide solubility: the entropy terms decide.
Saying solubility trends come only from lattice energy. It's the **balance** of lattice energy against hydration.

Topic 13 – Chemical Equilibria

SPEC 13.1 Writing Kc for homogeneous and heterogeneous equilibria

WHAT THIS SPEC POINT ASKS

Deduce an expression for Kc, in terms of equilibrium concentrations, for homogeneous and heterogeneous systems.

13.1 The rule

For aA + bB ⇌ cC + dD (all in solution or all gases):

$K_c = [C]^c[D]^d \div ([A]^a[B]^b)$

- **Products on top**, reactants underneath.
- Each concentration is raised to the **power of its coefficient** in the balanced equation.
- **Square brackets** = equilibrium concentration in mol dm^{-3}.
- **Homogeneous**: everything in the same phase. **Heterogeneous**: more than one phase.
- In a heterogeneous system, **leave out pure solids** (and pure liquids, such as water used as the solvent). Their "concentration" is fixed, so it's already part of K.

13.1 Examples

Equilibrium	Kc
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	$[HI]^2 \div ([H_2][I_2])$
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	$[NH_3]^2 \div ([N_2][H_2]^3)$
$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$	$[CH_3COOC_2H_5][H_2O] \div ([CH_3COOH][C_2H_5OH])$
$Fe^{2+}(aq) + Ag^+(aq) \rightleftharpoons Fe^{3+}(aq) + Ag(s)$	$[Fe^{3+}] \div ([Fe^{2+}][Ag^+])$ (solid silver left out)
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	$[CO_2]$

In the **ester** equilibrium water **is** included: it's a product mixed in with the other liquids, not a solvent present in huge excess.

13.1 Units of Kc

Substitute mol dm^{-3} for every concentration and cancel.

- $[HI]^2 \div ([H_2][I_2])$: $(\text{mol dm}^{-3})^2 \div (\text{mol dm}^{-3})^2 \rightarrow$ **no units**.
- $[NH_3]^2 \div ([N_2][H_2]^3)$: $(\text{mol dm}^{-3})^2 \div (\text{mol dm}^{-3})^4 = (\text{mol dm}^{-3})^{-2} =$ **$\text{mol}^{-2}\ \text{dm}^6$** ($\text{dm}^6\ \text{mol}^{-2}$).
- $[CO_2]$: **mol dm^{-3}**.

MARK-SCHEME LANGUAGE

- "Kc = $[NO]^2[Cl_2] \div [NOCl]^2$" (WCH14 Jun 2025 Q14). DO NOT AWARD round brackets, "p", or the 2 written inside the bracket.
- Heterogeneous: "Kc = $[CO_2]$" for a solid decomposing (WCH14 Oct 2025 Q14).
- Doubling every coefficient **squares** K (WCH14 Oct 2020 Q7).

Often asked as: write Kc (often as the first mark of a calculation); MCQ on Kc for a heterogeneous system; how K changes when the equation is reversed or multiplied.

BEYOND THE SPEC, BUT EXAMINED

- Change the equation, change K: **reverse** it → 1/K; **double** it → K^2; **halve** it → $\sqrt{K}$ (WCH14 Jan 2025 Q5: halving the coefficients turned 4×10^{24} into 2.0×10^{12}).

COMMON MISTAKE

Including a solid. Writing the equation instead of the expression. Putting reactants on top.

SPEC 13.2

Writing Kp in terms of partial pressures

WHAT THIS SPEC POINT ASKS

Deduce an expression for Kp for homogeneous and heterogeneous systems, in terms of equilibrium partial pressures in atm.

13.2 Partial pressure

- In a mixture of gases, the **partial pressure** of a gas is the pressure it would exert if it filled the container on its own.
- **Mole fraction**: x_A = moles of A ÷ total moles of gas.
- **Partial pressure**: $p_A = x_A \times$ total pressure.
- The partial pressures **add up to the total pressure**, and the mole fractions add up to 1.

13.2 Kp expressions

Like Kc, but with **partial pressures** (written p_{NH_3} or $p(NH_3)$) and **only gases**:

Equilibrium	Kp	Units
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	$p(NH_3)^2 \div (p(N_2) \times p(H_2)^3)$	atm^{-2}
$\frac{1}{2}N_2(g) + 1\frac{1}{2}H_2(g) \rightleftharpoons NH_3(g)$	$p(NH_3) \div (p(N_2)^{\frac{1}{2}} \times p(H_2)^{1\frac{1}{2}})$	atm^{-1}
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$	$p(NO_2)^2 \div p(N_2O_4)$	atm
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	$p(SO_3)^2 \div (p(SO_2)^2 \times p(O_2))$	atm^{-1}
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	$p(CO_2)$	atm
$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$	$p(CO) \times p(H_2) \div p(H_2O)$	atm

- Solids and liquids are left out; gaseous **products** still count.
- Units come from cancelling atm, exactly as for Kc.

MARK-SCHEME LANGUAGE

- "Kp = $p(NH_3) \div (p(N_2)^{\frac{1}{2}} \times p(H_2)^{1\frac{1}{2}})$" (WCH14 Jan 2026 Q15). Round brackets, "pp", "P" or "$p(NH_3)$" are all allowed.
- **DO NOT AWARD square brackets** in Kp (every mark scheme that asks says so), or an expression with no p at all.

Often asked as: write Kp (part of nearly every Kp calculation); give its units.

BEYOND THE SPEC, BUT EXAMINED

- **Fractional coefficients** give fractional powers and unusual units (atm^{-1} for the $\frac{1}{2}N_2 + 1\frac{1}{2}H_2$ equation) (WCH14 Jan 2026 Q15).

COMMON MISTAKE

Square brackets in Kp: examiner reports call this "disappointingly" common. Square brackets mean concentration.
Including a solid such as $CaCO_3$ in Kp.

SPEC 13.3

Calculating Kc and Kp, with units

WHAT THIS SPEC POINT ASKS

Calculate values of Kc and Kp, with units where needed, for homogeneous and heterogeneous reactions from experimental data.

13.3 Method (use an ICE table)

1. Write the balanced equation and the K expression.
2. **I**nitial moles; **C**hange (use the mole ratio); **E**quilibrium moles.
3. For **Kc**: divide each equilibrium amount by the **volume** (dm^3) to get concentrations.
   For **Kp**: find the **total equilibrium moles** of gas → mole fractions → partial pressures.
4. Substitute, calculate, and work out the units.

WORKED EXAMPLE 1: KC WITH NO UNITS (VOLUME CANCELS)

0.50 mol CH_3COOH and 1.00 mol C_2H_5OH reach equilibrium; 0.423 mol of ethyl ethanoate has formed. Find Kc.
$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$
Initial: 0.50, 1.00, 0, 0
Change: −0.423, −0.423, +0.423, +0.423
Equilibrium: 0.077, 0.577, 0.423, 0.423
Kc = (0.423/V × 0.423/V) ÷ (0.077/V × 0.577/V): the V's **cancel**, so moles can be used.
Kc = 0.1789 ÷ 0.04443 = **4.0 (no units)**

WORKED EXAMPLE 2: KC WITH UNITS

0.200 mol PCl_5 is heated in a 2.00 dm^3 flask: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. At equilibrium 0.120 mol PCl_5 remains.
Equilibrium moles: PCl_5 0.120; PCl_3 0.080; Cl_2 0.080
Concentrations: $[PCl_5] = 0.0600$, $[PCl_3] = 0.0400$, $[Cl_2] = 0.0400$ mol dm^{-3}
Kc = (0.0400 × 0.0400) ÷ 0.0600 = **0.0267 mol dm^{-3}**

WORKED EXAMPLE 3: KP FOR THE HABER PROCESS

1.00 mol N_2 and 3.00 mol H_2 reach equilibrium at a total pressure of 200 atm; 0.40 mol NH_3 has formed.
$N_2 + 3H_2 \rightleftharpoons 2NH_3$
Equilibrium moles: N_2 1.00 − 0.20 = 0.80; H_2 3.00 − 0.60 = 2.40; NH_3 0.40; **total 3.60**
Mole fractions: N_2 0.222; H_2 0.667; NH_3 0.111
Partial pressures (× 200): N_2 44.4 atm; H_2 133.3 atm; NH_3 22.2 atm
$K_p = 22.2^2 \div (44.4 \times 133.3^3) =$ **4.69×10^{-6} atm^{-2}**

WORKED EXAMPLE 4: KP FOR $N_2O_4 \rightleftharpoons 2NO_2$

1.00 mol N_2O_4 is 20% dissociated at 298 K and a total pressure of 1.00 atm.
Equilibrium moles: N_2O_4 0.80; NO_2 0.40; total 1.20
$p(N_2O_4) = (0.80/1.20) \times 1.00 = 0.667$ atm; $p(NO_2) = (0.40/1.20) \times 1.00 = 0.333$ atm
$K_p = 0.333^2 \div 0.667 =$ **0.167 atm**

WORKED EXAMPLE 5: EQUILIBRIUM AMOUNTS FROM KC (NO QUADRATIC)

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, $K_c = 49$ at a certain temperature. Start with 1.00 mol H_2 and 1.00 mol I_2. How much HI forms?
Equilibrium: H_2 (1.00 − x), I_2 (1.00 − x), HI 2x. Kc has no units, so volumes cancel.
$(2x)^2 \div (1.00 - x)^2 = 49$ → take the square root: $2x \div (1.00 - x) = 7$
$2x = 7 - 7x \rightarrow x = 0.778$ mol, so **HI = 1.56 mol**; $H_2 = I_2 = 0.222$ mol.

MARK-SCHEME LANGUAGE

- Typical mark pattern: equilibrium moles (1); mole fractions (1); partial pressures (1); substitution into Kp (1); value with units (1).
- Units: atm^{-1}, atm^{-2}, mol dm^{-3}, dm^3 mol^{-1}; "**no units**" when the powers cancel (WCH14 Oct 2022 Q18). SF: 2 or 3 usually.
- Ester equilibria: "a statement that the volumes cancel" earns the last mark (WCH14 Jan 2022 Q21, Jan 2024 Q19).

Often asked as: Kc from moles and a volume; Kp from moles → mole fractions → partial pressures; Kp from a percentage composition; show that a given mixture is not at equilibrium.

BEYOND THE SPEC, BUT EXAMINED

- **Converting Kp ↔ Kc.** For each gas, $p = (n/V)RT = cRT$, so **$K_p = K_c(RT)^{\Delta n}$**, where Δn = moles of gas products − moles of gas reactants. With p in atm and c in mol dm^{-3}, use **R = 0.0821 dm^3 atm K^{-1} mol^{-1}** and T in K (WCH14 Jan 2025 Q5, Oct 2025 Q14). For $N_2O_4 \rightleftharpoons 2NO_2$ (Δn = +1): $K_c = K_p \div RT = 0.167 \div (0.0821 \times 298) =$ **6.8×10^{-3} mol dm^{-3}**. For $CaCO_3 \rightleftharpoons CaO + CO_2$: $K_c = [CO_2] = p(CO_2) \div RT$.
- **pV = nRT** in SI units (p in Pa, V in **m^3**, R = 8.31 J K^{-1} mol^{-1}): e.g. 15 dm^3 = 0.015 m^3 (WCH14 Oct 2023 Q15).
- **The reaction quotient, Q**: same expression as K but with the concentrations or pressures you actually have. If Q = K the mixture is at equilibrium; if not, it isn't. Example: for $N_2O_4 \rightleftharpoons 2NO_2$ ($K_p = 0.167$ atm), a mixture with $p(NO_2) = p(N_2O_4) = 0.50$ atm has $Q_p = 0.50^2 \div 0.50 = 0.50$ atm > Kp, so it is **not** at equilibrium and will shift **left** (WCH14 Oct 2025 Q14: $Q_p = 4.5$ atm^{-1} against $K_p = 35.2$ atm^{-1}).
- Use the **total moles at equilibrium**, not at the start, for mole fractions (WCH14 Oct 2021 Q19 examiner report).

COMMON MISTAKE

Using initial moles in K. Using Mr(O) = 16 for O_2 (it's 32; about a third of candidates did this in one paper).
Not dividing by the volume when Kc has units.
Rounding in the middle of the calculation; 4 or more SF; missing units.

SPEC 13.4 How temperature, pressure and a catalyst affect equilibrium composition

WHAT THIS SPEC POINT ASKS

Understand how, if at all, changing the temperature or pressure, or adding a catalyst, affects the equilibrium composition of a homogeneous or heterogeneous system.

13.4 Summary

Change	Effect on position	Effect on K
Increase temperature	shifts in the **endothermic** direction	**changes** (13.6)
Decrease temperature	shifts in the **exothermic** direction	**changes**
Increase pressure	shifts towards the side with **fewer moles of gas**	no change
Decrease pressure	shifts towards **more moles of gas**	no change
Add a catalyst	**no change**: forward and backward rates go up **equally**; equilibrium is just reached sooner	no change
Add more of a reactant	shifts to use it up (towards products)	no change

- In a **heterogeneous** system, count **only gas moles** when you think about pressure. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$: 0 → 1 mol gas, so high pressure shifts it left. Adding more solid $CaCO_3$ doesn't shift the position.
- **Equal moles of gas on both sides** ($H_2 + I_2 \rightleftharpoons 2HI$): pressure doesn't shift the position.

13.4 Observations with coloured gases

- N_2O_4(g, colourless) $\rightleftharpoons 2NO_2$(g, brown), forward endothermic.
- **Compress** it: the colour **darkens at once** (NO_2 is squeezed into a smaller volume), then **fades a little** as the position shifts left (fewer moles of gas). It stays darker than at the start.
- **Heat** it: browner, because the endothermic forward reaction is favoured.
- For $H_2 + I_2 \rightleftharpoons 2HI$, compressing makes the purple colour more intense but **doesn't** shift the position.

13.4 Industrial compromise: the Haber process

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92$ kJ mol^{-1}

- **Temperature about 450 °C**: a low temperature would give a higher yield (exothermic forward), but the rate would be too slow. 450 °C gives a reasonable yield at a fast enough rate.
- **Pressure about 200 atm**: high pressure favours NH_3 (4 → 2 mol gas) and increases the rate, but very high pressures cost too much to build and run safely.
- **Iron catalyst**: faster, no effect on yield.

- Only about 15% conversion per pass: unreacted gases are **recycled**, so the overall conversion is much higher.

MARK-SCHEME LANGUAGE

- "increase in temperature would decrease yield / shift the equilibrium to the left, because the reverse reaction is endothermic"; "increase in pressure would increase the yield because there are fewer moles (of gas) on the right-hand side"; "a catalyst has no effect on the yield because the rates of the forward and backward reactions are increased (equally)" (WCH14 Jan 2026 Q15). "Molecules" for "moles" is allowed.
- "when the pressure is increased, equilibrium moves to the RHS and yield increases as fewer gas molecules on the RHS; **Kp remains constant**"; "temperature increase: equilibrium moves to LHS as the forward reaction is exothermic; **Kp decreases**" (WCH14 Jun 2021 Q20).
- Large K: "the equilibrium is (already) far to the right, so increased pressure would give only a small increase in yield" (WCH14 Oct 2025 Q14).

Often asked as: effect of each change on yield and on K, with reasons; colour change on compressing; justify compromise conditions.

BEYOND THE SPEC, BUT EXAMINED

- Compressing an equilibrium with **equal gas moles** on both sides: the position doesn't move, but the colour gets **more intense** because the same particles are in a smaller volume (WCH14 Jan 2025 Q10, the hardest MCQ on that paper).
- **Recycling** unreacted gases to get overall conversion above the equilibrium yield (WCH14 Oct 2021 Q19, Jan 2022 Q22).

COMMON MISTAKE

Saying "the equilibrium shifts left" without saying what happens to the **yield**.
Mixing up exothermic and endothermic directions.
Using Le Chatelier when the question says "in terms of Kp": then you must argue with Q and K (13.5).

SPEC 13.5 K doesn't change with concentration, pressure or a catalyst

WHAT THIS SPEC POINT ASKS

Understand that the value of the equilibrium constant is not affected by changes in concentration or pressure, or by adding a catalyst.

13.5 Only temperature changes K

- **Concentration**: adding a reactant makes the quotient Q smaller than K. The system reacts forwards (making more products, using reactants) until **Q = K** again. The position moves; K doesn't.
- **Pressure**: changing the total pressure changes every partial pressure by the same factor. If the gas moles differ on each side, Q changes; the system shifts until Q = Kp again.
- **Catalyst**: speeds up forward and reverse reactions by the same factor. Equilibrium arrives sooner with the same composition, so K is unchanged.

WORKED EXAMPLE: PRESSURE EXPLAINED WITH KP

$N_2 + 3H_2 \rightleftharpoons 2NH_3$: double the total pressure. Every partial pressure doubles at once.
$Qp = (2p(NH_3))^2 \div ((2p(N_2)) \times (2p(H_2))^3) = (4 \div 16) \times Kp =$ **$Kp \div 4$**
Qp is now smaller than Kp, so the system makes more NH_3 (top line up, bottom line down) until Qp = Kp again. **Position shifts right; Kp unchanged.**

$N_2O_4 \rightleftharpoons 2NO_2$: double the pressure.
$Qp = (2p(NO_2))^2 \div (2p(N_2O_4)) =$ **$2 \times Kp$**
Qp is too big, so NO_2 turns back into N_2O_4 until Qp = Kp. **Position shifts left.**

MARK-SCHEME LANGUAGE

- "the value of Kp would not change due to increased pressure or the presence of a catalyst, (but) temperature changes Kp" (WCH14 Jan 2026 Q15).
- Increased pressure "in terms of Kp": "Kp remains the same"; "the quotient / Qp / apparent value of Kp decreases (when pressure increases)"; "so equilibrium shifts to the right-hand side (to return to the value of Kp)" (WCH14 Oct 2020 Q19).

Often asked as: state and explain what happens to K; explain a pressure change using Kp.

BEYOND THE SPEC, BUT EXAMINED

- The **reaction quotient Q** argument is the only route credited when a question says "in terms of Kp" (WCH14 Oct 2020 Q19). The same idea shows whether a mixture is at equilibrium (WCH14 Oct 2025 Q14).

COMMON MISTAKE

Saying K increases because "more product is made at higher pressure". K is fixed; the position moves to keep it fixed.
Leaving out what happens to the value of K when asked (examiner report, Jan 2021).

SPEC 13.6 How temperature changes K: exothermic and endothermic reactions

WHAT THIS SPEC POINT ASKS

Know the effect of changing temperature on Kc and Kp for exothermic and endothermic reactions.

Forward reaction	Increase T	Decrease T
Exothermic ($\Delta H < 0$), e.g. $2SO_2 + O_2 \rightleftharpoons 2SO_3$; $N_2 + 3H_2 \rightleftharpoons 2NH_3$	**K decreases**: less product	K increases
Endothermic ($\Delta H > 0$), e.g. $N_2O_4 \rightleftharpoons 2NO_2$; $CaCO_3 \rightleftharpoons CaO + CO_2$	**K increases**: more product	K decreases

Memory aid: heating always favours the **endothermic** direction. If that's the forward direction, K goes up.

WORKED EXAMPLE: READING ΔH FROM K DATA

For $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, Kp is 3.4 atm^{-1} at 1000 K and 0.13 atm^{-1} at 1200 K (illustrative values).
Kp falls as T rises → less SO_3 at equilibrium → the forward reaction must be **exothermic**.
In terms of the expression: the partial pressure of SO_3 (top) has decreased and those of SO_2 and O_2 (bottom) have increased.

MARK-SCHEME LANGUAGE

- "(decreasing Kp means) the partial pressure of products decreases; the partial pressure of reactants increases; so equilibrium moves to the left (at higher temperature)"; "numerator decreases / denominator increases" allowed (WCH14 Oct 2025 Q14).
- "(forward) reaction is endothermic; at the lower temperature the reaction moves in the exothermic direction / backwards (making Kc smaller)" (WCH14 Jun 2025 Q14).
- "the equilibrium constant decreases (as temperature increases) because the forward reaction is exothermic". Both the yield and the change in K are needed (WCH14 Jan 2021 Q18).

Often asked as: predict the change in K; deduce the sign of ΔH from K at two temperatures; MCQ "which change alone increases K".

BEYOND THE SPEC, BUT EXAMINED

- Choosing the **one** change that increases K from a list (pressure, catalyst, heating, cooling): only a temperature change can, and which one depends on the sign of ΔH (WCH14 Jan 2025 Q10).

COMMON MISTAKE

Saying a catalyst or pressure changes K. Saying "K shifts to the right": K is a number; it increases or decreases.

SPEC 13.7 Temperature moves the equilibrium position by changing K

WHAT THIS SPEC POINT ASKS

Understand that the effect of temperature on the position of equilibrium is explained by a change in the value of the equilibrium constant.

The chain of reasoning, for an **exothermic** forward reaction heated up:

1. Raising the temperature makes **K smaller** (13.6, explained by entropy in 13.8).
2. At that instant the concentrations haven't changed, so the quotient Q is still the old value: now **Q > K**.
3. To make Q fall to the new K, products must turn back into reactants.
4. So the position shifts **left** and the yield falls.

For an endothermic forward reaction, heating makes K bigger, so Q < K and the position shifts right.

Compare with pressure and concentration: there **Q changes and K stays**; with temperature **K changes and Q has to follow**.

MARK-SCHEME LANGUAGE

- "temperature changes K" is the reason the position shifts (WCH14 Jan 2026 Q15).
- Linking to entropy: "ΔrH is negative (as increasing temperature moves the equilibrium to the left) … ΔSsurroundings becomes less positive as T increases … ΔStotal has decreased" (WCH14 Oct 2025 Q14).

Often asked as: explain in terms of K why heating changes the yield.

COMMON MISTAKE

Explaining a temperature change with Le Chatelier alone when the question asks for K.

SPEC 13.8 ΔStotal = R ln K: how temperature affects ΔStotal and K

WHAT THIS SPEC POINT ASKS

Understand the effect of a change in temperature on (i) ΔStotal and (ii) the size of the equilibrium constant, since ΔStotal = R ln K.

13.8 The link

ΔStotal = R ln K, so **ln K = ΔStotal ÷ R** and **$K = e^{\Delta S_{total}/R}$** ($R = 8.31$ J K^{-1} mol^{-1}, ΔStotal in J K^{-1} mol^{-1}, standard values). You have to remember this: it isn't given.

- ΔStotal large and positive → K very large → products dominate.
- ΔStotal = 0 → K = 1.
- ΔStotal negative → K < 1 → reactants dominate.

13.8 Temperature: the full chain

ΔStotal = ΔSsystem − ΔH/T. ΔSsystem and ΔH barely change with temperature, so only the −ΔH/T term changes.

Exothermic forward reaction (ΔH < 0):
increase T → ΔSsurroundings (= −ΔH/T, positive) gets **smaller** → ΔStotal **decreases** → ln K decreases → **K decreases**.

Endothermic forward reaction (ΔH > 0):
increase T → ΔSsurroundings (negative) gets **less negative** → ΔStotal **increases** → **K increases**.

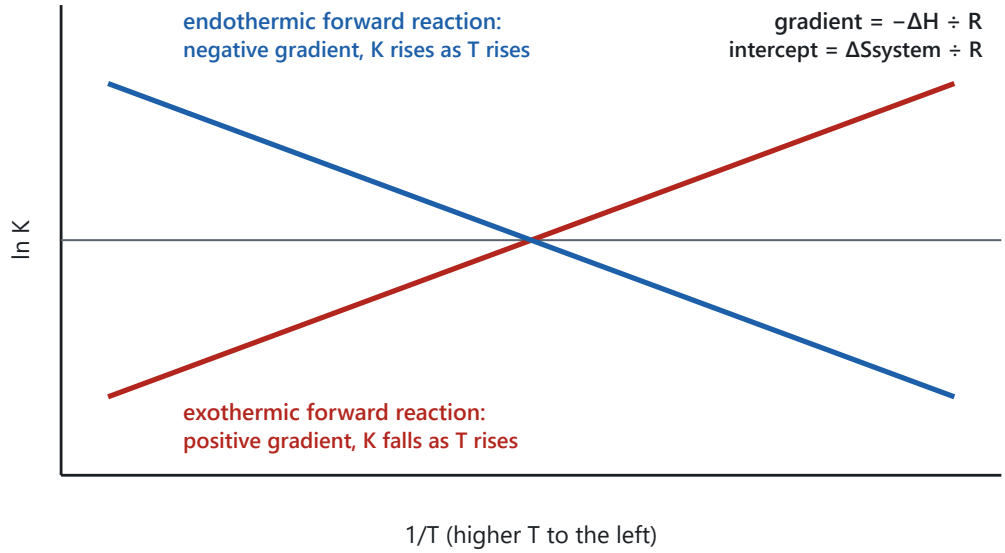
If ΔH ≈ 0, ΔSsurroundings hardly changes with T, so K hardly changes.

WORKED EXAMPLE: K FOR THE HABER PROCESS AT TWO TEMPERATURES

$N_2 + 3H_2 \rightleftharpoons 2NH_3$; ΔH = −92.2 kJ mol^{-1}; ΔSsystem = −198.8 J K^{-1} mol^{-1}
298 K: ΔStotal = −198.8 + 92 200/298 = +110.6 J K^{-1} mol^{-1}
ln K = 110.6 ÷ 8.31 = 13.31, so **$K = 6.0 \times 10^5$**: equilibrium lies far to the right.
700 K: ΔStotal = −198.8 + 92 200/700 = −198.8 + 131.7 = −67.1 J K^{-1} mol^{-1}
ln K = −67.1 ÷ 8.31 = −8.07, so **$K = 3.1 \times 10^{-4}$**: the yield at equilibrium is low.
That's why the plant needs high pressure and recycling to compensate for the high temperature it uses for rate.

WORKED EXAMPLE: ΔSTOTAL FROM K

K = 0.484 at 298 K. ΔStotal = R ln K = 8.31 × ln 0.484 = 8.31 × (−0.726) = **−6.03 J K^{-1} mol^{-1}** (K < 1, so ΔStotal is negative).



ln K = ΔSsystem/R − (ΔH/R)(1/T). The gradient is −ΔH/R, so an exothermic reaction gives a positive gradient.

MARK-SCHEME LANGUAGE

- "ΔStotal = R ln K" recalled; "ln K = ΔStotal ÷ R"; e.g. ln K = 1780.7 ÷ 8.31 = 214.2, K = 1.15×10^{93} (WCH14 Jan 2020 Q19). No TE from a wrong expression.
- Higher T, exothermic: "ΔSsystem and ΔH barely change with T, so ΔSsurroundings = −ΔH/T becomes less positive, so ΔStotal decreases" (WCH14 Jan 2022 Q22).
- ΔH close to zero: "ΔSsurroundings / ΔStotal only changes by a small amount with temperature", so "very little change in the equilibrium constant" (WCH14 Jan 2024 Q19).

Often asked as: calculate K from ΔStotal (or the reverse); explain how K changes with temperature using ΔStotal; MCQ on the relationship.

BEYOND THE SPEC, BUT EXAMINED

- **ln K against 1/T** (the van 't Hoff plot): ln K = −(ΔH/R)(1/T) + ΔSsystem/R, so **gradient = −ΔH/R** and **intercept = ΔSsystem/R** (WCH14 Oct 2022 Q18). Same shape of argument as ΔStotal against 1/T in 12.8.
- **ΔG = −RT ln K**, and ΔSsystem = (ΔH − ΔG) ÷ T (WCH14 Oct 2022 Q18).

WORKED EXAMPLE: ΔH AND ΔSSYSTEM FROM A LN K AGAINST 1/T GRAPH

The gradient of ln K against 1/T is +5000 K, and K = 50 at 500 K.

ΔH = −gradient × R = −5000 × 8.31 = −41 600 J mol^{-1} = **−41.6 kJ mol^{-1}** (positive gradient → exothermic)

ΔG = −RT ln K = −8.31 × 500 × ln 50 = −8.31 × 500 × 3.91 = −16 300 J mol^{-1}

ΔSsystem = (ΔH − ΔG) ÷ T = (−41 600 − (−16 300)) ÷ 500 = **−50.6 J K^{-1} mol^{-1}**

COMMON MISTAKE

Using kJ for ΔH in ΔStotal = ΔSsystem − ΔH/T (it must be J). Forgetting to take e^x to get from ln K to K.

Units of ΔG: J **mol^{-1}**, not J.

Describing ΔSsurroundings generically when the question is about a reaction with ΔH close to zero.

SPEC 13.9

Using K to judge how far a reaction goes

WHAT THIS SPEC POINT ASKS

Use the value of an equilibrium constant to predict the extent to which a reaction takes place.

Size of K	ΔStotal (= R ln K)	What's in the equilibrium mixture
K > about 10^{10}	more than about +200 J K^{-1} mol^{-1}	**essentially all products**: reaction "goes to completion"
K ≈ 1	≈ 0	**significant amounts of both** reactants and products
K < about 10^{-10}	below about −200 J K^{-1} mol^{-1}	**essentially no products**: reaction "doesn't go"

- A **large** K means the equilibrium lies **far to the right**; a small K means it lies far to the left.
- K tells you **how far**, never **how fast**. A reaction with K = 10^{50} can still be too slow to see if Ea is high (12.11).
- Where K is already huge, changing the pressure or concentration gains very little extra yield.

MARK-SCHEME LANGUAGE

- "the equilibrium constant is (very) large"; "the equilibrium position is (far) to the right / (heavily) favours the products". "Kp ≫ 1", "huge", "massive" allowed. IGNORE "quite large", "Kp > 1", "positive" (WCH14 Jun 2023 Q18).
- K < 1: reactants favoured (WCH14 Jun 2024 Q5).

Often asked as: what a very large K tells you about a reaction (2 marks); MCQ on what K < 1 means; why high pressure isn't worth it when K is already large.

COMMON MISTAKE

Saying "K is greater than 1" instead of commenting on how **large** it is. The mean mark on one 2-mark question was below 1 for this reason (examiner report, Jun 2023).
Describing the equilibrium "moving to the right" when the question is about the **final position**.
Assuming a large K means a fast reaction.

Topic 14 – Acid-base Equilibria

SPEC 14.1 Brønsted–Lowry acids and bases

WHAT THIS SPEC POINT ASKS

Know that a Brønsted–Lowry acid donates a proton, a base accepts one, and that an acid-base reaction is a proton transfer.

14.1 The definitions

- A **Brønsted–Lowry acid** is a **proton donor**. A **Brønsted–Lowry base** is a **proton acceptor**.
- "Proton" here means an **H^+ ion**: a hydrogen atom that has lost its electron. Never write "hydrogen atom".
- An acid-base reaction is a **proton transfer** from the acid to the base. You can't have one without the other: the acid only donates if something accepts.

14.1 Examples

Reaction	Acid (donates H^+)	Base (accepts H^+)
$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$	HCl	H_2O
$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$	H_2O	NH_3
$HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$	HCl	NH_3
$CH_3COOH(aq) + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O(l)$	CH_3COOH	OH^-

- In water, acids form the **oxonium ion H_3O^+** (also called hydronium or hydroxonium). $H^+(aq)$ is shorthand for $H_3O^+(aq)$; mark schemes accept either.
- **Water can act as an acid or a base**, depending on its partner: it accepts a proton from HCl but donates one to NH_3. Species like this are **amphoteric** (other examples: HCO_3^-, $H_2PO_4^-$, HSO_4^-).

MARK-SCHEME LANGUAGE

- "proton donor" / "proton acceptor"; "transfer of a proton / H^+".
- An acid "donates hydrogen ions and becomes an anion" (for a neutral acid like HCl or CH_3COOH).
- In an MCQ, check that the species you pick acts as an acid (or base) on the side of the equation the question means.

Often asked as: MCQ "which species is acting as a Brønsted–Lowry base?", identify the acid and base in a given equation, role of water.

COMMON MISTAKE

Writing "hydrogen atom" or "hydrogen" for proton.
Thinking water is always neutral in these reactions: in $NH_3 + H_2O$, water is the **acid**.
Putting a charge on only one product of a proton transfer when both change charge ($HCl + H_2O$ gives H_3O^+ **and** Cl^-).

BEYOND THE SPEC, BUT EXAMINED

- An acid-base **indicator** is itself a weak acid, $HIn \rightleftharpoons H^+ + In^-$, with different colours for HIn and In^- (WCH14 Oct 2024).
- A carboxylic acid can act as a **base** when mixed with a stronger acid: it accepts a proton to become $RCOOH_2^+$ (WCH14 Jun 2025). See 14.2.

SPEC 14.2

Conjugate acid-base pairs

WHAT THIS SPEC POINT ASKS

Identify the conjugate acid-base pairs in a Brønsted–Lowry reaction.

14.2 What a conjugate pair is

- When an acid loses a proton, what's left can accept one back: it is the **conjugate base** of that acid. When a base gains a proton, it becomes its **conjugate acid**.
- The two members of a pair **differ by exactly one H^+**. Every acid-base equilibrium contains **two** pairs:

$$\text{acid 1} + \text{base 2} \rightleftharpoons \text{base 1} + \text{acid 2}$$

14.2 Worked pairs

Equation	Pair 1 (acid / conjugate base)	Pair 2 (conjugate acid / base)
$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$	CH_3COOH / CH_3COO^-	H_3O^+ / H_2O
$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	H_2O / OH^-	NH_4^+ / NH_3
$HNO_3 + H_2SO_4 \rightleftharpoons H_2NO_3^+ + HSO_4^-$	H_2SO_4 / HSO_4^-	$H_2NO_3^+$ / HNO_3
$H_2PO_4^- + OH^- \rightleftharpoons HPO_4^{2-} + H_2O$	$H_2PO_4^-$ / HPO_4^{2-}	H_2O / OH^-

- In the third row, nitric acid is acting as a **base**, because sulfuric acid is the stronger acid. The **stronger acid** is always the one that donates.

14.2 How to label pairs in an answer

1. Find the two species on each side that differ by one H^+.
2. Label them **acid 1 / base 1** and **acid 2 / base 2**, or join each pair with a line. Lines or the words "conjugate acid" and "conjugate base" are safest.
3. Check the charges: removing H^+ lowers the charge by one (neutral acid → 1− base; 1+ acid → neutral base).

WORKED EXAMPLE: ONE WEAK ACID PROTONATING ANOTHER

Methanoic acid (pKa 3.75) is mixed with propanoic acid (pKa 4.87) in an unreactive solvent. Write the equation and label the pairs.

The **smaller pKa** means methanoic acid is the **stronger acid**, so it donates and propanoic acid accepts:

$HCOOH + CH_3CH_2COOH \rightleftharpoons HCOO^- + CH_3CH_2COOH_2^+$

acid 1 + base 2 ⇌ base 1 + acid 2

Pairs: HCOOH / $HCOO^-$ and $CH_3CH_2COOH_2^+$ / CH_3CH_2COOH.

MARK-SCHEME LANGUAGE

- Pairs credited when "labelled B1 A2 A1 B2 or joined by lines"; any clear indication of linkage.
- The equation must have correct charges: $CH_3COOH + CH_2ClCOOH \rightleftharpoons CH_3COOH_2^+ + CH_2ClCOO^-$ (chloroethanoic acid is stronger, so it donates).
- Each correct pair is often one mark; all four species identified correctly is the second.

Often asked as: identify both pairs in an equation; write the equation for one acid reacting with another (use the Ka/pKa table to decide which donates); MCQ on the conjugate base of a given species.

COMMON MISTAKE

Linking an acid with the **other** acid instead of with its own conjugate base.
Getting charges wrong: the conjugate base of a neutral acid is negative, and the conjugate base of a positive acid (like pyridinium, $C_5H_5NH^+$) is **neutral**.
Not using the Ka table that the question gives: it tells you which acid is stronger and so which one donates.

BEYOND THE SPEC, BUT EXAMINED

- **Carboxylic acids as bases**: protonated acids such as $CH_3COOH_2^+$ appear in equations when two acids are mixed, and this has come up on four papers (WCH14 Jun 2024).
- Conjugate pairs in **phosphate** systems: $H_2PO_4^-$ / HPO_4^{2-} (WCH14 Oct 2020).

SPEC 14.3

Definition of pH

WHAT THIS SPEC POINT ASKS

Define pH.

14.3 The definition

$$pH = -\log_{10}[H^+(aq)]$$

- In words: **pH is the negative logarithm, to base 10, of the hydrogen ion concentration** (in mol dm^{-3}). $[H_3O^+]$ in place of $[H^+]$ is equally correct.
- Square brackets mean "concentration of … in mol dm^{-3}".
- Because it is a **log scale**, a change of **1 pH unit = a factor of 10** in $[H^+]$. pH 3 has 10 times the $[H^+]$ of pH 4 and 100 times that of pH 5.
- Lower pH = higher $[H^+]$ = more acidic.

MARK-SCHEME LANGUAGE

- "$pH = -\log_{10}[H_3O^+]$"; ACCEPT lg or log, and $[H^+]$; or "negative log to the base 10 of the hydrogen ion / oxonium ion concentration".
- Square brackets are needed for concentration.

Often asked as: define pH (1 mark); as the first step of nearly every Topic 14 calculation.

COMMON MISTAKE

Missing the minus sign, or using ln (natural log) instead of $\log_{10}$.
Writing "the concentration of hydrogen" (it's hydrogen **ions**).

BEYOND THE SPEC, BUT EXAMINED

- **Negative pH**: if [H⁺] is greater than 1 mol dm⁻³, the pH is negative. 2.0 mol dm⁻³ HNO_3 has pH = −log 2.0 = **−0.30** (WCH14 Oct 2025). There is nothing special about pH 0: the 0–14 scale is just the common range.

SPEC 14.4

Calculating pH from [H⁺]

WHAT THIS SPEC POINT ASKS

Calculate pH from the hydrogen ion concentration.

14.4 Method

- Press **(−) log ([H⁺])** on the calculator. Enter powers of ten with the ×10ˣ / EXP key, not by typing "× 10 ^".
- Give pH to **2 decimal places** unless told otherwise (mark schemes accept 2 or 3 significant figures; 1 s.f. is never enough).

WORKED EXAMPLE: PH FROM [H⁺]

(a) [H⁺] = 3.2 × 10⁻³ mol dm⁻³. pH = −log(3.2 × 10⁻³) = **2.49**.
(b) [H⁺] = 2.0 mol dm⁻³. pH = −log(2.0) = **−0.30**. A negative pH is fine.
(c) [H⁺] = 1.0 × 10⁻⁹ mol dm⁻³. pH = **9.00**. This solution is alkaline; it still contains some H⁺.

EXAM TIP

Sense-check every answer. An acid must have pH below 7 (at 25 °C), and a stronger or more concentrated acid must have a lower pH. Mark schemes refuse any pH ≥ 7 for an acid even with error carried forward.

MARK-SCHEME LANGUAGE

- 2–3 s.f. acceptable (e.g. "2.68 or 2.7"); IGNORE s.f. except 1 s.f.
- "TE … but do not award a pH ≥ 7" for an acid solution.

Often asked as: the last step of weak acid, buffer and mixture calculations; MCQ with distractors that forget the minus sign.

COMMON MISTAKE

Forgetting to take the square root of Ka × [HA] **before** taking the log (weak acids, 14.9).
Rounding [H⁺] early, then taking the log: keep the full calculator value.

SPEC 14.5

Calculating [H⁺] from pH

WHAT THIS SPEC POINT ASKS

Calculate [H⁺] in mol dm⁻³ from pH using $[H^+] = 10^{-pH}$.

14.5 Method

$$[H^+] = 10^{-pH}$$

- On the calculator: **SHIFT log** (10ˣ) then **(−) pH**.

- The answer has units of **mol dm^{-3}**.

WORKED EXAMPLE: [H$^+$] FROM PH, AND DILUTING TO A TARGET PH

(a) pH = 3.40. [H$^+$] = $10^{-3.40}$ = **4.0×10^{-4} mol dm^{-3}**.
(b) 25.0 cm^3 of HCl has pH 1.00. What volume of water must be added to raise the pH to 2.00?
[H$^+$] goes from 0.100 to 0.0100 mol dm^{-3}: a tenfold dilution (HCl is fully dissociated, so [H$^+$] ∝ concentration).
Final volume = 25.0 × 10 = 250 cm^3, so add **225 cm^3** of water.

MARK-SCHEME LANGUAGE

- Evaluate the power: "[H$^+$] = 10^−1.125 = 0.075 mol dm^{-3}". Writing 10^−1.125 and stopping doesn't score.
- Dilution questions: final volume first, then subtract the starting volume for the water added (2–3 s.f.).

Often asked as: [H$^+$] from a measured pH as the first step of finding Ka (14.14); dilution to a given pH.

COMMON MISTAKE

Rounding $10^{-0.97}$ = 0.107 to "0.1" and then doubling for sulfuric acid. Use the value the pH gives you. Giving the final volume when the question asked for the volume of water **added**.

SPEC 14.6

Strong and weak acids: degree of dissociation

WHAT THIS SPEC POINT ASKS

Understand the difference between strong and weak acids in terms of how far they dissociate.

14.6 Strong v weak

- A **strong acid** is (almost) **completely dissociated** into ions in water. Use a one-way arrow:
  HCl(aq) → H$^+$(aq) + Cl$^-$(aq)
  Examples: HCl, HNO_3, HBr, HI, H_2SO_4 (first proton).
- A **weak acid** is only **partially dissociated** (slightly ionised). Most molecules stay as HA, so use ⇌:
  $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$
  Examples: carboxylic acids, HCN, H_2CO_3, NH_4^+, HF. In 0.1 mol dm^{-3} ethanoic acid only about 1 molecule in 80 is dissociated.
- For the same concentration, the weak acid has a **lower [H$^+$]**, so a **higher pH**, lower electrical conductivity and slower reactions with metals or carbonates. The total amount of acid that a base can neutralise is the same.

14.6 Strength is not concentration

	Meaning
Strong / weak	**how far** the acid dissociates (a fixed property at a given temperature)
Concentrated / dilute	**how much** acid is dissolved per dm^3

So 0.001 mol dm^{-3} HCl is a **dilute strong** acid; 10 mol dm^{-3} ethanoic acid is a **concentrated weak** acid.

14.6 Sulfuric acid

- $H_2SO_4 \rightarrow H^+ + HSO_4^-$ is **complete** (strong).

- HSO_4^- ⇌ H^+ + SO_4^{2-} is **partial**, and the high [H^+] from the first step pushes this second equilibrium to the left (suppresses it).
- So 0.100 mol dm^{-3} H_2SO_4 has [H^+] a little above 0.100 mol dm^{-3} (measured pH about 0.97), not 0.200.

MARK-SCHEME LANGUAGE

- "a weak acid is dissociated (into its ions) to a small extent and a strong acid is (almost) completely dissociated"; ALLOW "partially ionised" / "fully ionised".
- A weak acid solution "contains a significant proportion of undissociated acid".
- Sulfuric acid: "the first ionisation is complete … the high [H^+] from the first suppresses the second equilibrium".

Often asked as: define a weak acid (1–2 marks); compare the pH of equimolar strong and weak acids; explain why diluting changes their pH differently (14.13).

COMMON MISTAKE

Describing pH or "position of equilibrium" instead of **dissociation/ionisation** when asked to define strong and weak.
Mixing up "weak" with "dilute".
Vague sulfuric acid answers that don't say **which** of its two equilibria is complete.

BEYOND THE SPEC, BUT EXAMINED

- **Inductive effects**: an electronegative atom near the COOH group makes the acid **stronger**. In chloroethanoic acid the Cl is **electron-withdrawing**, so it pulls electron density away from the O–H bond (weakening it) and **stabilises the COO^- ion** by spreading its charge. Ka order: CH_3COOH < $CH_2ClCOOH$ < $CHCl_2COOH$ < CCl_3COOH. Alkyl groups push electrons the other way, which is why propanoic acid (pKa 4.87) is slightly weaker than ethanoic acid (pKa 4.76). "More hydrogen bonding" is not accepted as the reason (WCH14 Jun 2025).
- The extra OH in lactic acid is electron-withdrawing, which is why lactic acid is stronger than propanoic acid. The OH itself is **not** acidic (WCH14 Oct 2022).

SPEC 14.7 pH of a strong acid

WHAT THIS SPEC POINT ASKS

Calculate the pH of a strong acid.

14.7 Method

- A strong monoprotic acid is fully dissociated, so **[H^+] = concentration of the acid**. Then pH = −log[H^+].
- For a **mixture** of strong acid and strong base, work in **moles**:
  1. moles H^+ and moles OH^- at the start;
  2. subtract to find which is **in excess** and by how much;
  3. divide by the **total volume** (in dm^3) to get the concentration;
  4. take −log (acid in excess) or go via Kw (alkali in excess, 14.11).

WORKED EXAMPLE: STRONG ACID, AND A STRONG ACID–STRONG BASE MIXTURE

(a) 0.0250 mol dm^{-3} HCl: $[H^+]$ = 0.0250, pH = **1.60**.
(b) 25.0 cm^3 of 0.100 mol dm^{-3} HCl is mixed with 15.0 cm^3 of 0.100 mol dm^{-3} NaOH.
moles H^+ = 0.100 × 25.0/1000 = 2.50 × 10^{-3} mol
moles OH^- = 0.100 × 15.0/1000 = 1.50 × 10^{-3} mol
excess H^+ = 1.00 × 10^{-3} mol in 40.0 cm^3
$[H^+]$ = 1.00 × 10^{-3} ÷ 0.0400 = 0.0250 mol dm^{-3}, pH = **1.60**.

EXAM TIP

For sulfuric acid in a **neutralisation** (moles) calculation, 1 mol H_2SO_4 supplies **2 mol H^+**. For the pH of the acid **on its own**, read the question: it may tell you to assume both protons dissociate, or give you a measured pH.

MARK-SCHEME LANGUAGE

- Mixtures: "moles of H^+ – moles of OH^-", "÷ total volume", then pH.
- With H_2SO_4 and NaOH: moles H^+ = 2 × moles H_2SO_4.

Often asked as: MCQ on the pH after a tenfold dilution or concentration change; pH of a mixture; pH after part of an acid has reacted.

COMMON MISTAKE

Forgetting to subtract the moles that reacted, or subtracting moles of H_2SO_4 instead of moles of H^+. Dividing by the volume of one solution instead of the **total** volume.

BEYOND THE SPEC, BUT EXAMINED

- Strong acid–strong base mixtures and the dibasic nature of H_2SO_4 (WCH14 Oct 2024). Mixtures with excess acid (WCH14 Oct 2025).

SPEC 14.8

The acid dissociation constant, Ka

WHAT THIS SPEC POINT ASKS

Write the expression for Ka for a weak acid.

14.8 The expression

For a weak acid HA(aq) ⇌ H^+(aq) + A^-(aq):

$$K_a = [H^+][A^-] \div [HA] \text{ units: mol dm}^{-3}$$

- For ethanoic acid: Ka = $[H^+][CH_3COO^-] \div [CH_3COOH]$.
- **Water does not appear** (it is the solvent; its concentration is effectively constant).
- All concentrations are **equilibrium** concentrations.
- **Bigger Ka = stronger acid** (more dissociated). Ka depends only on **temperature**.

Acid	Ka / mol dm^{-3}	pKa
methanoic, HCOOH	1.8×10^{-4}	3.75
benzoic, C_6H_5COOH	6.3×10^{-5}	4.20
ethanoic, CH_3COOH	1.7×10^{-5}	4.76
propanoic, CH_3CH_2COOH	1.3×10^{-5}	4.87
carbonic, H_2CO_3	4.3×10^{-7}	6.37

(Values vary slightly between data sources: use whatever the question gives.)

MARK-SCHEME LANGUAGE

- "Ka = $[H_3O^+][CH_3COO^-] \div [CH_3COOH]$"; ACCEPT $[H^+]$.
- DO NOT AWARD HA and A^- unless you have defined them for that acid; round or curly brackets; the charge written outside the square bracket.

Often asked as: write Ka for a named or unfamiliar acid (hexanoic, gluconic, the pyridinium ion, Tris-H^+), usually 1 mark before a calculation.

COMMON MISTAKE

Writing the dissociation **equation** when the **expression** is asked (examiners call it "the most common error").
Including water, or using round brackets.

BEYOND THE SPEC, BUT EXAMINED

- Ka for **positive weak acids**, such as the pyridinium ion: $C_5H_5NH^+ \rightleftharpoons C_5H_5N + H^+$, so Ka = $[C_5H_5N][H^+] \div [C_5H_5NH^+]$ (WCH14 Jun 2024).

SPEC 14.9

pH of a weak acid

WHAT THIS SPEC POINT ASKS

Calculate the pH of a weak acid from Ka or pKa, stating the assumptions (no quadratics needed).

14.9 Method

1. If you are given pKa, find **Ka = 10^{-pKa}**.
2. Make **two assumptions**:
   - **$[H^+] = [A^-]$**: all the H^+ comes from the acid; the tiny amount from water is ignored.
   - **[HA]equilibrium = [HA]initial**: the acid dissociates so little that its concentration hardly changes.
3. Ka = $[H^+]^2 \div [HA]$, so **$[H^+] = \sqrt{(Ka \times [HA])}$**.
4. pH = $-\log[H^+]$.

WORKED EXAMPLE: PH OF WEAK ACIDS

(a) 0.100 mol dm^{-3} ethanoic acid, Ka = 1.7 × 10^{-5} mol dm^{-3}.
$[H^+] = \sqrt{(1.7 \times 10^{-5} \times 0.100)} = 1.30 \times 10^{-3}$ mol dm^{-3}, pH = **2.88**.
(b) 0.0500 mol dm^{-3} propanoic acid, pKa = 4.87.
Ka = $10^{-4.87}$ = 1.35 × 10^{-5} mol dm^{-3}; $[H^+] = \sqrt{(1.35 \times 10^{-5} \times 0.0500)} = 8.21 \times 10^{-4}$; pH = **3.09**.
(c) What mass of ethanoic acid (Mr 60.0) must be dissolved to make 250 cm^3 of solution of pH 3.00?
$[H^+] = 1.00 \times 10^{-3}$; $[HA] = [H^+]^2 \div Ka = 1.00 \times 10^{-6} \div 1.7 \times 10^{-5} = 0.0588$ mol dm^{-3}
moles = 0.0588 × 0.250 = 0.0147 mol; mass = 0.0147 × 60.0 = **0.88 g**.

14.9 When the assumptions fail

- The second assumption fails when the acid is **not very weak** (Ka bigger than about 10^{-3}) or very **dilute**: then a significant fraction dissociates, so [HA]eqm is noticeably **less** than [HA]initial.
- The calculated $[H^+]$ is then **too high**, so the calculated pH is **lower** than the measured pH. The error is bigger for the stronger acid.

MARK-SCHEME LANGUAGE

- "$[H^+] = \sqrt{(Ka \times [HA])}$", then −log; correct answer with no working scores full marks.
- Assumptions (any one per mark): "degree of dissociation is very small / negligible"; "$[CH_3COOH]_{eqm} = [CH_3COOH]_{initial}$"; "$[H^+] = [CH_3COO^-]$"; "all the hydrogen ions come from the acid / none from water".
- Questioning validity: "[HA]equilibrium is lower than [HA]initial because dissociation is significant", so the calculated pH is lower than the measured one.

Often asked as: pH from Ka/pKa (2–3 marks); state one or two assumptions; MCQ on which assumption is invalid; 6-mark comparing calculated and measured pH; mass of acid for a given pH.

COMMON MISTAKE

Giving the **same assumption twice** in different words: give one about water ($H^+ = A^-$) and one about the degree of dissociation.
Forgetting to convert pKa to Ka, or g dm^{-3} to mol dm^{-3} (divide by Mr).
Blaming the dissociation of **water** for a difference between measured and calculated pH: the real reason is that dissociation of the acid is not negligible.

BEYOND THE SPEC, BUT EXAMINED

- An acid whose dissociation is **not** negligible: dichloroethanoic acid, Ka = 4.5 × 10^{-2} mol dm^{-3}. The simple method gives a pH that is too low (WCH14 Jun 2024).
- Working backwards to the **mass of acid** needed for a given pH (WCH14 Oct 2022). With the assumption, the mass is an underestimate, because [HA]eqm < [HA]initial means more acid is really needed.

SPEC 14.10 The ionic product of water, Kw

WHAT THIS SPEC POINT ASKS

Define Kw.

14.10 Where Kw comes from

- Water ionises very slightly: $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$
- $[H_2O]$ is effectively constant, so it is absorbed into the constant:

$$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}\ \text{mol}^2\ \text{dm}^{-6} \text{ at } 298\ \text{K}$$

- In **pure water** $[H^+] = [OH^-]$, so $[H^+] = \sqrt{Kw} = 1.00 \times 10^{-7}$ mol dm^{-3} and pH = 7.00 at 298 K.
- **Neutral** means $[H^+] = [OH^-]$, not "pH 7". Acidic: $[H^+] > [OH^-]$. Alkaline: $[OH^-] > [H^+]$.

14.10 Kw changes with temperature

- The ionisation of water is **endothermic** (it breaks bonds). Raising the temperature shifts the equilibrium **right**, so **Kw increases**.
- At higher temperature $[H^+]$ is bigger, so the **pH of pure water falls below 7**, but the water is **still neutral** because $[H^+]$ still equals $[OH^-]$.

Temperature	Kw / mol^2 dm^{-6}	pH of pure water
283 K (10 °C)	2.9×10^{-15} (pKw 14.53)	7.27
298 K (25 °C)	1.00×10^{-14}	7.00
313 K (40 °C)	2.9×10^{-14}	6.77
323 K (50 °C)	5.5×10^{-14}	6.63

WORKED EXAMPLE: PURE WATER AT 313 K

Kw at 313 K = 2.92×10^{-14} mol^2 dm^{-6}.
$[H^+] = \sqrt{(2.92 \times 10^{-14})} = 1.71 \times 10^{-7}$ mol dm^{-3}, pH = **6.77**. The water is neutral: $[H^+] = [OH^-]$.

MARK-SCHEME LANGUAGE

- "Kw = $[H^+][OH^-]$" with units mol^2 dm^{-6}.
- Temperature: "the dissociation of water is endothermic, so the concentration of hydrogen ions is higher at 75 °C"; "in pure water the concentrations of hydrogen ions and hydroxide ions are always equal".
- At a temperature other than 298 K use the pKw you are given: pH = pKw − pOH.

Often asked as: MCQ on the pH of pure water at 40 °C or 75 °C; "is water at pH 6.6 acidic?"; pH of NaOH at 283 K using pKw = 14.53.

COMMON MISTAKE

Saying hot water is "acidic" because its pH is below 7.
Using 14 for pKw when the question gives a different value.
Including $[H_2O]$ in the Kw expression.

BEYOND THE SPEC, BUT EXAMINED

- **Neutral water that isn't pH 7** above 25 °C, and the endothermic explanation (WCH14 Jun 2025). At 100 °C Kw is about 5×10^{-13}, so pure water has pH about 6.1 (WCH14 Jan 2023).

SPEC 14.11 pH of a strong base

WHAT THIS SPEC POINT ASKS

Calculate the pH of a strong base from its concentration, using Kw or pKw.

14.11 Method

1. **[OH^-]** from the concentration. NaOH and KOH give [OH^-] = c. $Ba(OH)_2$ and other Group 2 hydroxides give **[OH^-] = 2c**.
2. Either **[H^+] = Kw ÷ [OH^-]**, then pH = −log[H^+];
   or **pOH = −log[OH^-]**, then **pH = pKw − pOH** (pKw = 14.00 at 298 K).

WORKED EXAMPLE: PH OF STRONG BASES

(a) 0.0500 mol dm^{-3} NaOH at 298 K: [H^+] = $1.00 \times 10^{-14} \div 0.0500 = 2.00 \times 10^{-13}$; pH = **12.70**.
(b) 0.0200 mol dm^{-3} $Ba(OH)_2$: [OH^-] = 0.0400; pOH = 1.40; pH = 14.00 − 1.40 = **12.60**.
(c) 0.100 mol dm^{-3} NaOH at 283 K, where pKw = 14.53: pOH = 1.00, pH = **13.53**.
(d) 25.0 cm^3 of 0.100 mol dm^{-3} HCl + 30.0 cm^3 of 0.100 mol dm^{-3} NaOH:
excess OH^- = $3.00 \times 10^{-3} - 2.50 \times 10^{-3} = 5.0 \times 10^{-4}$ mol in 55.0 cm^3
[OH^-] = 9.09×10^{-3} mol dm^{-3}; [H^+] = $1.00 \times 10^{-14} \div 9.09 \times 10^{-3} = 1.10 \times 10^{-12}$; pH = **11.96**.

MARK-SCHEME LANGUAGE

- "[OH^-] = 0.4 mol dm^{-3}" for 0.2 mol dm^{-3} $Ba(OH)_2$, pOH = 0.40, pH = 13.6.
- After a reaction that uses up OH^- (e.g. ester hydrolysis): moles NaOH − moles used, new concentration, "pH = 14 − (−log[OH^-])".

Often asked as: pH of NaOH or $Ba(OH)_2$; pH of an alkali left over after a reaction or titration; MCQ at a non-standard temperature.

COMMON MISTAKE

Taking −log[OH^-] and calling it the pH.
Forgetting the 2 for $Ba(OH)_2$, or forgetting to subtract the OH^- that has been used up.
Not dividing by the final (combined) volume.

BEYOND THE SPEC, BUT EXAMINED

- pH of the excess NaOH after an ester hydrolysis: subtract the moles that reacted first; "13.99" from the original concentration scored nothing (WCH14 Jan 2020).

SPEC 14.12 pKa and pKw

WHAT THIS SPEC POINT ASKS

Define pKa and pKw.

14.12 Definitions

$$pK_a = -\log_{10} K_a \quad K_a = 10^{-pKa}$$

$$pK_w = -\log_{10} K_w = 14.00 \text{ at } 298 \text{ K}$$

- The "p" means "$-\log_{10}$ of".
- **Smaller pKa = larger Ka = stronger acid.** A difference of 1 in pKa is a factor of 10 in Ka.
- pKa values are easier to compare than Ka values with negative powers.
- At 298 K, pH + pOH = 14.00 (more generally pH + pOH = pKw).

WORKED EXAMPLE: CONVERTING

(a) pKa = 5.25 → Ka = $10^{-5.25}$ = **5.6×10^{-6} mol dm^{-3}**.
(b) Ka = 4.5×10^{-2} mol dm^{-3} → pKa = **1.35**.
Which acid is stronger? (b), because its pKa is smaller.

MARK-SCHEME LANGUAGE

- "inverse log(−5.25) to 2 s.f." (5.6×10^{-6}) and "−log(4.5×10^{-2}) to 3 s.f." (1.35). DO NOT AWARD 6×10^{-6} or 1.4.
- pKa and pKw have **no units**.

Often asked as: conversions inside longer calculations; rank acids by pKa.

COMMON MISTAKE

Rounding to 1 s.f., or giving s.f. inconsistent with the data.
Thinking a bigger pKa means a stronger acid.

SPEC 14.13

Analysing pH data: acids, bases, salts and dilution

WHAT THIS SPEC POINT ASKS

Interpret pH measurements of equimolar strong and weak acids and bases and of salts, and compare the pH of strong and weak acids after 10, 100 and 1000-fold dilution.

14.13 Equimolar acids and bases

At the **same concentration and temperature**, lower pH = stronger acid; higher pH = stronger base. Typical values for 0.100 mol dm^{-3} solutions at 298 K:

Solution (0.100 mol dm^{-3})	pH	Why
HCl	1.00	strong acid, [H$^+$] = 0.100
CH_3COOH	2.9	weak acid, ~1% dissociated
NH_3	11.1	weak base, partly protonated
NaOH	13.0	strong base, [OH$^-$] = 0.100

14.13 Salts are not all neutral

The pH of a salt solution depends on which acid and base it came from:

Salt from	Example	pH (0.1 mol dm^{-3})	Reason
strong acid + strong base	NaCl, KNO_3	7	neither ion reacts with water
weak acid + strong base	CH_3COONa	about 8.9	$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$
strong acid + weak base	NH_4Cl	about 5.1	$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$
weak acid + weak base	CH_3COONH_4	about 7	both ions react, about equally

- The anion of a weak acid is a weak **base**; the cation of a weak base is a weak **acid**. That's why the equivalence point of a titration is not always at pH 7 (14.15).

14.13 Dilution: strong v weak acid

Dilution of 0.100 mol dm^{-3}	×1	×10	×100	×1000
HCl, pH	1.00	2.00	3.00	4.00
CH_3COOH, pH	2.88	3.38	3.88	about 4.4

- **Strong acid**: each tenfold dilution divides $[H^+]$ by 10, so the pH rises by **1 unit**.
- **Weak acid**: the pH rises by only about **0.5 units** per tenfold dilution. Two ways to see why:
  - $[H^+] = \sqrt{(Ka \times c)}$, so dividing c by 10 divides $[H^+]$ by $\sqrt{10} \approx 3.2$, and $\log 3.2 \approx 0.5$.
  - Le Chatelier: diluting shifts $HA \rightleftharpoons H^+ + A^-$ to the right (more particles side), so the acid **dissociates more** and partly makes up for the dilution.
- At great dilution the weak acid's dissociation is no longer small (about 12% at 0.0001 mol dm^{-3}), so the simple formula slightly underestimates the pH.
- Diluting an acid can never take its pH above 7: at around 10^{-8} mol dm^{-3} the H^+ from water dominates and the pH just approaches 7.

MARK-SCHEME LANGUAGE

- Dilution (4 points): HCl "fully dissociated"; "a decrease in concentration by a factor of 10 decreases $[H^+]$ by a factor of 10" so pH **increases** by 1; ethanoic acid "not fully dissociated (so its pH is much higher)"; "when diluted, ethanoic acid dissociates more (so the pH does not change so much)". IGNORE "pH increases by 0.5" on its own.
- Ammonium chloride: "$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$".

Often asked as: explain the different pH changes on dilution; rank the pH of several solutions (MCQ); explain why a salt solution is acidic or alkaline.

COMMON MISTAKE

Saying dilution "decreases the pH". Diluting an acid **raises** its pH.
Writing NH_4Cl, rather than the NH_4^+ ion, in the hydrolysis equation ("the majority used ammonium chloride").
Comparing acids of **different** concentrations as if pH alone showed strength.

BEYOND THE SPEC, BUT EXAMINED

- **Salt solutions with pH ≠ 7**, with the hydrolysis equation for the ion that reacts (WCH14 Jun 2022).

SPEC
14.14 Ka from experimental data

WHAT THIS SPEC POINT ASKS

Calculate Ka of a weak acid from the pH of a solution made from a known mass of the acid.

14.14 Method

1. **Concentration** of the acid: moles = mass ÷ Mr; c = moles ÷ volume in dm^3.
2. **$[H^+] = 10^{-pH}$** from the measured pH.
3. Assume **$[A^-] = [H^+]$** and **$[HA] \approx c$** (dissociation small).
4. **$K_a = [H^+]^2 \div c$**, units mol dm^{-3}. Then pKa = −log Ka if asked.
5. If dissociation is not small, the better value uses **$[HA] = c - [H^+]$**.

WORKED EXAMPLE: KA FROM MASS AND PH

0.600 g of ethanoic acid (Mr 60.0) is dissolved in water and made up to 250 cm^3. The pH is 3.08.
moles = 0.600 ÷ 60.0 = 0.0100 mol; c = 0.0100 ÷ 0.250 = 0.0400 mol dm^{-3}
$[H^+] = 10^{-3.08} = 8.32 \times 10^{-4}$ mol dm^{-3}
$K_a = (8.32 \times 10^{-4})^2 \div 0.0400 =$ **1.7×10^{-5} mol dm^{-3}** (pKa = 4.76)
Percentage dissociated = $8.32 \times 10^{-4} \div 0.0400 \times 100$ = 2.1%, so the assumption is fine.

WORKED EXAMPLE: CHECKING THE ASSUMPTION

0.244 g of benzoic acid (Mr 122) in 100 cm^3 of solution has pH 2.94.
c = 0.00200 mol ÷ 0.100 dm^3 = 0.0200 mol dm^{-3}; $[H^+] = 10^{-2.94} = 1.15 \times 10^{-3}$ mol dm^{-3}
With [HA] = c: $K_a = (1.15 \times 10^{-3})^2 \div 0.0200 =$ **6.6×10^{-5} mol dm^{-3}**
Percentage dissociated = 5.7%. Using [HA] = 0.0200 − 0.00115 = 0.01885 gives $K_a = 7.0 \times 10^{-5}$ mol dm^{-3}, a better value.

MARK-SCHEME LANGUAGE

- Steps credited: concentration of acid from mass ÷ Mr ÷ volume; $[H^+]$ from pH; Ka expression; Ka with units.
- Phosphoric acid type: "$[H^+] = 10^{-1.2} = 0.0631$", "assume $[H_2PO_4^-] = [H_3O^+]$", K_{a1} with units.

Often asked as: Ka from a mass and a pH; Ka of a polyprotic acid's first dissociation; part of Core Practical 11 questions.

COMMON MISTAKE

Using the mass concentration (g dm^{-3}) as if it were mol dm^{-3}.
Using the Mr of the **ion** instead of the acid or salt.
Using the initial concentration without comment when the acid is quite strong: the best answers subtract $[H^+]$.

BEYOND THE SPEC, BUT EXAMINED

- For a relatively strong acid such as H_3PO_4, the best answer uses the equilibrium concentration $[H_3PO_4] = c - [H^+]$; "only a very few" did this (WCH14 Oct 2020).

SPEC 14.15

Titration curves

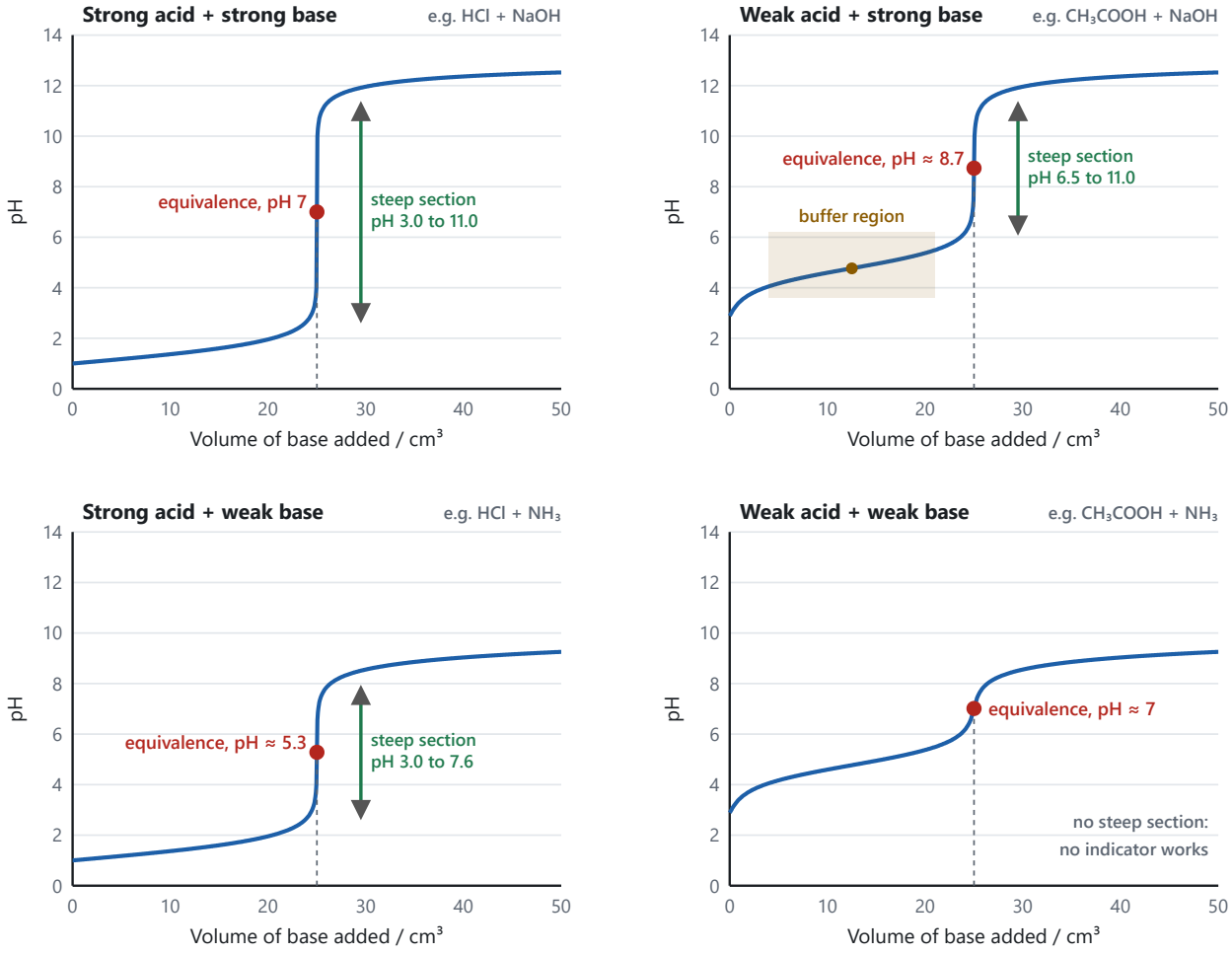
WHAT THIS SPEC POINT ASKS

Draw and interpret pH titration curves for all combinations of strong and weak monoprotic acids and bases, and apply the same ideas to diprotic acids and bases.

14.15 Reading a pH curve

A pH curve plots **pH (y-axis)** against **volume of solution added (x-axis)**. Things to read off:

- **Starting pH**: tells you what is in the flask (strong acid about 1; weak acid about 3).
- **Equivalence point**: where the moles of acid and base match the equation. It is the **middle of the steep (vertical) section**. Its volume gives you the concentration; its pH tells you which salt is present.
- **Steep section**: near equivalence, almost no H^+ (or OH^-) is left, so one drop of titrant is a large fraction of what remains and the pH jumps several units.
- **Final pH**: approaches (but stays below) the pH of the solution being added, because it is diluted.



Calculated curves: 25.0 cm^3 of 0.100 mol dm^{-3} acid with 0.100 mol dm^{-3} base. The steep section is shown from 24.5 to 25.5 cm^3 of base added.

14.15 The four combinations

Acid + base	Start pH	Equivalence pH	Steep section	End pH
Strong + strong (HCl + NaOH)	about 1	**7**	about 3 to 11	about 12–13
Weak + strong (CH_3COOH + NaOH)	about 3	**above 7** (about 8.7)	about 6.5 to 11	about 12–13
Strong + weak (HCl + NH_3)	about 1	**below 7** (about 5.3)	about 3 to 7.5	about 9–10
Weak + weak (CH_3COOH + NH_3)	about 3	about 7	**none**	about 9–10

- **Why the equivalence pH differs**: at equivalence only the salt is present. CH_3COO^- is a weak base (pH > 7); NH_4^+ is a weak acid (pH < 7). See the salt table in 14.13.
- **Weak acid + strong base** has an initial small rise, then a gently sloping **buffer region** (14.21), then the steep section.

14.15 Why the equivalence point of a weak acid is above 7 (extension)

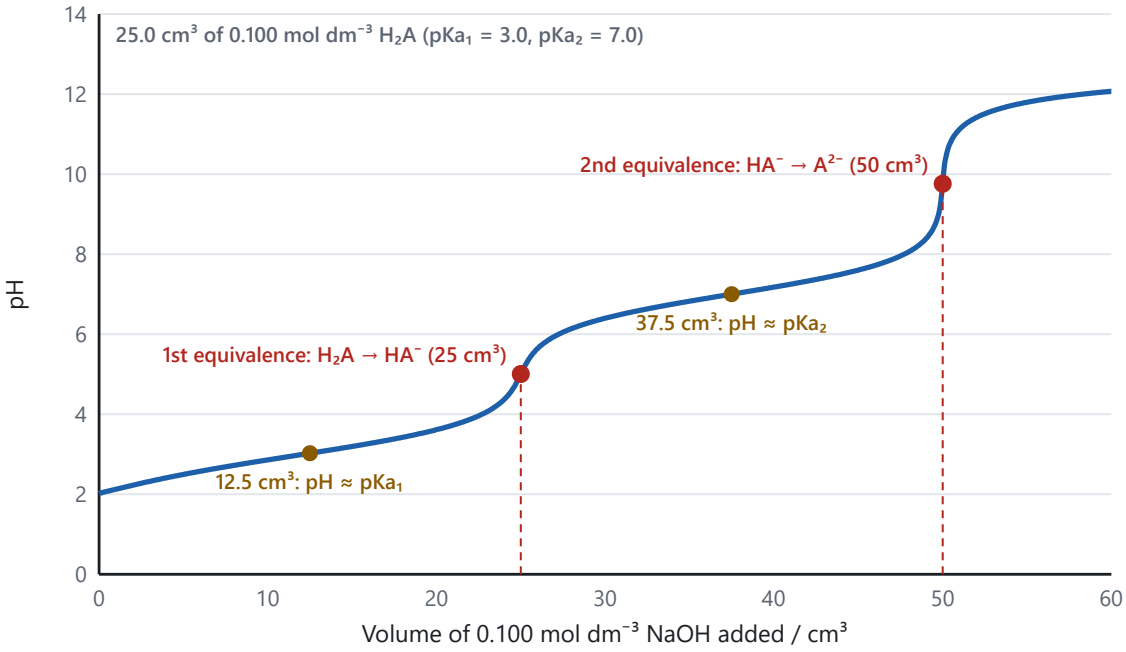
At equivalence, 25.0 cm^3 of 0.100 mol dm^{-3} CH_3COOH + 25.0 cm^3 of NaOH gives 0.0500 mol dm^{-3} CH_3COO^-, which reacts slightly with water: $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$. Using Kb = Kw ÷ Ka = 5.9×10^{-10}: $[OH^-] = \sqrt{(Kb \times c)} = 5.4 \times 10^{-6}$ mol dm^{-3}, so pOH = 5.27 and pH = **8.73**. You won't be asked for this calculation, but you must be able to say that "the salt of a weak acid is basic".

14.15 Acid added to alkali

If the **acid is in the burette** and the alkali in the flask, the curve is the same shape **upside down**: it starts high and falls. Check which solution is being added before you sketch.

14.15 Diprotic acids and bases

- A **diprotic acid** H_2A gives up two protons in turn: $H_2A + OH^- \rightarrow HA^- + H_2O$, then $HA^- + OH^- \rightarrow A^{2-} + H_2O$.
- If the two pKa values are well apart, the curve has **two steep sections**: the second equivalence volume is **twice** the first.
- Half-way to each equivalence point, pH ≈ pKa_1 and pH ≈ pKa_2.
- A **diprotic base** behaves the same way. Sodium carbonate titrated with HCl shows two steps: $CO_3^{2-} \rightarrow HCO_3^-$ (near pH 8.3) and then $HCO_3^- \rightarrow H_2CO_3$ ($CO_2 + H_2O$, near pH 3.7).
- With a strong diprotic acid such as H_2SO_4 titrated with NaOH, both protons are removed together and there is one steep section, at the volume for 2 mol NaOH per mol H_2SO_4.



A diprotic weak acid with NaOH: two steep sections, at 25 and 50 cm³. At 12.5 and 37.5 cm³ the pH is close to pKa_1 and pKa_2.

WORKED EXAMPLE: READING A CURVE

25.0 cm³ of propanoic acid is titrated with 0.100 mol dm⁻³ NaOH. The steep section is centred on 20.0 cm³. At 10.0 cm³ the pH is 4.87.

(a) Concentration of the acid: moles NaOH = 0.100 × 0.0200 = 2.00 × 10^{-3} mol = moles acid (1 : 1). c = 2.00 × 10^{-3} ÷ 0.0250 = **0.0800 mol dm⁻³**.

(b) Ka: 10.0 cm³ is half-way to equivalence, so pH = pKa = 4.87. Ka = $10^{-4.87}$ = **1.3 × 10^{-5} mol dm⁻³**.

(c) The acid is weak: the start pH is about 3 and the equivalence point is above 7.

MARK-SCHEME LANGUAGE

- Sketch of weak acid + strong base (any three): "start pH between 2 and 4 and finish at 12–14"; "vertical section of at least 3 pH units starting at or above 6 and finishing at or below 11"; "S-shaped curve with equivalence at the right volume".
- Diprotic: "vertical section at 25 cm³ with height 1–4 pH units"; a second vertical section at 50 cm³.
- Weak acid + strong base: equivalence pH is above 7 because "the salt of a weak acid is basic".

Often asked as: sketch a curve on a grid; pick which curve matches a pair of solutions; read the equivalence volume; sketch a diprotic curve; calculate the pH at a point on the curve.

COMMON MISTAKE

Curves that don't start or end at a sensible pH, or a steep section that isn't vertical.

Reading the equivalence volume as the last volume plotted (e.g. 46 cm³) instead of the middle of the steep section (22 cm³).

Drawing the curve going up when acid is added to alkali.

Only one steep section for a diprotic acid whose pKa values are well apart.

BEYOND THE SPEC, BUT EXAMINED

- **pH at a point on the curve** by moles (e.g. after excess alkali is added), and showing why the curve levels off below the pH of the alkali added: it is diluted and partly neutralised (WCH14 Oct 2023).
- Choosing the right curve for **acid added to ammonia** (WCH14 Jun 2022).

SPEC 14.16

Choosing an indicator

WHAT THIS SPEC POINT ASKS

Choose a suitable indicator for a titration from its pH curve and indicator data.

14.16 How indicators work

- An indicator is a **weak acid** whose undissociated form and conjugate base have different colours: $HIn \rightleftharpoons H^+ + In^-$.
- In acid, the equilibrium lies left (colour of HIn); in alkali, right (colour of In^-).
- The colour changes over a range of about **pKIn ± 1**. At the **end point**, [HIn] = [In^-] and the solution shows the **mixed colour** (orange for methyl orange, green for bromothymol blue, pale pink for phenolphthalein).

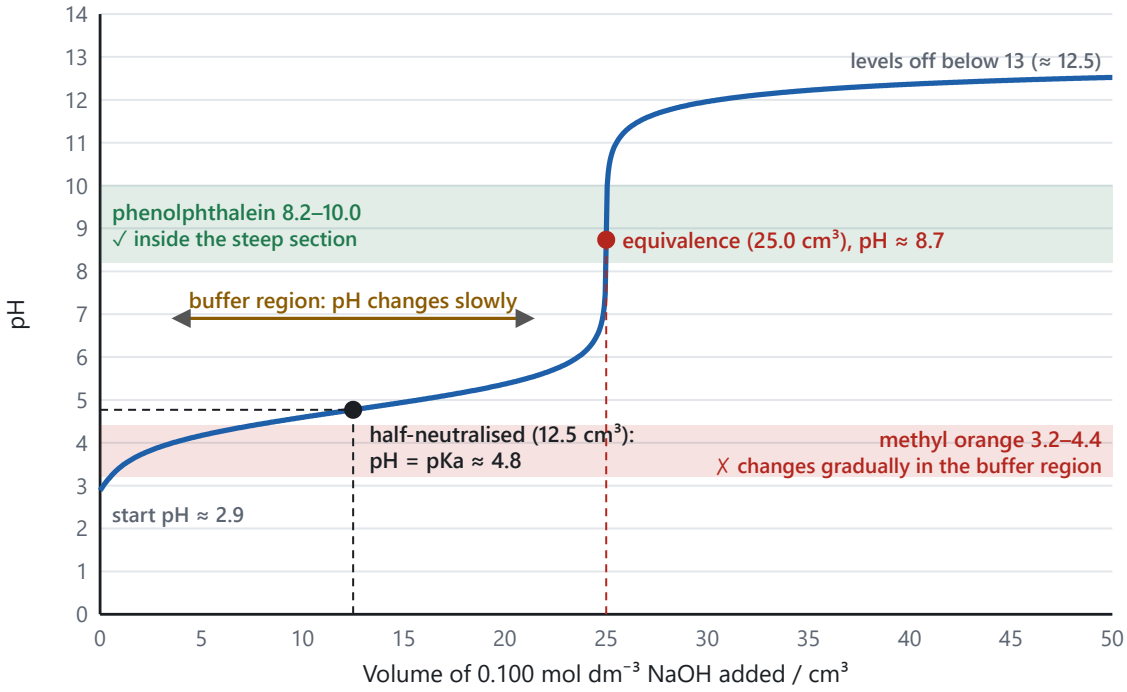
14.16 The rule

Choose an indicator whose whole colour-change range lies within the steep (vertical) section of the curve. Then one drop of titrant at the equivalence point changes the colour completely.

Indicator (Data Booklet)	pH range	Colour: acid → alkali
methyl orange	3.2–4.4	red → yellow
bromophenol blue	2.8–4.6	yellow → blue
bromocresol green	3.8–5.4	yellow → blue
methyl red	4.2–6.3	red → yellow
bromothymol blue	6.0–7.6	yellow → blue
phenol red	6.8–8.4	yellow → red
thymol blue (base)	8.0–9.6	yellow → blue
phenolphthalein	8.2–10.0	colourless → pink

(Check the exact ranges in your own Data Booklet.)

Titration	Steep section	Suitable indicator
strong acid + strong base	about 3–11	**methyl orange or phenolphthalein** (or anything in between)
weak acid + strong base	about 6.5–11	**phenolphthalein** (also thymol blue, phenol red)
strong acid + weak base	about 3–7.5	**methyl orange** (also methyl red, bromocresol green)
weak acid + weak base	none	**no indicator** works; use a pH meter



Ethanoic acid (25.0 cm^3, 0.100 mol dm^{-3}) with 0.100 mol dm^{-3} NaOH. Phenolphthalein changes inside the steep section; methyl orange would change slowly in the buffer region, long before equivalence.

MARK-SCHEME LANGUAGE

- "phenolphthalein / bromothymol blue / phenol red" "because the indicator pH range / pKIn ± 1 lies (completely) within the vertical section of the graph". Two points: name, then reason.
- End-point colours: methyl orange **orange**; bromophenol blue, bromothymol blue, bromocresol green **green**; phenol red **orange**; phenolphthalein **(pale) pink**.
- Methyl orange in a weak acid titration "starts red and turns orange across the buffer region", so the end point can't be seen sharply.

Often asked as: choose from a Data Booklet list and justify; explain why an indicator is unsuitable; state the colour at the end point; colour changes during a titration or on dilution.

COMMON MISTAKE

Saying the indicator "changes colour at the equivalence point": the mark is for its **range lying within the vertical section**.
Quoting the Data Booklet colours (e.g. "red to yellow") as the **end-point** colour: the end point is the **in-between** colour.
Describing whether an indicator is suitable when the question asked you to **describe the colours** seen.

BEYOND THE SPEC, BUT EXAMINED

- Indicators with two ranges, e.g. thymol blue is listed twice ("acid" 1.2–2.8 and "base" 8.0–9.6); use the right one (WCH14 Jun 2025).
- A diprotic titration needs **one indicator per equivalence point**, and two indicators in one flask is a bad idea because "the colours will mix" (WCH14 Jan 2025).

SPEC 14.17 What a buffer solution is

WHAT THIS SPEC POINT ASKS

Know what a buffer solution is.

14.17 Definition

A **buffer solution resists changes in pH** when **small** amounts of **acid or alkali** are added (or when it is diluted).

- "Resists" does not mean "prevents": the pH changes a little.
- Add a lot of acid or alkali and the buffer is overwhelmed.

14.17 What buffers are made of

Type	Made from	Example	pH
Acidic buffer	a weak acid + a salt of that weak acid (its conjugate base)	CH_3COOH + CH_3COONa	below 7
	or: excess weak acid partly neutralised by a strong base	2 mol CH_3COOH + 1 mol NaOH	
Basic buffer	a weak base + a salt of that weak base (its conjugate acid)	NH_3 + NH_4Cl	above 7
	or: excess weak base partly neutralised by a strong acid	2 mol NH_3 + 1 mol HCl	

- A strong acid and its salt (HCl + NaCl) is **not** a buffer: Cl^- is too weak a base to take up added H^+.
- If the strong acid or base is **in excess** when you mix, there is no weak acid/conjugate base pair left, so no buffer.

MARK-SCHEME LANGUAGE

- "a buffer resists change in pH on the addition of small amounts of acid and / or of alkali"; ACCEPT "maintains a fairly / nearly constant pH".
- Both "small" and "acid and alkali" are needed.

Often asked as: define a buffer (1–2 marks); MCQ "which mixture forms a buffer?".

COMMON MISTAKE

Leaving out "small amounts", or saying the pH "does not change".
Picking a strong acid + its salt, or a mixture with the strong reagent in excess, as a buffer.

SPEC 14.18 How a buffer works

WHAT THIS SPEC POINT ASKS

Explain the action of a buffer solution.

14.18 The acidic buffer CH_3COOH / CH_3COO^-

$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$

- The buffer contains a **large reservoir** of both **CH_3COOH** (from the acid, barely dissociated) and **CH_3COO^-** (from the fully dissociated salt).
- **Add a little acid (H^+)**: the H^+ reacts with the ethanoate ions:
  $CH_3COO^- + H^+ \rightarrow CH_3COOH$
  The equilibrium shifts **left**; almost all the added H^+ is removed.
- **Add a little alkali (OH^-)**: OH^- reacts with H^+ (and with ethanoic acid directly):
  $H^+ + OH^- \rightarrow H_2O$, or $CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$
  The equilibrium shifts **right**: more CH_3COOH dissociates to replace the H^+ used.
- **Why the pH hardly changes**: $[H^+] = Ka \times [HA] \div [A^-]$. Because [HA] and $[A^-]$ are both large, a small addition changes the **ratio [HA]/$[A^-]$** only slightly, so $[H^+]$ stays almost constant.

14.18 The basic buffer NH_3 / NH_4^+

$NH_3(aq) + H^+(aq) \rightleftharpoons NH_4^+(aq)$

- Add H^+: $NH_3 + H^+ \rightarrow NH_4^+$ (large reservoir of NH_3 removes it).
- Add OH^-: $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$ (large reservoir of NH_4^+ removes it).

14.18 Buffer capacity

- A buffer only works while **both** reservoirs last. Once most of the A^- (or HA) has been used up, the ratio changes fast and so does the pH.
- More concentrated buffers have a bigger capacity. A buffer works best when [HA] ≈ $[A^-]$, i.e. pH ≈ pKa (within about ±1).

MARK-SCHEME LANGUAGE

- M1 "(large) reservoir of CH_3COOH and CH_3COO^-"; M2 "OH^- ions react with CH_3COOH / H^+ ions"; M3 an equation, "$CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$"; M4 "$H^+$ ions react with CH_3COO^-" with "$CH_3COO^- + H^+ \rightarrow CH_3COOH$".
- "base and conjugate acid present in high concentrations"; "the ratio [base]/[acid] only changes very slightly so the pH changes only slightly".

Often asked as: explain with equations how a named buffer works (3–5 marks); write the equation for the reaction with added acid or alkali.

COMMON MISTAKE

Writing equations with no words to explain them, or only giving the definition.
Adding H^+ to the wrong member of the pair (H^+ reacts with the **conjugate base**).
Forgetting to mention the **large reservoir** of the weak acid.
Writing a species that isn't in the buffer (e.g. H_2SO_4 in an HSO_4^-/SO_4^{2-} buffer).

BEYOND THE SPEC, BUT EXAMINED

- Explaining buffer action **through the ratio** [salt]/[acid], not just "resists change" (WCH14 Jan 2026).
- Buffers made from less familiar pairs: **HSO_4^- / SO_4^{2-}** (WCH14 Jan 2021), **$H_2PO_4^-$ / HPO_4^{2-}** and the base **Tris** with its conjugate acid (WCH14 Jun 2023).

SPEC
14.19 Calculating the pH of a buffer

WHAT THIS SPEC POINT ASKS

Calculate the pH of a buffer solution from given data.

14.19 The method (Ka route)

1. Write Ka = $[H^+][A^-] \div [HA]$ and rearrange:

$$[H^+] = K_a \times [HA] \div [A^-]$$

2. Assume **[HA] = concentration of the acid added** and **[A^-] = concentration of the salt added** (the acid barely dissociates and the salt is fully dissociated).
3. pH = $-\log[H^+]$.

- Because both are in the same volume, you can use **moles** instead of concentrations: the volume cancels.
- The equivalent log form (Henderson–Hasselbalch) is: **pH = pKa + log([A^-] ÷ [HA])**. Examiners repeatedly see it used with the ratio **upside down**, so the Ka route is safer.
- A buffer is **not** a weak acid on its own: $[H^+] \neq [A^-]$ here.
- **Basic buffers** use Ka of the conjugate acid in exactly the same way. 0.100 mol dm^{-3} NH_3 with 0.200 mol dm^{-3} NH_4Cl, Ka(NH_4^+) = 5.6×10^{-10}: $[H^+] = 5.6 \times 10^{-10} \times 0.200 \div 0.100 = 1.12 \times 10^{-9}$ mol dm^{-3}, pH = 8.95.

WORKED EXAMPLE: ACID + SALT

4.10 g of sodium ethanoate (Mr 82.0) is dissolved in 250 cm^3 of 0.300 mol dm^{-3} ethanoic acid. Ka = 1.7×10^{-5} mol dm^{-3}.

$[CH_3COO^-] = (4.10 \div 82.0) \div 0.250 = 0.200$ mol dm^{-3}; $[CH_3COOH] = 0.300$ mol dm^{-3}

$[H^+] = 1.7 \times 10^{-5} \times 0.300 \div 0.200 = 2.55 \times 10^{-5}$ mol dm^{-3}; pH = **4.59**.

WORKED EXAMPLE: BUFFER MADE BY PARTIAL NEUTRALISATION

50.0 cm^3 of 0.200 mol dm^{-3} CH_3COOH is mixed with 30.0 cm^3 of 0.200 mol dm^{-3} NaOH.

moles CH_3COOH = 0.0100; moles NaOH = 0.00600

$CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$, so after reaction: CH_3COOH left = 0.00400 mol; CH_3COO^- formed = 0.00600 mol

$[H^+] = 1.7 \times 10^{-5} \times 0.00400 \div 0.00600 = 1.13 \times 10^{-5}$ mol dm^{-3}; pH = **4.95**.

WORKED EXAMPLE: ADDING ACID OR ALKALI TO A BUFFER

A buffer contains 0.0100 mol CH_3COOH and 0.0100 mol CH_3COO^- in 100 cm^3. Its pH = pKa = 4.77.

(a) Add 1.00 cm^3 of 1.00 mol dm^{-3} HCl (0.00100 mol H^+). The H^+ turns CH_3COO^- into CH_3COOH:

CH_3COOH = 0.0110 mol; CH_3COO^- = 0.0090 mol

$[H^+] = 1.7 \times 10^{-5} \times 0.0110 \div 0.0090 = 2.08 \times 10^{-5}$; pH = **4.68** (a fall of only 0.09).

(b) Instead add 0.00100 mol NaOH. The OH^- turns CH_3COOH into CH_3COO^-:

CH_3COOH = 0.0090 mol; CH_3COO^- = 0.0110 mol; pH = **4.86**.

(c) Compare: the same HCl added to 100 cm^3 of pure water gives $[H^+] = 0.00100 \div 0.101 = 9.9 \times 10^{-3}$, pH = **2.00**, a fall of 5 units.

MARK-SCHEME LANGUAGE

- "$[H^+] = Ka \times [HA] \div [A^-]$" then pH; the moles route gives the same answer; Henderson–Hasselbalch is allowed if quoted and used correctly. "If the final answer is above pH 7 do not award" (acidic buffer).
- Partial neutralisation: use the moles **remaining** after reaction, not the starting amounts.
- Adding acid: recalculate **both** members of the pair.

Often asked as: pH of a buffer from masses or volumes (3–5 marks); pH after mixing a weak acid with less than its equivalent of NaOH; compare the pH change on adding acid to the buffer and to water; MCQ with distractors that invert the ratio or use ln.

COMMON MISTAKE

Putting $[H^+]$ = [salt] or $[H^+] = [A^-]$ in a buffer.
Inverting the ratio in $pH = pKa + \log([A^-]/[HA])$ or getting the sign wrong.
Using the starting moles of acid when some has been neutralised.
Treating the buffer as a plain weak acid.

BEYOND THE SPEC, BUT EXAMINED

- **Partial neutralisation** buffers (WCH14 Jan 2020): when the weak acid is in twofold excess, half is neutralised and pH = pKa (WCH14 Oct 2025).
- A **phosphate** buffer, $H_2PO_4^-$ / HPO_4^{2-}, with pH about 7.2 (WCH14 Jan 2026).

SPEC 14.20

Preparing a buffer of a given pH

WHAT THIS SPEC POINT ASKS

Calculate the concentrations needed to prepare a buffer with a chosen pH.

14.20 Method

1. Choose a weak acid with **pKa close to the target pH** (within about 1).
2. $\mathbf{[H^+] = 10^{-pH}}$ for the target pH.
3. Rearrange Ka: $\mathbf{[A^-] \div [HA] = Ka \div [H^+]}$.
4. Fix one concentration (or the volume) and calculate the other; convert to moles or mass if asked.

WORKED EXAMPLE: MASS OF SALT FOR A TARGET PH

Make 500 cm^3 of a pH 5.00 buffer using 0.100 mol dm^{-3} ethanoic acid ($Ka = 1.7 \times 10^{-5}$). What mass of sodium ethanoate (Mr 82.0) is needed?
$[H^+] = 1.0 \times 10^{-5}$ mol dm^{-3}
$[A^-] = Ka \times [HA] \div [H^+] = 1.7 \times 10^{-5} \times 0.100 \div 1.0 \times 10^{-5} = 0.170$ mol dm^{-3}
moles in 500 cm^3 = $0.170 \times 0.500 = 0.0850$ mol; mass = $0.0850 \times 82.0 =$ **6.97 g**.

WORKED EXAMPLE: MIXING VOLUMES

You have 0.100 mol dm^{-3} ethanoic acid and 0.100 mol dm^{-3} sodium ethanoate. What volumes give 100 cm^3 of the pH 5.00 buffer?
Needed ratio $[A^-] : [HA] = 1.7 : 1$. The solutions have equal concentrations, so the volumes must be in the same ratio.
Volume of acid = $100 \div 2.7 =$ **37.0 cm^3**; volume of salt solution = **63.0 cm^3**.

MARK-SCHEME LANGUAGE

- [H^+] from pH; rearranged Ka; [salt]; scaled to the volume to give moles; × Mr for mass.
- Ratio questions: give the ratio **in the order asked** (acid : salt or salt : acid).

Often asked as: moles or mass of salt to add for a target pH; acid : salt ratio for a given pH.

COMMON MISTAKE

Leaving out the scaling step (concentration → moles in the actual volume).
Giving salt : acid when acid : salt was asked.
Not noticing that the answer makes no sense (e.g. an "acidic" buffer needing a pH above 7).

SPEC 14.21

Using titration curves: buffer action and Ka

WHAT THIS SPEC POINT ASKS

Use a weak acid–strong base (or strong acid–weak base) titration curve to show buffer action and to find Ka from the pH at half-neutralisation.

14.21 The buffer region

- In a weak acid–strong base titration, after the first few cm^3 the flask contains both unreacted **HA** and the **A^-** made by the reaction. That mixture is a **buffer**.
- So the curve is **flat and gently sloping** over a wide range of volume: adding OH^- changes the pH only slowly. This is the **buffer region**.

14.21 Half-neutralisation: pH = pKa

- When **half** the acid has been neutralised (half the equivalence volume), **[HA] = [A^-]**.
- Ka = [H^+][A^-] ÷ [HA], and the two concentrations cancel, so **Ka = [H^+]** and **pH = pKa**.
- So: find the equivalence volume, halve it, read the pH there, and **Ka = 10^{-pH}**.
- For a weak **base** titrated with strong acid, the pH at half-neutralisation equals the pKa of the **conjugate acid** (e.g. 9.25 for NH_4^+).
- For a diprotic acid, read pH at ¼ and ¾ of the second equivalence volume: these give pKa_1 and pKa_2.

WORKED EXAMPLE: KA FROM THE CURVE

A weak acid is titrated with NaOH. The equivalence point is at 18.0 cm^3. At 9.0 cm^3 the pH is 4.20.
At 9.0 cm^3, half the acid is neutralised, so [HA] = [A^-] and pH = pKa = 4.20.
Ka = $10^{-4.20}$ = **6.3×10^{-5} mol dm^{-3}**. (Show the 9.0 cm^3 line and the pH reading on the graph.)

MARK-SCHEME LANGUAGE

- "(when half the acid has been neutralised) [CH_3CH_2COOH] = [$CH_3CH_2COO^-$]" and "pH at half neutralisation = pKa". ALLOW in words.
- Read-off tolerance: e.g. "pH 3.8 ± 0.1", then Ka = 10^{-pH}; TE from your equivalence volume.
- Buffer region: "this region is a buffer"; "large reservoir of undissociated acid (and its anion)"; added OH^- reacts with H^+ and more acid dissociates.

Often asked as: find Ka (or pKa) from a curve; identify and explain the flat region; Ka_1 and Ka_2 of a diprotic acid.

COMMON MISTAKE

Not realising that the pH at half-neutralisation gives pKa directly.
Reading the pH at the equivalence point or at the end of the curve (e.g. 8 or 13) instead.
Giving Ka to 1 s.f. ("rarely acceptable") or leaving no working on the graph.

BEYOND THE SPEC, BUT EXAMINED

- **Ka_1 and Ka_2 of a diprotic acid** from the two half-equivalence points (WCH14 Jan 2025).
- Doing a full buffer calculation at the half-neutralisation point gives the same answer and scores (WCH14 Oct 2023).

SPEC 14.22

Buffers in living things and food

WHAT THIS SPEC POINT ASKS

Understand why buffers matter in cells and blood (H_2CO_3/HCO_3^-) and in food (preventing spoilage from pH changes caused by bacteria or fungi).

14.22 Blood: the carbonic acid–hydrogencarbonate buffer

- Blood pH must stay within **7.35–7.45**. Enzymes and proteins (e.g. haemoglobin) only work properly in a narrow pH range.
- Two linked equilibria:
  $CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$
  $H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$
- **Extra H^+** (e.g. lactic acid during exercise, or extra CO_2 from respiration): the large reservoir of **HCO_3^-** removes it, $HCO_3^- + H^+ \rightarrow H_2CO_3$; the H_2CO_3 breaks down to CO_2, which is **exhaled**.
- **Loss of H^+** (or OH^- added): H_2CO_3 dissociates to replace it, and more CO_2 dissolves to make more H_2CO_3.
- The ratio $[HCO_3^-] : [H_2CO_3]$ stays nearly constant, so $[H^+]$ does too. The pH can still shift slightly (it falls a little during exercise).
- Inside **cells**, phosphate ($H_2PO_4^-$ / HPO_4^{2-}) and proteins act as buffers too.

WORKED EXAMPLE: THE RATIO IN BLOOD

Ka of H_2CO_3 = 4.3×10^{-7} mol dm^{-3}. What ratio $[HCO_3^-] : [H_2CO_3]$ gives pH 7.40?
$[H^+] = 10^{-7.40} = 3.98 \times 10^{-8}$ mol dm^{-3}
$[HCO_3^-] \div [H_2CO_3] = Ka \div [H^+] = 4.3 \times 10^{-7} \div 3.98 \times 10^{-8}$ = **10.8 : 1**.
There is far more HCO_3^- than H_2CO_3, so blood is better at mopping up added acid than added alkali, which suits the body (metabolism produces acids).

14.22 Food

- Bacteria and fungi growing in food produce acids (or bases), which change the pH and spoil flavour and texture; a changed pH can also help some microbes grow.
- Adding a buffer such as **citric acid + sodium citrate** keeps the pH almost constant, which helps **prevent deterioration**. Buffers are also used in drinks, jams and some medicines.

MARK-SCHEME LANGUAGE

- Blood (5 marks): equations for $CO_2 + H_2O \rightleftharpoons H_2CO_3$ and $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$; "addition of CO_2 (from respiration)"; "removal of CO_2 (by exhalation)"; "there is a large excess of hydrogencarbonate"; "the ratio of salt / acid remains (almost) the same".
- "(large concentration of) HCO_3^- reacts with (extra) H^+ ions".
- Food: "(sodium citrate) forms a buffer (with citric acid) which maintains an (almost) constant pH to prevent deterioration (due to pH change caused by bacteria or fungal activity)".
- Why pH matters: "pH is important for enzyme function".

Often asked as: explain how blood pH is controlled (5–6 marks); effect of exercise on blood pH (MCQ); why sodium citrate is added to a food or drink.

COMMON MISTAKE

Saying a buffer keeps the pH **exactly** constant: during exercise blood pH falls slightly.
Picking the wrong one of the two linked equilibria to apply.
Saying food buffers are there to improve taste, colour or appearance.

BEYOND THE SPEC, BUT EXAMINED

- The **HCO_3^- : H_2CO_3 ratio** needed to hold a given blood pH, calculated from Ka (WCH14 Jun 2021).

SPEC 14.23

Core Practical 11: finding Ka for a weak acid

WHAT THIS SPEC POINT ASKS

Determine Ka for a weak acid (Core Practical 11).

14.23 Principle

- At **half-neutralisation**, [HA] = [A^-], so **pH = pKa** (14.21).
- You make a half-neutralised solution by titrating one portion of the acid to the end point and then adding an **equal** portion of the same acid. You don't need to know the acid's concentration at all.

CORE PRACTICAL 11: METHOD IN BRIEF

1. **Calibrate the pH meter** with buffer solutions of known pH (e.g. pH 4.00 and 7.00), rinsing the electrode with distilled water between solutions.
2. **Pipette 25.0 cm^3** of the weak acid (e.g. ethanoic acid, about 0.1 mol dm^{-3}) into a conical flask and add a few drops of **phenolphthalein**.
3. **Titrate** with about 0.1 mol dm^{-3} **NaOH** from a burette until the solution just turns **pale pink**. The acid is now exactly neutralised (all A^-).
4. Pipette **another 25.0 cm^3** of the same acid into the flask and swirl. Now half the total acid is HA and half is A^-.
5. Measure the **pH** of the mixture with the pH meter. **pKa = pH**, and **Ka = 10^{-pH}**.
6. Repeat to get concordant values and take a mean.

Alternative: record pH every 1 cm^3 (every 0.1 cm^3 near the end point) while adding NaOH, using a pH meter or data logger; plot the titration curve and read the pH at half the equivalence volume.

14.23 Variables and data

Measured	Controlled	Calculated
pH of the half-neutralised mixture	temperature (Ka depends on it); equal volumes of acid in steps 2 and 4	$Ka = 10^{-pH}$, with units mol dm^{-3}

WORKED EXAMPLE: PROCESSING THE RESULTS

pH readings of the half-neutralised mixture: 4.78, 4.75, 4.76. Mean pH = 4.76.
pKa = 4.76; Ka = $10^{-4.76}$ = **1.7×10^{-5} mol dm^{-3}**.
A second method: a 0.100 mol dm^{-3} solution of the same acid had pH 2.88. Ka = $(10^{-2.88})^2 \div 0.100 = 1.74 \times 10^{-5}$ mol dm^{-3}. The two methods agree.

14.23 Safety

- **NaOH** is corrosive/irritant to eyes: wear eye protection. Ethanoic acid at this concentration is an irritant.
- Handle the glass pH electrode carefully; store it in its storage solution, not dry.

14.23 Sources of error and improvements

- **pH meter**: must be calibrated; reads to ±0.01 or ±0.02. A 0.02 error in pH is about a 5% error in Ka. Calibrate with a buffer close to the expected pH.
- **Temperature**: Ka (and the meter reading) changes with temperature; do the experiment at a constant, recorded temperature.
- **End point** judgement: overshooting with NaOH makes the mixture more than half-neutralised, so the pH (and pKa) reads high. Add NaOH dropwise near the end point.
- Rinse the electrode between readings and allow the reading to stabilise.

MARK-SCHEME LANGUAGE

- "(when half the acid has been neutralised) [HA] = [A^-]" so "pH = pKa"; $Ka = 10^{-pH}$.
- Choice of indicator for the first titration: phenolphthalein, because its range lies in the vertical section of a weak acid–strong base curve.

Often asked as: explain why the pH at half-neutralisation equals pKa; read Ka from a recorded curve; select an indicator; evaluate the method (calibration, temperature).

COMMON MISTAKE

Not recognising that the half-neutralisation pH gives pKa, even after doing the practical (examiners noted this).
Using methyl orange for the first titration.
Forgetting units for Ka (mol dm^{-3}) or giving it to 1 s.f.

Topic 15 – Organic Chemistry: Carbonyls, Carboxylic Acids and Chirality

SPEC 15.1 Optical isomerism and the chiral centre

WHAT THIS SPEC POINT ASKS

Know that optical isomerism comes from chirality in molecules with a single chiral centre.

15.1 What makes a molecule chiral

- A **chiral centre** (asymmetric carbon) is a carbon atom bonded to **four different atoms or groups**. Mark it with an asterisk (*).
- A molecule with one chiral centre is **chiral**: it is **not superimposable on its mirror image**, like your left and right hands.
- So it exists as **two optical isomers** (enantiomers, 15.2). Optical isomerism is a type of **stereoisomerism**: same structural formula, different arrangement in space.

15.1 Finding the chiral centre

1. Ignore any carbon with two identical groups: **CH_3, CH_2** and **C=C or C=O carbons** (only three groups) can't be chiral.
2. For each remaining carbon, list the four groups. Look along each branch until the groups differ: CH_2CH_3 and $CH_2CH_2CH_3$ are different.
3. If all four are different, it's a chiral centre.

Molecule	Carbon checked	Its four groups	Chiral?
butan-2-ol, $CH_3CH(OH)CH_2CH_3$	C2	H, OH, CH_3, C_2H_5	**yes**
propan-2-ol, $CH_3CH(OH)CH_3$	C2	H, OH, CH_3, CH_3	no (two CH_3)
2-hydroxypropanoic acid (lactic acid), $CH_3CH(OH)COOH$	C2	H, OH, CH_3, COOH	**yes**
2-hydroxypropanenitrile, $CH_3CH(OH)CN$	C2	H, OH, CH_3, CN	**yes**
pentan-3-ol, $CH_3CH_2CH(OH)CH_2CH_3$	C3	H, OH, C_2H_5, C_2H_5	no (two C_2H_5)

MARK-SCHEME LANGUAGE

- "it is non-superimposable on its mirror image"; ALLOW "there are four different atoms / groups attached to a carbon (atom)", "it has a chiral centre".
- Not chiral: "no carbon atom has 4 different groups" / "the carbon is bonded to two CH_3 / the same groups" / "it is superimposable on its mirror image".

Often asked as: explain why a compound shows optical isomerism (1–2 marks); MCQ "which compound has a chiral centre?"; explain why one product is chiral and another isn't (e.g. the HCN products of pentan-2-one and pentan-3-one).

COMMON MISTAKE

"Four different **molecules**" attached to the carbon: it's four different **atoms or groups**.
Listing several reasons when one is wrong: a wrong statement cancels the right one.
Confusing a chiral centre with a **planar** molecule or a planar carbocation.

BEYOND THE SPEC, BUT EXAMINED

- Counting several chiral centres in a bigger molecule: **menthol has 3**. With n chiral centres there can be up to **2^n** optical isomers, so menthol has 8 (WCH14 Jun 2021). The 2^n rule over-counts if the molecule is symmetrical.
- Chiral carbons "hidden" in rings, such as in the cyclic dimer of lactic acid (lactide) (WCH14 Jan 2023).

SPEC 15.2

Enantiomers and 3D diagrams

WHAT THIS SPEC POINT ASKS

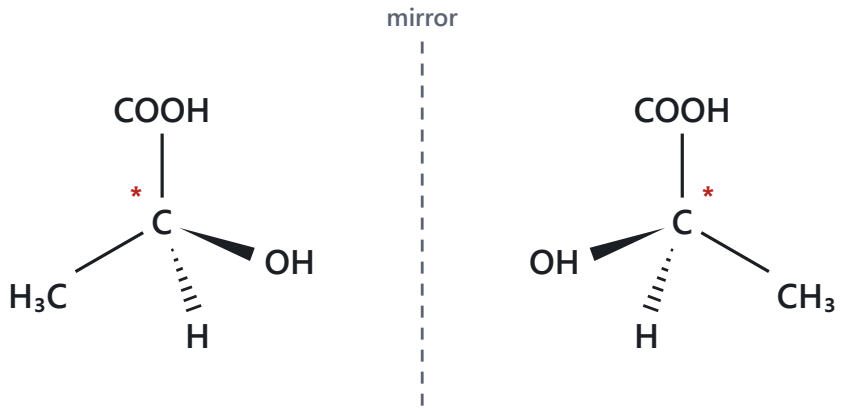
Understand that optical isomers (enantiomers) are object and non-superimposable mirror image, and draw them in 3D.

15.2 Enantiomers

- The two optical isomers are called **enantiomers**. They are **mirror images** of each other that can't be superimposed: however you turn one, it never matches the other.
- They have the **same** boiling point, melting point, density and chemical reactions with non-chiral reagents. They differ in:
  - the **direction** they rotate plane-polarised light (15.3);
  - how they react with **other chiral molecules**, such as enzymes and receptors. That's why one enantiomer of a drug can be useful and the other inactive or harmful, and why two enantiomers can smell different.

15.2 Drawing them

1. Draw the chiral carbon with a **tetrahedral** arrangement: two bonds in the plane of the paper (plain lines), one **wedge** (coming towards you) and one **hashed** bond (going away). Put the wedge and hashed bonds **next to each other**.
2. Draw a dashed **mirror line**.
3. Draw the **mirror image**: reflect every group across the line. Don't just swap two labels at random and don't draw the same isomer twice.



The enantiomers of 2-hydroxypropanoic acid (lactic acid). Each is the mirror image of the other.

MARK-SCHEME LANGUAGE

- "a three-dimensional diagram … showing at least one dotted bond and at least one wedged bond which are next to each other", then "the mirror image of the first structure".
- Groups in any order; CO_2H for COOH is fine.

Often asked as: draw both enantiomers of lactic acid, 2-hydroxybutanenitrile or another product (2 marks).

COMMON MISTAKE

Drawing the **same** enantiomer twice. The safest way is to draw a true reflection across a mirror line.
Flat (2D) structures with no wedge or hashed bonds, or bonds at 90° with no 3D shape.
Wrong atoms joined (e.g. the bond going to the H of OH): draw the bond to the atom actually attached.

SPEC 15.3

Optical activity

WHAT THIS SPEC POINT ASKS

Know that optical activity is the ability of one optical isomer to rotate the plane of polarisation of plane-polarised monochromatic light.

15.3 Plane-polarised light

- Ordinary light vibrates in all planes at right angles to its direction. A **polarising filter** lets through only light vibrating in **one plane**: **plane-polarised light**.
- **Monochromatic** light (one wavelength, e.g. from a sodium lamp) is used because the angle of rotation depends on wavelength.

15.3 Rotating the plane

- A solution of **one** enantiomer **rotates the plane** of plane-polarised light. It is **optically active**.
- The other enantiomer rotates it by the **same angle** in the **opposite direction**: one clockwise (+), one anticlockwise (−).
- A **polarimeter** measures this: light source → polariser → tube of sample solution → second polarising filter (analyser) that you rotate until the light passing through is brightest (or darkest); the angle turned is the rotation.

MARK-SCHEME LANGUAGE

- "one isomer rotates the plane of (monochromatic) plane-polarised light in one direction and the other in the opposite direction"; ALLOW clockwise and anticlockwise.
- "Optically active" means the molecules "rotate the plane of plane-polarised light".

Often asked as: how the two enantiomers differ (1–2 marks); what "optically active" means; IP1 of a 6-mark answer on optical isomerism.

COMMON MISTAKE

Saying the light is "reflected", "deflected" or "bent": it is the **plane of polarisation** that is **rotated**.
Mentioning plane-polarised light but not saying the two isomers rotate it in **opposite directions**.

BEYOND THE SPEC, BUT EXAMINED

- An SN2 product made from one enantiomer is optically active and "rotates plane-polarised light in the opposite direction" (inversion) (WCH14 Oct 2024). See 15.5.

SPEC 15.4 Racemic mixtures

WHAT THIS SPEC POINT ASKS

Know what a racemic mixture is.

15.4 Definition

- A **racemic mixture** (racemate) contains **equal amounts (equal moles) of the two enantiomers**: a 50 : 50 mixture.
- It is **optically inactive**: the rotation caused by one enantiomer is **cancelled** by the equal and opposite rotation of the other.
- A reaction that makes a chiral centre from **non-chiral** starting materials gives a racemic mixture, because there is an equal chance of forming each enantiomer (15.5). Separating them is difficult and expensive, which matters for drugs where only one enantiomer works.

MARK-SCHEME LANGUAGE

- "(mixture containing) equal amounts of (both) enantiomers"; ACCEPT "equimolar / 50 : 50 mixture".
- Why inactive: "the rotations caused by both enantiomers cancel".
- Just stating that equal amounts of each enantiomer form scores the racemic-mixture mark.

Often asked as: define a racemic mixture; explain why a product is not optically active although it has a chiral centre.

COMMON MISTAKE

Saying only that a racemic mixture "has no optical activity" without saying what it **contains**.
Saying "two enantiomers" without "**equal amounts**".

BEYOND THE SPEC, BUT EXAMINED

- **Partial racemisation**: a product with more of one enantiomer than the other is optically active, but less so than the starting material. It shows the reaction went partly by SN1 and partly by SN2 (WCH14 Oct 2025).

SPEC 15.5 Optical activity as evidence for mechanisms

WHAT THIS SPEC POINT ASKS

Use the optical activity of reactants and products as evidence for SN1, SN2 and nucleophilic addition to carbonyl compounds.

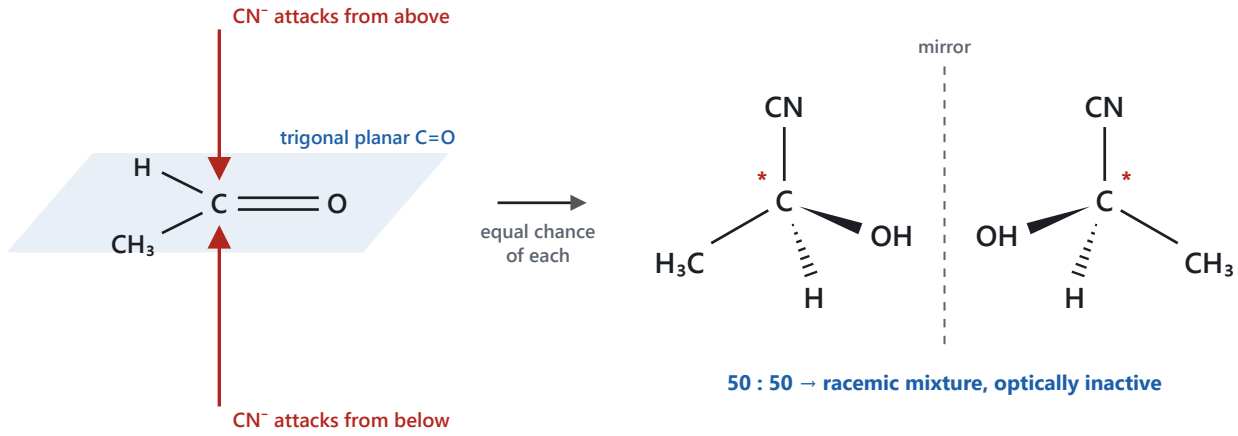
15.5 Start with a single enantiomer, then look at the product

	SN1	SN2	Nucleophilic addition to C=O
Typical substrate	tertiary halogenoalkane	primary (or secondary) halogenoalkane	aldehyde or unsymmetrical ketone + HCN/CN^-
Key feature	C–X breaks first → **planar carbocation** intermediate	nucleophile attacks from the side **opposite** the halogen, in one step	the C=O group is **trigonal planar**
Attack	from **either side** of the planar carbocation, equally	only from the back	CN^- from **above or below** the plane, equally
Product from one enantiomer	**racemic mixture**	**one enantiomer**, with the arrangement **inverted**	(starting material isn't chiral) **racemic mixture**
Optical activity of product	**inactive** (activity lost)	**active** (often rotates the opposite way)	**inactive**

- So if an optically active halogenoalkane gives an **inactive** (racemic) product, the mechanism was **SN1**. If the product is still **optically active**, it was **SN2**. This backs up the rate-equation evidence in 11.9.

15.5 Nucleophilic addition: why HCN gives a racemic product

- The carbonyl carbon and the three atoms attached to it are in **one plane** (trigonal planar).
- The cyanide ion can approach that plane from **above or below** with **equal probability**.
- Attack from one side gives one enantiomer; from the other side, the mirror image. So equal amounts form: a **racemic mixture**, which is **optically inactive** even though each product molecule is chiral.
- This only matters if the product has a chiral centre: **methanal** and **symmetrical ketones** (propanone, pentan-3-one) give products with two identical groups on that carbon, so no optical isomers.



CN^- attacks the planar C=O of ethanal equally from above and below, giving equal amounts of the two enantiomers of 2-hydroxypropanenitrile.

MARK-SCHEME LANGUAGE

- Carbonyl: "propanal is planar around the CHO / reaction site / C=O / carbonyl"; "CN^- / nucleophile can attack on either side / both sides / above and below (giving a racemic / 50 : 50 mixture)". DO NOT AWARD "propanal is planar" alone or "planar intermediate / carbocation".
- "cyanide attacks from above and below; (as) trigonal planar around the C(δ+)". IGNORE "from both directions".
- SN2 from one enantiomer: "product is optically active; stereochemistry is inverted / product rotates plane-polarised light in the opposite direction".

Often asked as: explain why the product of HCN + an aldehyde is not optically active (2–3 marks); why one ketone gives optical isomers and another doesn't; what optical activity shows about an SN1 or SN2 reaction (6-mark or MCQ).

COMMON MISTAKE

Writing "SN1" or "SN2" in an answer about **nucleophilic addition**: it automatically loses the marks.
Saying "a planar **carbocation**" forms in the HCN reaction. There is no carbocation; the **C=O group** is planar.
Saying "the molecule is planar": only the arrangement around the carbonyl carbon is.
Giving the stock SN1 answer (racemic) for an SN2 reaction.

BEYOND THE SPEC, BUT EXAMINED

- Which product shows optical isomerism: the HCN product of **pentan-2-one** has a chiral carbon; the product of **pentan-3-one** (two ethyl groups) doesn't. Both ketones still react (WCH14 Jan 2025).

SPEC 15.6

Naming and drawing aldehydes and ketones

WHAT THIS SPEC POINT ASKS

Name aldehydes and ketones and draw their structural, displayed and skeletal formulae.

15.6 The functional groups

- Both contain the **carbonyl group, C=O**.
- **Aldehyde**: C=O at the **end** of the chain, written **–CHO**. Name ends **-al**. The CHO carbon is always C1, so no number is needed: methanal HCHO, ethanal CH_3CHO, propanal CH_3CH_2CHO.
- **Ketone**: C=O **inside** the chain, joined to two carbons, written **–CO–**. Name ends **-one**, with a number when needed: propanone CH_3COCH_3, butanone $CH_3COCH_2CH_3$, pentan-2-one $CH_3COCH_2CH_2CH_3$, pentan-3-one $CH_3CH_2COCH_2CH_3$.
- Aldehydes and ketones with the same number of carbons are **functional group isomers**: propanal and propanone are both C_3H_6O.

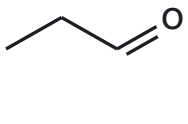
15.6 Names with substituents

- Number the chain so the C=O gets the **lowest** number, then add substituents: 2-methylpropanal $(CH_3)_2CHCHO$; 3-methylbutan-2-one $CH_3COCH(CH_3)_2$; 1-hydroxy-4-methylpentan-2-one $HOCH_2COCH_2CH(CH_3)_2$.

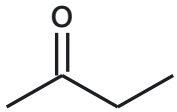
15.6 Formulae

Type	Propanal	Butanone
Molecular	C_3H_6O	C_4H_8O
Structural	CH_3CH_2CHO	$CH_3COCH_2CH_3$
Displayed	every atom and every bond drawn, including C=O and each C–H	same
Skeletal	zig-zag line ending in =O	zig-zag with =O on the second carbon

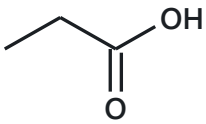
- In **skeletal** formulae every line end and corner is a carbon; H atoms on carbon are left out; O, Cl, N and the H of OH are shown.



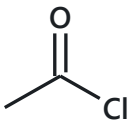
propanal
CH_3CH_2CHO



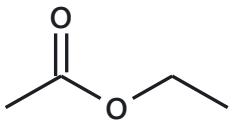
butanone
$CH_3COCH_2CH_3$



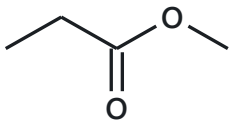
propanoic acid
CH_3CH_2COOH



ethanoyl chloride
CH_3COCl



ethyl ethanoate
$CH_3COOCH_2CH_3$



methyl propanoate
$CH_3CH_2COOCH_3$

Skeletal formulae for Topic 15. Ethyl ethanoate and methyl propanoate are isomers (both $C_4H_8O_2$): see 15.13 and 15.20.

MARK-SCHEME LANGUAGE

- Names must have the right stem and locant. Hyphens and spaces are usually not penalised; "hydroxyl" is allowed for "hydroxy".
- Displayed formulae: "penalise missing H's only once"; structural or skeletal formulae score at most 2 when displayed was asked.
- "If more than one structure type is given, both must be correct."

Often asked as: name a carbonyl compound or its product; draw displayed formulae of products from a reaction sequence; draw skeletal formulae of isomers.

COMMON MISTAKE

Writing propanal as CH_3CH_2COH, which looks like an alcohol: write **–CHO**.
Ignoring the instruction "displayed" or "skeletal"; leaving out H atoms; five bonds on a carbon.
Not writing the H of an OH group on a skeletal formula.

BEYOND THE SPEC, BUT EXAMINED

- **Hydroxynitrile names**: the nitrile carbon counts in the chain and the nitrile is the main group, so ethanal + HCN gives **2-hydroxypropanenitrile** and propanal + HCN gives **2-hydroxybutanenitrile** (WCH14 Jun 2022). Don't call them "cyanides".

SPEC 15.7

Physical properties of aldehydes and ketones

WHAT THIS SPEC POINT ASKS

Understand that aldehydes and ketones don't hydrogen bond with each other (which affects their physical properties) but can hydrogen bond with water (which affects their solubility).

15.7 Forces between carbonyl molecules

- The **C=O bond is polar** (O is more electronegative): $C^{\delta+}=O^{\delta-}$. So carbonyl molecules have **permanent dipole–dipole forces** as well as **London forces**.
- They have **no H on an O (or N or F)**, so they **cannot hydrogen bond to each other**.
- Result: boiling temperatures are **higher than alkanes** of similar Mr (extra dipole–dipole forces) but **lower than alcohols** (no hydrogen bonds).

Compound (Mr ≈ 58–60)	Strongest force between molecules	Boiling temperature / °C
butane, C_4H_{10}	London	−0.5
propanal, CH_3CH_2CHO	permanent dipole–dipole	49
propanone, CH_3COCH_3	permanent dipole–dipole	56
propan-1-ol, $CH_3CH_2CH_2OH$	hydrogen bonds	97
ethanoic acid, CH_3COOH	hydrogen bonds	118

15.7 Solubility in water

- The **lone pairs on the carbonyl O** can accept a hydrogen bond from the **δ+ H of a water molecule**. So small aldehydes and ketones dissolve well (propanone mixes with water in all proportions).
- As the carbon chain gets longer, solubility **falls**: the non-polar chain can't hydrogen bond with water, and London forces between the bigger molecules get stronger.

MARK-SCHEME LANGUAGE

- "pentan-2-one does not form hydrogen bonds (between molecules)"; "hydrogen bonds require more energy to break (so pentan-2-one has a lower boiling temperature)". ALLOW "(only) permanent dipole–dipole and London forces". IGNORE "pentan-2-ol has stronger London forces". DO NOT AWARD "more hydrogen bonds".
- "propanal forms permanent dipole–permanent dipole forces" (not hydrogen bonds).

Often asked as: compare the boiling temperatures of a carbonyl and an alcohol of similar Mr (2 marks); 6-mark comparing an aldehyde, an alcohol and a carboxylic acid; MCQ on forces.

COMMON MISTAKE

Saying an aldehyde or ketone hydrogen bonds **with itself**.
Naming the forces in each compound without **comparing their strengths** and linking to the **energy needed**.
Talking about hydrogen bonding with **water** when the question is about boiling temperature.
Saying covalent bonds break on boiling.

BEYOND THE SPEC, BUT EXAMINED

- A hydrogen bond needs an H on N/O/F **on one side only**: an ester or ketone O can accept one from water even though the molecule has no O–H (WCH14 Jan 2022).
- London forces are similar when molecules have a similar number of electrons, so you can compare the other forces directly (WCH14 Jan 2021).

SPEC 15.8

Reactions of carbonyl compounds

WHAT THIS SPEC POINT ASKS

Know the reactions of aldehydes and ketones with: (i) Fehling's/Benedict's, Tollens' and acidified dichromate(VI); (ii) $LiAlH_4$ in dry ether; (iii) HCN with KCN (mechanism, including optical activity); (iv) 2,4-DNPH, including identification from derivative melting temperatures; (v) iodine in alkali (iodoform test).

15.8 (i) Telling aldehydes from ketones: oxidation

- **Aldehydes** have an H on the carbonyl carbon, so they are **easily oxidised** to carboxylic acids: RCHO + [O] → RCOOH.
- **Ketones** have no such H and are **not oxidised** by these mild reagents (it would need a C–C bond to break).
- The aldehyde **reduces** the reagent; the colour change shows it.

Reagent and conditions	Aldehyde	Ketone	What is reduced
Fehling's or **Benedict's** solution, warm	blue solution → **brick-red precipitate**	stays blue	Cu^{2+} → Cu_2O (copper(I) oxide)
Tollens' reagent (ammoniacal silver nitrate), warm in a water bath	**silver mirror** (or black/grey precipitate)	no change	Ag^+ → Ag: $[Ag(NH_3)_2]^+ + e^- \rightarrow Ag + 2NH_3$
Acidified potassium dichromate(VI) ($K_2Cr_2O_7$ + dilute H_2SO_4), warm	**orange → green**	stays orange	$Cr_2O_7^{2-} \rightarrow Cr^{3+}$

- Equation with [O]: $CH_3CH_2CHO + [O] \rightarrow CH_3CH_2COOH$. (Fehling's and Tollens' are alkaline, so the carboxylate ion actually forms.)
- Acidified dichromate also oxidises primary and secondary **alcohols**, so it is not a test for aldehydes if alcohols might be present. Fehling's and Tollens' don't react with alcohols.

MARK-SCHEME LANGUAGE

- Test + positive result for the aldehyde + "no change" for the ketone; M2 depends on M1. ALLOW "ammoniacal silver nitrate", "$Cr_2O_7^{2-}$ and H^+". DO NOT AWARD hydrochloric acid.
- Observations: "brick-red / red / brownish red precipitate (from blue solution)"; "silver mirror" (ALLOW black/grey precipitate); "goes green / blue (from orange)".
- "No change", "no result" and "no observation" are equivalent.

Often asked as: one test to distinguish an aldehyde from a ketone (3 marks); part of a multi-test identification (6-mark); equation using [O].

COMMON MISTAKE

Mixing up Tollens' (silver mirror) and Fehling's/Benedict's (red precipitate).
Leaving out "**acidified**" before dichromate.
Adding water as a product in RCHO + [O] → RCOOH.
Writing just "positive" or "it reacts": give the **observation**.

15.8 (ii) Reduction with $LiAlH_4$

- **Reagent**: lithium tetrahydridoaluminate(III), **$LiAlH_4$**, in **dry ether** (ethoxyethane), at room temperature. Then add water or dilute acid to finish. Water must be absent during the reduction because $LiAlH_4$ reacts violently with it.
- Use **[H]** in equations:
  - **aldehyde → primary alcohol**: $CH_3CH_2CHO + 2[H] \rightarrow CH_3CH_2CH_2OH$ (propan-1-ol)
  - **ketone → secondary alcohol**: $CH_3COCH_3 + 2[H] \rightarrow CH_3CH(OH)CH_3$ (propan-2-ol)
- $LiAlH_4$ supplies **hydride ions, H^-**, which act as nucleophiles: H^- attacks the δ+ carbonyl carbon and the O^- formed is then protonated. It attacks C=O but not C=C, because C=C is **non-polar** (no δ+ carbon to attract H^-).

MARK-SCHEME LANGUAGE

- "lithium tetrahydridoaluminate / lithium aluminium hydride / $LiAlH_4$"; "dry and ether / ethoxyethane". ALLOW "anhydrous ether". DO NOT AWARD heat or reflux. Name the product with its number (propan-1-ol, not "propanol").
- Why C=O and not C=C: "the C=O bond is polarised by the (electronegative) oxygen"; "H^- is attracted to the δ+ carbon"; "the C=C bond is non-polar".

Often asked as: reagent and conditions (2 marks); name the products from an aldehyde and a ketone; balanced equation with [H]; MCQ picking the reducing agent.

COMMON MISTAKE

"Dry ester" for dry ether, or adding heat/reflux.
Calling the reaction "nucleophilic addition" when asked for the type: the spec classes it as **reduction**.
Unbalanced equations: two [H] per C=O reduced.

BEYOND THE SPEC, BUT EXAMINED

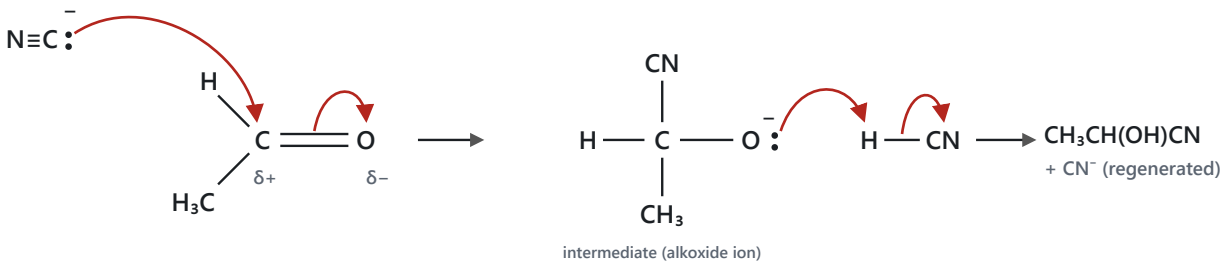
- The **mechanism** of hydride addition: dipole on C=O, lone pair on H^-, arrow from H^- to the carbonyl C, arrow from the C=O bond to O; then the O^- is protonated by acid (WCH14 Jan 2023).

15.8 (iii) Nucleophilic addition of HCN

- **Reagents**: **HCN** in the presence of **KCN** (or KCN with a little dilute acid), at room temperature. Both are **highly toxic**: use a fume cupboard.
- **Why KCN?** HCN is a very weak acid, so it hardly dissociates and gives very little CN^-. KCN supplies plenty of **CN^-**, which is the real **nucleophile**. CN^- is regenerated in step 2, so it acts as a **homogeneous catalyst**.
- **Product**: a **2-hydroxynitrile** (hydroxynitrile). The carbon chain **grows by one carbon**, which is useful in synthesis.
  - ethanal + HCN → **2-hydroxypropanenitrile**, $CH_3CH(OH)CN$
  - propanone + HCN → **2-hydroxy-2-methylpropanenitrile**, $(CH_3)_2C(OH)CN$

15.8 The mechanism, step by step

1. The C=O bond is **polar**: show **δ+ on C** and **δ− on O**.
2. **Step 1**: a **lone pair on the carbon of CN^-** attacks the δ+ carbonyl carbon (curly arrow from the lone pair to the C). At the same time the **π bond of C=O breaks**, with both electrons moving to the O (curly arrow from the C=O bond to the O).
3. This gives a negatively charged **intermediate** (an alkoxide ion, with O^-).
4. **Step 2**: a **lone pair on O^-** attacks the **H of an HCN molecule** (or H^+ / H_2O); the H–C bond in HCN breaks, re-forming CN^-.
5. Product: the 2-hydroxynitrile.



Step 1: CN^- attacks the δ+ carbon; the π bond breaks onto O

Step 2: O^- takes H^+ from HCN (or H_2O / H^+)

Nucleophilic addition of HCN to ethanal. Four curly arrows: CN^- lone pair → C; C=O bond → O; O^- lone pair → H of HCN; H–CN bond → C of CN.

- Why "nucleophilic addition": the attacking species is a **nucleophile** (CN^-, an electron-pair donor attracted to the δ+ carbon), and two species **add** across the C=O to give **one** product, with nothing removed.
- With an aldehyde (other than methanal) or an unsymmetrical ketone the product is a **racemic mixture**: see 15.5.

MARK-SCHEME LANGUAGE

- Mechanism marks: dipole on C=O; arrow from the C=O bond to O; **lone pair on the C of CN^-** and the arrow from it to the carbonyl carbon; intermediate with its negative charge; lone pair on O^- and arrow to the H of HCN (or H^+); product.
- Role of KCN: "catalyst"; "homogeneous" = same phase as the reactants.
- Role of NaOH (if used instead of KCN): "reacts with HCN to produce CN^- / moves the dissociation equilibrium of HCN to the right / increases $[CN^-]$". IGNORE "speeds up the reaction".
- "Justify nucleophilic addition": "the (negative) cyanide ion attacks a δ+ centre / region of low electron density"; "nothing is lost / only one product".

Often asked as: the full mechanism (4 marks, very common); reagents for adding one carbon; role of KCN or NaOH; the rate equation; why the product is not optically active.

COMMON MISTAKE

Arrows that start in empty space or end on a bond instead of an atom.
No lone pair on CN^-, or the lone pair drawn on **N** with the bond made from C.
Missing negative charges on CN^- and on the O of the intermediate; a **positive** charge on the carbonyl C (it's only δ+).
Using the wrong carbonyl compound (read the stem carefully).
Explaining "nucleophilic addition" by just repeating the words "nucleophile" and "add".

BEYOND THE SPEC, BUT EXAMINED

- **Rate equation**: rate = $k[CH_3CHO][CN^-]$. HCN is **not** in it, because step 1 (CN^- attack) is the slow step (WCH14 Oct 2024).
- A **diketone** adds HCN at **both** C=O groups: $CH_3C(OH)(CN)CH_2CH_2C(OH)(CN)CH_3$ from hexane-2,5-dione (WCH14 Jun 2023).

15.8 (iv) 2,4-Dinitrophenylhydrazine (2,4-DNPH, Brady's reagent)

- **Test for a carbonyl group (C=O in aldehydes and ketones)**: add 2,4-DNPH solution. An **orange (or yellow/red) precipitate** forms. Both aldehydes and ketones give it.
- Carboxylic acids and esters also contain C=O but **don't** react.
- The reaction is a **condensation** (addition, then loss of water). You don't need the equation.

15.8 Identifying the carbonyl compound from the derivative

1. **Filter** off the orange precipitate (Buchner funnel, reduced pressure).
2. **Recrystallise** it to purify:
   - dissolve in the **minimum volume of hot solvent** (so the solution is saturated and as much as possible crystallises on cooling);
   - **cool** (then in ice) so crystals of the derivative form while soluble impurities stay in solution;
   - **filter** and **wash with a little ice-cold solvent** (removes the impure solution without dissolving many crystals);
   - **dry** (between filter papers, in a low oven or a desiccator).
3. Measure the **melting temperature** and compare it with a **data table** of known 2,4-DNPH derivatives. A pure compound melts **sharply** at a value matching the table.

WORKED EXAMPLE: IDENTIFYING A CARBONYL FROM ITS DERIVATIVE

An unknown C_4H_8O gives an orange precipitate with 2,4-DNPH and no change with Tollens' reagent. After recrystallisation the derivative melts at 115–116 °C.
Melting temperatures of 2,4-DNPH derivatives (example data): butanal 123 °C; butanone 116 °C; 2-methylpropanal 182 °C.
The negative Tollens' test shows a **ketone**, and the melting temperature matches **butanone**.

MARK-SCHEME LANGUAGE

- "2,4-dinitrophenylhydrazine / 2,4-DNPH" (ALLOW Brady's reagent); "orange precipitate" (ALLOW yellow / red; DO NOT AWARD reddish-brown).
- "the solids need to be recrystallised"; "their melting temperatures determined and compared with data / reference books". DO NOT AWARD **boiling** temperature. IGNORE the colour of the precipitate as a way of identifying.
- Full outline: filter; dissolve in minimum hot solvent and cool; rinse with ice-cold solvent; dry; melting temperature compared with data.

Often asked as: test for a carbonyl; how to identify which carbonyl it is (2–3 marks, or a 6-mark on recrystallisation); link a positive 2,4-DNPH and a negative Tollens' to a ketone.

COMMON MISTAKE

Boiling temperature instead of **melting** temperature: very common.
"Purify it" without saying **recrystallise**; "fractional distillation" for a solid.
Using the colour of the precipitate to tell carbonyls apart.
Thinking 2,4-DNPH reacts only with aldehydes.
Suggesting NMR or IR when the question asks for a school-lab chemical method.

BEYOND THE SPEC, BUT EXAMINED

- The **purification and identification routine**: filter, recrystallise, measure the melting temperature, never the boiling temperature, then compare with data. This has been asked on four papers (WCH14 Oct 2024).

15.8 (v) The iodoform (triiodomethane) test

- **Reagents**: **iodine** solution and **sodium hydroxide** solution (alkaline iodine), warm gently.
- **Positive result**: a **pale yellow precipitate** of **triiodomethane, CHI_3** (iodoform), which has an **antiseptic smell**.
- **Positive for**: compounds with a **$CH_3CO–$** group (**methyl ketones** and **ethanal**, the only aldehyde).
- Also positive for alcohols with a **$CH_3CH(OH)–$** group (ethanol and secondary alcohols such as propan-2-ol and butan-2-ol), because the alkaline iodine first oxidises them to $CH_3CO–$ compounds.
- **Negative for**: methanal and other aldehydes; ketones without a methyl next to the C=O (e.g. pentan-3-one); methanol; primary alcohols other than ethanol; tertiary alcohols.

Equation (propanone): $CH_3COCH_3 + 3I_2 + 4NaOH \rightarrow CH_3COONa + CHI_3 + 3NaI + 3H_2O$

- The CH_3 group becomes CHI_3; the rest of the molecule becomes the **sodium salt of a carboxylic acid with one fewer carbon**.

Compound	Iodoform test
ethanal, CH_3CHO	positive
propanal, CH_3CH_2CHO	negative
propanone, CH_3COCH_3	positive
pentan-2-one, $CH_3COCH_2CH_2CH_3$	positive
pentan-3-one, $CH_3CH_2COCH_2CH_3$	negative
ethanol, CH_3CH_2OH	positive
propan-2-ol, $CH_3CH(OH)CH_3$	positive
propan-1-ol, $CH_3CH_2CH_2OH$	negative

MARK-SCHEME LANGUAGE

- "iodine / I_2 and alkali / NaOH / KOH"; "(pale) yellow precipitate / crystals" for the methyl ketone and no change for the other; ALLOW "iodoform / CHI_3 / triiodomethane" and "antiseptic smell". DO NOT AWARD "yellow-orange".
- Equation marks: CHI_3; CH_3COONa; the rest balanced with state symbols.
- "S contains a methyl secondary alcohol group … as it reacts with iodine in sodium hydroxide".

Often asked as: distinguish pentan-2-one from pentan-3-one (2 marks); observation; balanced equation; which alcohols give a positive result (MCQ).

COMMON MISTAKE

"Yellow solution" or "turns yellow": iodine is already yellow-brown. You need a **precipitate**.
Giving **CHI_3** as the group that is present: the group is **$CH_3CO–$** (or $CH_3CH(OH)–$).
Two observations when one is wrong (e.g. "yellow precipitate, blue-black colour"): the wrong one cancels the mark.

BEYOND THE SPEC, BUT EXAMINED

- The test is positive for **$CH_3CH(OH)–$** alcohols too, which lets you tell propan-2-ol from propan-1-ol (WCH14 Oct 2024).
- **Methanal** is an aldehyde, so it gives a positive Fehling's/Benedict's test, but it has no CH_3CO group, so no iodoform. Ethanal is the only aldehyde that gives both (WCH14 Oct 2022).

SPEC 15.9

Naming and drawing carboxylic acids

WHAT THIS SPEC POINT ASKS

Name carboxylic acids and draw their structural, displayed and skeletal formulae.

15.9 Naming

- The **carboxyl group, –COOH**, is always at the **end** of a chain. Its carbon is **C1** and is counted in the stem.
- Name ends **-oic acid**: methanoic acid HCOOH, ethanoic acid CH_3COOH, propanoic acid CH_3CH_2COOH, butanoic acid $CH_3CH_2CH_2COOH$. General formula **$C_nH_{2n+1}COOH$**.
- Substituents are numbered from the COOH carbon: 2-hydroxypropanoic acid (lactic acid) $CH_3CH(OH)COOH$; 2-methylpropanoic acid $(CH_3)_2CHCOOH$; 3-hydroxy-2-methylbutanoic acid

$CH_3CH(OH)CH(CH_3)COOH$.

- Two COOH groups: **-dioic acid**, e.g. ethanedioic acid HOOC–COOH; benzene-1,4-dicarboxylic acid HOOC–C_6H_4–COOH.
- Unsaturated acids: the COOH carbon is still C1, so CH_2=$CHCH_2COOH$ is **but-3-enoic acid**.

15.9 Formulae

- Structural: CH_3CH_2COOH (or $CH_3CH_2CO_2H$).
- Displayed: show C=O and the O–H bond separately.
- Skeletal: see the figure in 15.6; the =O and OH are drawn on the last carbon.

MARK-SCHEME LANGUAGE

- "but-3-enoic acid"; "2-methylpropenoic acid"; IGNORE missing hyphens, spaces, commas, capitals; ACCEPT "3-butenoic acid".
- "2-hydroxypropanoic acid"; ALLOW "propaneoic"; DO NOT AWARD "hydroxo".

Often asked as: name an acid from its structure; draw or name isomers of a C_4 acid.

COMMON MISTAKE

Numbering the alkene as position 1: the **COOH carbon** must get the lowest number.
Forgetting that the COOH carbon is part of the stem (CH_3CH_2COOH is propanoic, not ethanoic, acid).
Drawing carbons with five bonds when listing isomers.

BEYOND THE SPEC, BUT EXAMINED

- Isomers of unsaturated acids, including **E/Z** isomers and a ring isomer such as cyclopropanecarboxylic acid (WCH14 Jun 2024).

SPEC 15.10 Hydrogen bonding in carboxylic acids

WHAT THIS SPEC POINT ASKS

Understand how hydrogen bonding affects the boiling temperatures and solubility of carboxylic acids.

15.10 Boiling temperatures

- The –COOH group has an **O–H** bond (δ+ H) and two O atoms with lone pairs, so carboxylic acid molecules form **hydrogen bonds** with each other, as well as London and dipole–dipole forces.
- So their boiling temperatures are **high**: ethanoic acid (Mr 60) boils at 118 °C, well above propanal (49 °C) and propan-1-ol (97 °C) of similar Mr.
- More energy is needed to overcome the hydrogen bonds between the molecules.

15.10 Solubility

- The –COOH group hydrogen bonds with **water** (both through its O–H and through the lone pairs on its O atoms), so the small acids (C_1–C_4) **mix completely** with water.
- As the **non-polar hydrocarbon chain** gets longer, solubility **falls**: the chain can't hydrogen bond with water, and London forces between the large molecules are stronger.

15.10 Drawing a hydrogen bond to water

- Show the **δ+ H** of one molecule lined up with a **lone pair on an O** of the other, the H···O bond **straight (about 180°)**, labelled "hydrogen bond", with **δ+ and δ−** on the atoms involved and on the atoms bonded to them.

MARK-SCHEME LANGUAGE

- Hydrogen bond diagram (3 marks for five points): water drawn correctly and aligned at "160–200°"; "hydrogen bond" labelled; "lone pair on the relevant O lined up with the hydrogen bond"; δ+ and δ− on the two atoms in the hydrogen bond; and on the atoms bonded to them.
- Solubility: "can form hydrogen bonds (with water)"; "the larger the (non-polar hydrocarbon) chain the lower the solubility" because "London forces between larger acid molecules are stronger (more electrons)".
- Boiling: the acid "forms hydrogen bonds but the others do not / hydrogen bonds are the strongest intermolecular force".

Often asked as: draw a hydrogen bond between an acid and water; explain the trend in solubility; rank the boiling temperatures of an acid, an alcohol and an aldehyde.

COMMON MISTAKE

Hydrogen bond drawn to the H of water from the H of the acid, or at an angle; no lone pair shown.
Writing water as HO_2.
Explaining boiling temperature with hydrogen bonding to water.

EXAM TIP

You may meet the idea that carboxylic acids form **dimers** (pairs held by two hydrogen bonds), which helps explain why their boiling temperatures are even higher than alcohols of similar Mr. It hasn't been examined: the marks are for "hydrogen bonds between molecules need more energy to break".

SPEC 15.11 Preparing carboxylic acids

WHAT THIS SPEC POINT ASKS

Know that carboxylic acids can be made by oxidising primary alcohols or aldehydes, and by hydrolysing nitriles.

15.11 Oxidation of primary alcohols and aldehydes

- **Reagents and conditions**: **potassium dichromate(VI)** and **dilute sulfuric acid** (acidified $K_2Cr_2O_7$), in **excess**, **heat under reflux**. Colour change **orange** → **green**.
- Reflux lets the reaction run hot for a long time without losing volatile material, so the aldehyde formed first is oxidised all the way to the acid. (If you **distil** the mixture as it forms instead, you collect the **aldehyde**.)
- Equations:
  - $CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$ (primary alcohol → acid)
  - $CH_3CHO + [O] \rightarrow CH_3COOH$ (aldehyde → acid)
- Secondary alcohols give **ketones** (no further oxidation); tertiary alcohols are **not oxidised**.

15.11 Hydrolysis of nitriles

- **Acid hydrolysis**: heat under reflux with **dilute hydrochloric (or sulfuric) acid**:
  $CH_3CH_2CN + 2H_2O + HCl \rightarrow CH_3CH_2COOH + NH_4Cl$
  (ionic version: $RCN + 2H_2O + H^+ \rightarrow RCOOH + NH_4^+$)
- **Alkaline hydrolysis**: reflux with NaOH(aq) gives the **salt** and ammonia: $CH_3CH_2CN + H_2O + NaOH \rightarrow CH_3CH_2COONa + NH_3$. Then add a strong acid to release the free carboxylic acid.
- Combined with 15.8(iii), this gives a two-step route from a carbonyl to a **2-hydroxy acid** with one extra carbon:
  $CH_3CHO \rightarrow$ (HCN, KCN) $\rightarrow CH_3CH(OH)CN \rightarrow$ (dilute HCl, reflux) $\rightarrow CH_3CH(OH)COOH$ (lactic acid)

MARK-SCHEME LANGUAGE

- "potassium dichromate((VI)) / $K_2Cr_2O_7$ and sulfuric acid / H_2SO_4; (heat under) reflux". ALLOW $Cr_2O_7^{2-}$ and H^+, sodium dichromate. DO NOT AWARD hydrochloric acid or another reagent.
- Nitrile: "hydrolysis; (dilute) hydrochloric acid / HCl(aq); heat (under reflux)"; equation $(CH_3)_2C(OH)CH_2CN + 2H_2O + HCl \rightarrow (CH_3)_2C(OH)CH_2COOH + NH_4Cl$. Leaving out water or HCl, or giving HCN as a product, loses the mark.

Often asked as: reagents and conditions for oxidation; equation for nitrile hydrolysis; a two-step synthesis via a hydroxynitrile.

COMMON MISTAKE

Calling the sulfuric acid a **catalyst** (it's a reagent here); using HCl or $KMnO_4$ with dichromate.
Distillation instead of reflux when the acid is wanted.
Nitrile hydrolysis products such as HCN, or no NH_4Cl.
For the second step of a synthesis, giving an oxidising agent instead of dilute acid.

BEYOND THE SPEC, BUT EXAMINED

- **Two-step synthesis**: carbonyl → hydroxynitrile → hydroxy acid, with reagents for each step (WCH14 Jan 2024).

SPEC 15.12 Reactions of carboxylic acids

WHAT THIS SPEC POINT ASKS

Know the reactions of carboxylic acids with $LiAlH_4$ in dry ether, with bases to make salts, with PCl_5, and with alcohols (acid catalyst).

15.12 (i) Reduction with $LiAlH_4$

- **$LiAlH_4$ in dry ether**, room temperature, then water/dilute acid. The acid is reduced to a **primary alcohol**:
  $CH_3COOH + 4[H] \rightarrow CH_3CH_2OH + H_2O$
- Four [H] are needed (C=O and C–OH both changed), and water is formed.

15.12 (ii) With bases: salts

- Carboxylic acids are **weak acids**, but strong enough to react with:
  - **alkalis**: $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$
  - **carbonates and hydrogencarbonates**, with **effervescence** of CO_2 (turns limewater milky):
    $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$

$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + CO_2 + H_2O$

- **reactive metals**: $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$

- Fizzing with Na_2CO_3 or $NaHCO_3$ is a **test for –COOH**: alcohols and phenols don't do it.
- The salts are **ionic**: write $CH_3COO^-Na^+$ or CH_3COONa, never a covalent O–Na bond. Name: sodium ethanoate.

15.12 (iii) With phosphorus(V) chloride, PCl_5

- Add solid **PCl_5** to the dry acid at room temperature: **steamy (misty) fumes of HCl**. The product is an **acyl chloride**:
  $CH_3COOH + PCl_5 \rightarrow CH_3COCl + POCl_3 + HCl$
- PCl_5 reacts with **any –OH** group, so alcohols give the same steamy fumes ($ROH + PCl_5 \rightarrow RCl + POCl_3 + HCl$). It is a test for OH, not specifically for COOH.

15.12 (iv) With alcohols: esterification

- Heat the carboxylic acid and alcohol under **reflux** with a few drops of **concentrated sulfuric acid** as a **catalyst**:
  $CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOCH_2CH_3 + H_2O$ (ethyl ethanoate)
- It is a **condensation** (water lost) and is **reversible**, so the yield is limited. To push the equilibrium right: use an **excess** of one reactant, or remove the ester or water as it forms. Conc. H_2SO_4 also absorbs some water.
- Esters have **sweet, fruity smells**: used in flavourings, perfumes and as solvents.

MAKING AND PURIFYING AN ESTER (SUGGESTED PRACTICAL)

1. Mix ethanol and ethanoic acid in a round-bottomed flask, add a few drops of concentrated H_2SO_4 and anti-bumping granules.
2. **Heat under reflux** for about 20 minutes.
3. **Distil** off the crude ester (with some ethanol, acid and water).
4. Shake the distillate with **sodium carbonate solution** in a **separating funnel** to remove acids (release the pressure: CO_2 forms). Run off and discard the lower aqueous layer.
5. Shake with **calcium chloride solution** to remove ethanol; discard the aqueous layer.
6. **Dry** the ester layer with **anhydrous calcium chloride** or **anhydrous magnesium sulfate** until clear.
7. **Redistil**, collecting the fraction that boils near the ester's boiling temperature (ethyl ethanoate 77 °C).

MARK-SCHEME LANGUAGE

- $LiAlH_4$: "dry and ether"; product a primary alcohol.
- Salt: $CH_3CH_2CH_2COOH + NaHCO_3 \rightarrow CH_3CH_2CH_2COONa + CO_2 + H_2O$; ALLOW the ionic form and H_2CO_3 for $CO_2 + H_2O$. DO NOT AWARD a covalent O–Na bond.
- PCl_5: "PCl_5 / phosphorus(V) chloride / phosphorus pentachloride"; "steamy / misty fumes"; DO NOT AWARD $PCl_5(aq)$ or "fruity smell".
- Esterification: "(concentrated) sulfuric acid"; its role is "catalyst" (DO NOT AWARD just "proton donor").

Often asked as: equations for each reaction; test results that identify a carboxylic acid (fizzing with carbonate); reagents to make a named ester; MCQ choosing the reducing agent; identify the acid and alcohol that make a given ester.

COMMON MISTAKE

O–Na drawn as a covalent bond in a sodium carboxylate.
"Sodium" or "potassium" as the metal that gives fizzing in a test: use magnesium or a carbonate.
Forgetting that PCl_5 also reacts with an alcohol OH elsewhere in the molecule.
Unbalanced PCl_5 equations (a diacid needs $2PCl_5$).

BEYOND THE SPEC, BUT EXAMINED

- A **hydroxy acid** reacts with $NaHCO_3$ at the COOH only; the alcohol OH doesn't react: $CH_3CH(OH)COOH + NaHCO_3 \rightarrow CH_3CH(OH)COO^-Na^+ + H_2O + CO_2$ (WCH14 Jun 2021).
- With **excess PCl_5**, both the COOH and an alcohol OH in the same molecule are converted to –Cl (WCH14 Jun 2024). A **dicarboxylic acid** gives a diacyl chloride: $HOOCCOOH + 2PCl_5 \rightarrow ClOCCOCl + 2POCl_3 + 2HCl$ (WCH14 Jan 2025).

SPEC 15.13

Naming acyl chlorides and esters

WHAT THIS SPEC POINT ASKS

Name acyl chlorides and esters and draw their structural, displayed and skeletal formulae.

15.13 Acyl chlorides

- The –OH of the carboxylic acid is replaced by –Cl: functional group **–COCl**.
- Name: change **-oic acid** to **-oyl chloride**. Ethanoyl chloride CH_3COCl; propanoyl chloride CH_3CH_2COCl; benzoyl chloride C_6H_5COCl.

15.13 Esters

- Functional group **–COO–** (written RCOOR').
- Name = **alkyl** group from the **alcohol** + **-oate** from the **acid**: "alkyl alkanoate".
- To name an ester from its formula, split it at the **single-bonded O**:
  - the part **with the C=O** comes from the acid → "...anoate";
  - the part joined to the other O comes from the alcohol → "...yl", written **first**.

Ester	Acid part	Alcohol part	Name
$CH_3COOCH_2CH_3$	ethanoic	ethanol	ethyl ethanoate
$CH_3CH_2COOCH_3$	propanoic	methanol	methyl propanoate
$HCOOCH_2CH_2CH_3$	methanoic	propan-1-ol	propyl methanoate
$CH_3COOCH_2CH_2CH(CH_3)_2$	ethanoic	3-methylbutan-1-ol	3-methylbutyl ethanoate

- Watch the order: CH_3COOCH_3 is methyl ethanoate, but $CH_3OOCCH_2CH_3$ (= $CH_3CH_2COOCH_3$) is methyl propanoate. Ethyl ethanoate and methyl propanoate are isomers ($C_4H_8O_2$); see the figure in 15.6.

MARK-SCHEME LANGUAGE

- "3-methylbutyl ethanoate" (ALLOW "butanyl" for butyl; DO NOT AWARD "butanoyl").
- "2,2-dimethylpropyl propanoate" scores 2; "any name with '-propyl propanoate' scores 1".
- The alcohol named from an ester: non-systematic names are sometimes allowed if the structure is clear.

Often asked as: name an ester from its structure; name the alcohol and acid that make it; MCQ on ester names.

COMMON MISTAKE

Writing the names the wrong way round (acid part first).
Numbering the alkyl group from the wrong end: 3-methylbutyl, not 2-methylbutyl, when the branch is three carbons from the O.
Calling $CH_3COO–$ "methanoate" (it's **ethanoate**: count the C=O carbon).

SPEC 15.14

Reactions of acyl chlorides

WHAT THIS SPEC POINT ASKS

Know the reactions of acyl chlorides with water, alcohols, concentrated ammonia and amines.

15.14 Why acyl chlorides are so reactive

- The carbonyl carbon is bonded to **two electronegative atoms (O and Cl)**, so it is strongly **δ+** and attacked readily by nucleophiles with a lone pair (O in water and alcohols, N in ammonia and amines).
- The overall change is **nucleophilic addition–elimination**: the nucleophile adds to the C=O, then **HCl is eliminated** (the Cl becomes Cl^-). In each case the –Cl is replaced.
- The reactions are fast and exothermic **at room temperature**, and give **steamy fumes of HCl** (with water and alcohols).

Reacts with	Equation (ethanoyl chloride)	Organic product
water	$CH_3COCl + H_2O \rightarrow CH_3COOH + HCl$	carboxylic acid
alcohol	$CH_3COCl + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + HCl$	ester
concentrated ammonia	$CH_3COCl + 2NH_3 \rightarrow CH_3CONH_2 + NH_4Cl$	primary amide (ethanamide)
primary amine	$CH_3COCl + 2CH_3NH_2 \rightarrow CH_3CONHCH_3 + CH_3NH_3^+Cl^-$	N-substituted amide (N-methylethanamide)

- With ammonia and amines, **two** molecules are needed: one reacts, and the second reacts with the HCl formed (giving NH_4Cl, a **white smoke/solid**, or an alkylammonium chloride).
- Making an ester from an acyl chloride is better than from the carboxylic acid: it is **fast**, needs no catalyst or heating, and goes to **completion** (not an equilibrium).

MARK-SCHEME LANGUAGE

- 6-mark: equations for water → ethanoic acid + HCl, ethanol → ethyl ethanoate + HCl, ammonia → ethanamide + NH_4Cl, an amine → N-substituted amide; the **similarity**: all are addition–elimination (nucleophilic substitution) with HCl (or its salt) eliminated.
- Ammonia: "NH_3 reactant" and "NH_4Cl product and balanced equation", $CH_3COCl + 2NH_3 \rightarrow CH_3CONH_2 + NH_4Cl$.
- With concentrated ammonia: an amide forms, the reaction is exothermic, white smoke is seen.

Often asked as: equations for the four reactions; the product with a named nucleophile; the structure of an amide product; 6-mark comparing the reactions.

COMMON MISTAKE

Writing HCl as the by-product with ammonia: it reacts with the excess NH_3, so write **NH_4Cl**.
Giving NaOH when asked for the reagent that turns an acyl chloride into an acid (it's just **water**).
Careless amide structures: two H on the N of an N-methyl amide, or two methyl groups.
Not spotting the **similarities** between the reactions when a question asks for them.

BEYOND THE SPEC, BUT EXAMINED

- **Other nucleophiles** do the same thing: a thiol (CH_3SH) reacts like an alcohol to give a thioester (WCH14 Oct 2022).
- A **carboxylic acid** with an amine or ammonia gives an **ammonium salt**, not an amide (acid–base reaction) (WCH14 Jun 2024).

SPEC 15.15

Hydrolysis of esters

WHAT THIS SPEC POINT ASKS

Understand ester hydrolysis in acidic and in alkaline solution.

15.15 Acid hydrolysis

- **Heat under reflux** with **dilute** sulfuric or hydrochloric acid (the acid is a catalyst).
- It is the **reverse of esterification** and is **reversible**, so it doesn't go to completion: $CH_3COOCH_2CH_3 + H_2O \rightleftharpoons CH_3COOH + CH_3CH_2OH$
- Products: the **carboxylic acid** and the **alcohol**.

15.15 Alkaline hydrolysis

- **Heat under reflux** with **sodium hydroxide solution**: $CH_3COOCH_2CH_3 + NaOH \rightarrow CH_3COONa + CH_3CH_2OH$
- It goes **to completion** (one-way arrow). The carboxylic acid formed is immediately turned into the **carboxylate ion**, which can't react with the alcohol to re-form the ester.
- Products: the **sodium salt** of the acid and the **alcohol**. To get the free acid, add excess strong acid: $CH_3COONa + HCl \rightarrow CH_3COOH + NaCl$.

15.15 Splitting an ester

Break the bond between the C=O carbon and the single-bonded O. Add **OH** to the C=O side (acid) and **H** to the O side (alcohol). Methyl propanoate gives propanoic acid and methanol.

WORKED EXAMPLE: PURITY OF AN ESTER BY BACK TITRATION

1.60 g of impure ethyl ethanoate (Mr 88.0) is refluxed with 50.0 cm^3 of 0.500 mol dm^{-3} NaOH (an excess). The leftover NaOH needs 16.0 cm^3 of 0.500 mol dm^{-3} HCl.
NaOH added = 0.500 × 0.0500 = 0.0250 mol; NaOH left = 0.500 × 0.0160 = 0.00800 mol
NaOH used by ester = 0.0170 mol = moles of ester (1 : 1)
mass of ester = 0.0170 × 88.0 = 1.50 g; purity = 1.50 ÷ 1.60 × 100 = **93.5%**.

MARK-SCHEME LANGUAGE

- Triester (a fat or oil) with excess NaOH: "correct structure of propane-1,2,3-triol; correct structure of sodium ethanoate (or the salt); mole ratio" (3 NaOH, 3 salt). DO NOT AWARD a molecular formula.
- To release the acid: "add (excess) HCl(aq)" or any named strong acid.

Often asked as: products of acid and alkaline hydrolysis (MCQ); the reagent to free the acid from its salt; back-titration to find the percentage of ester; products of hydrolysing a triester.

COMMON MISTAKE

Using ⇌ for alkaline hydrolysis, or → for acid hydrolysis.
Giving the free acid (not the salt) as the product of alkaline hydrolysis.
Sodium ethanoate drawn with a covalent O–Na bond.
Answering "acid hydrolysis" when asked for a reagent: name the acid.

BEYOND THE SPEC, BUT EXAMINED

- **Back titration** of the excess NaOH to find the percentage of ester (WCH14 Jan 2020).
- **Triesters** (fats and oils) with excess alkali give propane-1,2,3-triol (glycerol) and three carboxylate salts (soaps) (WCH14 Oct 2022).

SPEC 15.16 Polyesters

WHAT THIS SPEC POINT ASKS

Understand how polyesters such as Terylene form by condensation polymerisation.

15.16 Condensation polymerisation

- In **condensation polymerisation**, monomers join and a **small molecule** (usually water, or HCl from an acyl chloride) is **lost** at each link.
- Each monomer needs **two** functional groups so the chain can keep growing at both ends.
- A **polyester** forms from:
  - a **dicarboxylic acid** (or diacyl chloride) + a **diol**, or
  - a single monomer with **–OH at one end and –COOH at the other** (a hydroxy acid, e.g. lactic acid).
- Each new link is an **ester link, –COO–**.

15.16 Terylene (PET)

- Monomers: **benzene-1,4-dicarboxylic acid**, HOOC–C_6H_4–COOH, and **ethane-1,2-diol**, $HOCH_2CH_2OH$.
- n HOOC–C_6H_4–COOH + n $HOCH_2CH_2OH$ → –[–OC–C_6H_4–COO–CH_2CH_2–O–]$_n$– + 2n H_2O
- Repeat unit: **–OC–C_6H_4–COO–CH_2–CH_2–O–** (one acid residue + one diol residue).

15.16 Poly(lactic acid)

- From 2-hydroxypropanoic acid, $CH_3CH(OH)COOH$: repeat unit **–O–CH(CH_3)–CO–**.
- Polyesters can be broken down by **hydrolysis** of the ester links (acid or alkali), so some (e.g. poly(lactic acid)) are **biodegradable**, unlike addition polymers such as poly(ethene).

15.16 Drawing and reading polymers

- Draw the repeat unit with **continuation bonds** at both ends (through the brackets) and **no –OH or –H** left on the ends.
- To find the monomers from a polymer: break each ester link; put **–OH** back on the C=O and **–H** on the O.

MARK-SCHEME LANGUAGE

- "ester linkage"; "remainder of the structure including continuation bonds". ALLOW more than one repeat unit; IGNORE brackets and n.
- Poly(lactic acid) repeat unit "–[OCH(CH_3)CO]–" or "–[CH(CH_3)COO]–"; reaction type "esterification" (ALLOW condensation).
- Monomers "in either order"; OH groups written correctly on the monomers.

Often asked as: draw the repeat unit (or two repeat units) from given monomers; identify the monomers from a polymer section; MCQ on the type of polymerisation; hydrolysis of a polyester.

COMMON MISTAKE

Ether, ketone or plain carbon links in place of the **ester** link.
Leaving –OH or –COOH groups in the repeat unit, or extra OH groups branching off the chain.
No continuation bonds; carbons with three or five bonds.
Using a dialdehyde instead of a diacid as a monomer.

BEYOND THE SPEC, BUT EXAMINED

- Polyesters from a **single hydroxy-acid** monomer (WCH14 Jan 2024), and the cyclic **lactide** formed from two lactic acid molecules (WCH14 Jan 2023). Terylene itself hasn't been named in a paper: questions give you the monomers.

SPEC 15.17 Mass spectra: accurate relative molecular mass

WHAT THIS SPEC POINT ASKS

Use mass spectra to suggest structures from accurate relative molecular masses, and calculate accurate Mr from relative atomic masses given to four decimal places.

15.17 High-resolution mass spectrometry

- The **molecular ion peak, M^+**, is the peak with the highest m/z (ignoring small isotope peaks). Its m/z = Mr.
- A **high-resolution** spectrometer measures m/z to **4 decimal places**. Because accurate isotope masses aren't whole numbers (^1H = 1.0078, ^{16}O = 15.9949, ^{14}N = 14.0031, ^{12}C = 12.0000 exactly), molecules with the **same nominal Mr** have **different accurate Mr**.
- So the accurate Mr fixes the **molecular formula**. It can't tell **isomers** apart: they have identical formulae.

WORKED EXAMPLE: WHICH FORMULA?

A compound has M^+ at m/z = 88.0524. Which is it: $C_4H_8O_2$, $C_5H_{12}O$ or $C_3H_8N_2O$?
$C_4H_8O_2$ = 4(12.0000) + 8(1.0078) + 2(15.9949) = **88.0522**
$C_5H_{12}O$ = 60.0000 + 12.0936 + 15.9949 = 88.0885
$C_3H_8N_2O$ = 36.0000 + 8.0624 + 28.0062 + 15.9949 = 88.0635
The closest match is **$C_4H_8O_2$** (e.g. ethyl ethanoate, methyl propanoate or butanoic acid). Other evidence (IR, NMR) is needed to choose between isomers.

15.17 Fragments tell isomers apart

- In the spectrometer, molecular ions break into **fragments**; only the **positive** fragments are detected.
- Propanal and propanone both have M^+ = 58, but **propanal** gives a peak at **m/z 29** ($C_2H_5^+$ or CHO^+), while **propanone** gives a big peak at **m/z 43** (CH_3CO^+) and none at 29.

MARK-SCHEME LANGUAGE

- Each fragment needs its m/z, its formula and a **positive charge**: "a fragment m/z = 136 (with a positive charge)". DO NOT AWARD a molecular formula or a negative charge.
- Fragmentation happens because "(high energy) electrons are fired at the molecules", "breaking bonds".
- "propanal and propanone have the same molecular ion peak m/z = 58"; "propanal has a peak at m/z 29 ($C_2H_5^+$ or CHO^+) but propanone does not".

Often asked as: MCQ choosing the compound from an accurate Mr; identify a fragment; use M^+ in a combined-spectra problem.

COMMON MISTAKE

Saying two isomers have "the same mass spectrum": the M^+ is the same, but the **fragments** differ.
Writing fragments with no + charge.
Rounding accurate masses before comparing.

BEYOND THE SPEC, BUT EXAMINED

- Distinguishing **propanal from propanone** using the fragment peaks at m/z 29 and 43 (WCH14 Jun 2025).

SPEC 15.18 Carbon-13 NMR: what it shows

WHAT THIS SPEC POINT ASKS

Understand that ^{13}C NMR gives information about the positions of ^{13}C atoms in a molecule.

15.18 How it works (in outline)

- About 1% of carbon atoms are **^{13}C**. Their nuclei behave like tiny magnets: in a strong magnetic field they absorb **radio waves** of a particular frequency (^{12}C nuclei don't).
- The exact frequency depends on the **electron environment** of each carbon. Electronegative atoms (O, N, Cl) nearby pull electrons away, so the carbon absorbs at a higher **chemical shift, δ**.
- δ is measured in **ppm** relative to the reference **TMS**, tetramethylsilane $Si(CH_3)_4$, set at δ = 0. TMS is used because it's inert, volatile (easily removed) and gives a single peak away from most others.

- Each **different carbon environment** gives **one peak**. Carbons in identical environments (related by symmetry) give the same peak.
- Samples are dissolved in a solvent with no interfering signals, such as $CDCl_3$.

MARK-SCHEME LANGUAGE

- Answers must refer to **carbon environments**, not to hydrogen atoms.

Often asked as: as part of 15.19 questions; the idea isn't usually asked on its own.

COMMON MISTAKE

Confusing ^{13}C and ^{1}H NMR, e.g. discussing splitting or peak areas for a ^{13}C spectrum.
Mixing up IR ranges and NMR shifts.

SPEC 15.19

Using ^{13}C NMR spectra

WHAT THIS SPEC POINT ASKS

Use ^{13}C NMR data to predict the carbon environments in a molecule from chemical shifts, and justify the number of peaks.

15.19 Chemical shifts

Carbon environment	Typical δ / ppm
C–C (alkyl)	5–60
C–Cl, C–Br	30–70
C–N	35–60
C–O (alcohols, ethers, esters)	50–90
C=C, aromatic C	110–160
C≡N	110–125
C=O in acids and esters	160–185
C=O in aldehydes and ketones	190–220

(Use the ranges in your Data Booklet in the exam.)

15.19 Counting peaks

- **Number of peaks = number of different carbon environments.**
- Look for **symmetry**: carbons that can be swapped by a mirror plane or rotation are equivalent.

Compound	Environments (peaks)
propanone, CH_3COCH_3	2 (CH_3 × 2 equivalent; C=O)
propanal, CH_3CH_2CHO	3
butanone, $CH_3COCH_2CH_3$	4
pentan-3-one, $CH_3CH_2COCH_2CH_3$	3
2-methylpropanal, $(CH_3)_2CHCHO$	3
ethyl ethanoate, $CH_3COOCH_2CH_3$	4
2-methylpropan-2-ol, $(CH_3)_3COH$	2

WORKED EXAMPLE: CHOOSING AN ISOMER

A ketone $C_5H_{10}O$ has **three** peaks in its ^{13}C spectrum, one at about 210 ppm. Which isomer is it? Pentan-2-one, $CH_3COCH_2CH_2CH_3$, has 5 environments. 3-Methylbutan-2-one, $CH_3COCH(CH_3)_2$, has 4. **Pentan-3-one**, $CH_3CH_2COCH_2CH_3$, is symmetrical: CH_3 (×2), CH_2 (×2) and C=O, so **3 peaks**. The 210 ppm peak is the ketone C=O.

MARK-SCHEME LANGUAGE

- "three peaks in its carbon-13 NMR spectrum; the other three have four peaks; labelling of the two equivalent carbon atoms". ACCEPT "three carbon environments".
- Shift answers need **both** the range and the environment: "190–225 ppm for C=O, 0–60 for C–C" (a value within the range is fine).

Often asked as: number of peaks for given structures; which isomer fits a given number of peaks; draw an isomer with a stated number of environments; assign a peak to a carbon.

COMMON MISTAKE

Drawing a structure with more environments than the question allows (only about one in five got this right in one paper).
Quoting a shift outside the allowed range, or with no environment.
Not saying which carbon gives which peak.

BEYOND THE SPEC, BUT EXAMINED

- Choosing between isomers from the **number of environments alone**, using symmetry (WCH14 Jan 2025).

SPEC 15.20

Proton (^1H) NMR

WHAT THIS SPEC POINT ASKS

Use low- and high-resolution proton NMR to predict proton environments from chemical shifts, relate peak areas to numbers of protons, deduce splitting with the n + 1 rule and suggest structures, and predict the spectrum of a given molecule.

15.20 What a ^{1}H spectrum tells you

Feature	Information
Number of peaks	number of different **proton environments**
Chemical shift, δ	the **type** of environment (what's nearby)
Relative peak areas (integration trace)	**ratio of the numbers of H** in each environment
Splitting (high resolution only)	number of H on the **adjacent** carbon(s)

- **Low resolution**: one peak per environment, no splitting. **High resolution**: peaks are split into multiplets.
- Areas give a **ratio**, not absolute numbers: 2 : 3 : 3 might be 2H : 3H : 3H.

15.20 Chemical shifts

Proton environment	Typical δ / ppm
alkyl H–C–C (CH_3, CH_2, CH)	0.8–1.8
H–C–C=O (e.g. CH_3CO–)	1.6–2.8
H–C–O (alcohol, ether, ester O–CH)	2.8–4.2
H–C–Cl, H–C–Br	3.0–4.2
O–H in alcohols	1–5 (variable)
H–C=C (alkene)	4.5–6.5
H on a benzene ring	6.5–8.0
aldehyde, –CHO	9–10
carboxylic acid, –COOH	10–12

(Use the Data Booklet ranges in the exam.)

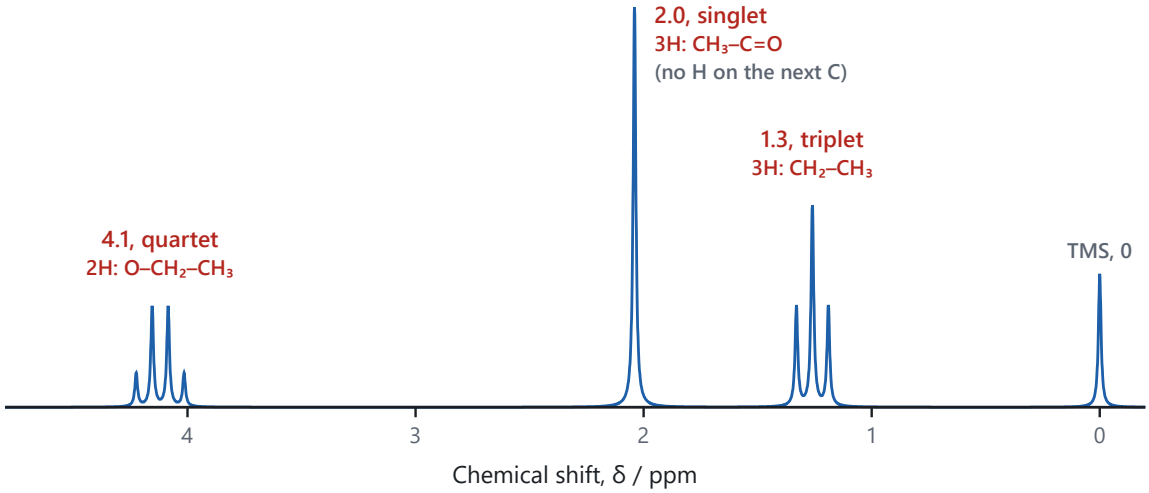
15.20 Splitting: the n + 1 rule

- A peak is split by the H atoms on the **neighbouring carbon(s)**. With **n** such H atoms, the peak splits into **n + 1** lines.
- H atoms in the **same** environment don't split each other. H on O (–OH) usually gives a **singlet** and doesn't split its neighbours.

H on adjacent carbons (n)	Lines	Name	Intensities
0	1	singlet	1
1	2	doublet	1 : 1
2	3	triplet	1 : 2 : 1
3	4	quartet	1 : 3 : 3 : 1
4	5	quintet (pentet)	1 : 4 : 6 : 4 : 1
6	7	heptet (septet)	

- Coupled groups come in pairs: an **ethyl group, CH_3CH_2–**, gives a **triplet (3H) + quartet (2H)**; an **isopropyl group, $(CH_3)_2CH$–**, gives a **doublet (6H) + heptet (1H)**.

Ethyl ethanoate, $CH_3COOCH_2CH_3$: peak areas 2 : 3 : 3



High-resolution ^{1}H NMR of ethyl ethanoate (sketch). The CH_2 is split into a quartet by the CH_3 next to it; the CH_3 of the ethyl group is a triplet; the CH_3CO has no neighbouring H, so it's a singlet.

WORKED EXAMPLE: ETHYL ETHANOATE OR METHYL PROPANOATE?

Both are $C_4H_8O_2$ and both give a singlet (3H), a quartet (2H) and a triplet (3H). The **shifts** decide:
- **Ethyl ethanoate, $CH_3COOCH_2CH_3$**: quartet at **4.1** (CH_2 joined to O, H–C–O), singlet at **2.0** (CH_3 next to C=O), triplet at 1.3.
- **Methyl propanoate, $CH_3CH_2COOCH_3$**: singlet at **3.7** (CH_3 joined to O), quartet at **2.3** (CH_2 next to C=O), triplet at 1.1.

So a quartet near 4.1 ppm and a singlet near 2.0 ppm (as in the figure) means ethyl ethanoate.

WORKED EXAMPLE: PROPANAL OR PROPANONE?

Propanal, CH_3CH_2CHO: 3 environments, areas **3 : 2 : 1**. CHO at 9–10 ppm (triplet, next to CH_2); CH_2 at about 2.4 ppm (split by 3 + 1 = 4 neighbours: a quintet/multiplet); CH_3 at about 1.1 ppm (triplet, next to CH_2).

Propanone, CH_3COCH_3: the two CH_3 groups are equivalent, so **one** peak (6H) at about 2.1 ppm, a **singlet** (no H on the C=O carbon next door).

WORKED EXAMPLE: DEDUCING A STRUCTURE

C_3H_7Br gives two peaks: a doublet at 1.7 ppm and a heptet at 4.2 ppm, area ratio 6 : 1.
Two environments, ratio 6 : 1 → two equivalent CH_3 groups and one CH. The heptet means the CH has 6 neighbouring H; the doublet means each CH_3 has 1 neighbour. The CH at 4.2 ppm is next to Br.
Structure: **2-bromopropane, $CH_3CHBrCH_3$**. (1-Bromopropane would give three peaks, 3 : 2 : 2: a triplet, a sextet and a triplet.)

15.20 Sketching a low-resolution spectrum

One line per environment, heights in the **area ratio**, at sensible shifts, **no splitting**.

MARK-SCHEME LANGUAGE

- Explaining splitting must be **specific**: "4.2 ppm quartet due to 3 hydrogens on the adjacent carbon"; "1.3 ppm triplet due to 2 hydrogens on the adjacent carbon". General statements of the n + 1 rule don't score.
- Shift ranges quoted by mark schemes: "2.8–4.2 ppm (H–C–O)", "1.6–2.8 ppm (H–C–C=O)"; "CH_3CO … will be a singlet".
- "Peak areas show there are three proton environments in the ratio 3 : 2 : 1"; low-resolution sketch: correct number of peaks, heights in the ratio, sensible δ, **no splitting**.
- ALLOW pentet/quintet for 5 lines and hextet/sextet for 6.

Often asked as: assign each peak to a set of H; fill in a table of shift, splitting and relative area; explain a given spectrum; sketch a low-resolution spectrum; 6-mark deducing an isomer (often with mass spectrum and IR data too).

COMMON MISTAKE

"Height" instead of "**area**"; "chemical environment" instead of "**proton** environment".
Giving "3 protons in one environment and 1 in the other" instead of the **ratio**.
Linking the shift to the functional group in general, not to the specific protons.
Drawing split peaks for a **low-resolution** spectrum.
Missing symmetry: quoting 3 : 2 : 1 for a symmetrical molecule.
Treating the CH_3 next to C=O as the same environment as the CH_3 of an ethyl group.

BEYOND THE SPEC, BUT EXAMINED

- **Quintets** from 4 neighbouring H, e.g. the middle CH_2 of 4-hydroxybutanal (WCH14 Oct 2023). The n + 1 rule still applies; "quintet", "pentet" and "multiplet" are all accepted.
- Combined questions using M^+, IR, ^{13}C and ^1H data in one 6-mark answer (WCH14 Oct 2022). State which data confirm which feature.

SPEC 15.21

Chromatography: mobile and stationary phases

WHAT THIS SPEC POINT ASKS

Know that chromatography separates a mixture using a mobile phase and a stationary phase.

15.21 The principle

- The **stationary phase** stays in place: the solid silica or alumina layer on a TLC plate, the paper (and water held in its fibres) in paper chromatography, the packed solid in HPLC, the liquid coating in GC.
- The **mobile phase** moves through or over it, carrying the mixture: the solvent in paper chromatography, TLC and HPLC; an inert gas in GC.
- Each component is **shared (distributed) between the two phases**:
  - **stronger attraction to the stationary phase** (adsorption) → spends longer stuck → **moves less** (lower Rf, longer retention time);
  - **more soluble in the mobile phase** → **moves further** (higher Rf, shorter retention time).
- Different components have different balances of these, so they **separate**.

15.21 Polarity

- **Silica** (TLC stationary phase) is very **polar** (Si–OH groups). Polar molecules, especially ones that can hydrogen bond, are held strongly and move slowly.
- A **non-polar** solvent (e.g. hexane) moves non-polar compounds well, but leaves polar ones near the baseline; a more polar solvent moves polar compounds further.

MARK-SCHEME LANGUAGE

- A compound moves a short distance because "only London forces form between it and hexane / it is polar and hexane is non-polar" and "weaker interaction between hexane and the compound / it is less soluble in hexane". DO NOT AWARD a mention of hydrogen bonds with hexane.
- Stationary phase with amine groups: the molecule with more OH groups is held more strongly because it "can form more / stronger hydrogen bonds (with the amine groups)".

Often asked as: MCQ on why one spot moves furthest; explain the position of a spot in terms of the two phases.

COMMON MISTAKE

Discussing the solubility of the wrong pair (e.g. the solvent in the solvent) instead of the **compound in the mobile phase**.
Just writing the Rf formula when the question asks you to **explain** a position.

SPEC 15.22 Rf values in paper and thin-layer chromatography

WHAT THIS SPEC POINT ASKS

Calculate Rf values from one-way paper and TLC chromatograms and understand why Rf values differ.

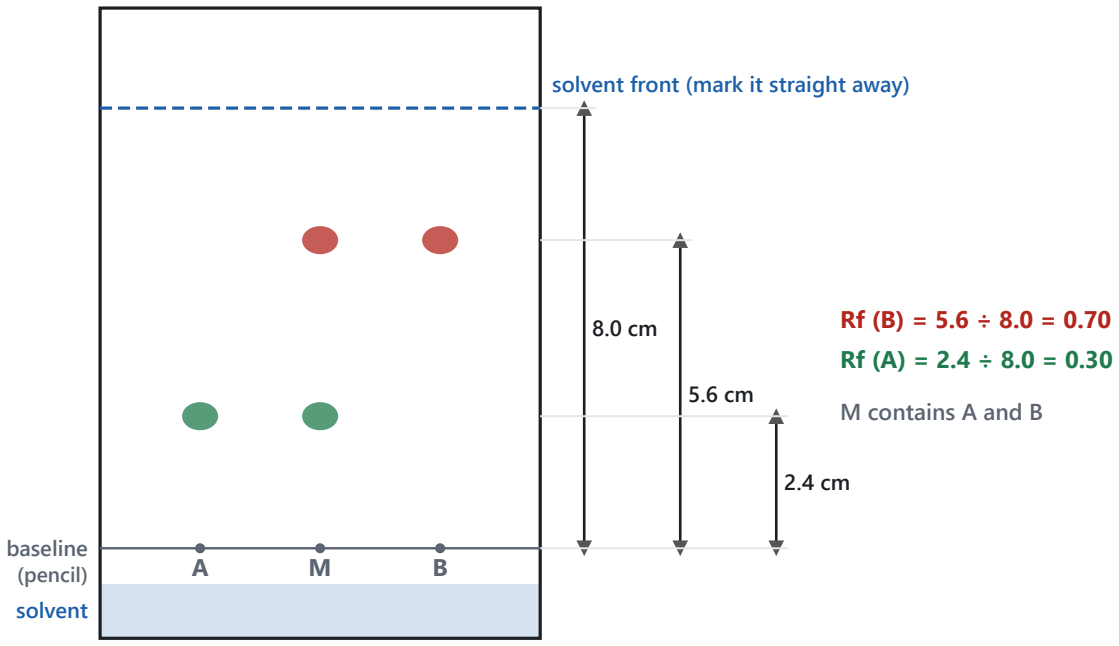
15.22 Running a chromatogram

1. Draw a **baseline in pencil** (ink would dissolve and run) about 1–2 cm from the bottom.
2. Put small, concentrated **spots** of the mixture and of pure reference compounds on the line.
3. Stand the plate/paper in a beaker with solvent **below the baseline** (so the spots don't dissolve into the solvent pool). **Cover** the beaker so the air is saturated with solvent vapour and the solvent doesn't evaporate off the plate.
4. Remove the plate when the solvent is near the top and **mark the solvent front** in pencil straight away (before it evaporates).
5. Dry the plate and **locate colourless spots**: UV light (TLC plates often have a fluorescent coating), iodine vapour, or ninhydrin spray for amino acids.

15.22 Calculating Rf

$$R_f = \text{distance moved by the spot} \div \text{distance moved by the solvent front}$$

- Measure **both distances from the baseline**, to the **centre** of the spot. Rf has **no units** and is always **between 0 and 1**.



A one-way TLC plate. Mixture M contains A and B, because its spots have the same Rf values as the reference spots run on the same plate.

WORKED EXAMPLE: RF

The solvent front is 8.0 cm above the baseline. Spot B has moved 5.6 cm.
Rf = 5.6 ÷ 8.0 = **0.70**.
Spot A (2.4 cm) has Rf = 0.30: it is held **more strongly by the stationary phase** (or is less soluble in the solvent) than B.

15.22 Why Rf values differ, and how to use them

- Rf depends on the compound's relative attraction to the stationary and mobile phases (polarity, hydrogen bonding, solubility).
- It also depends on **conditions**: the solvent, the type of plate or paper, and the temperature. So compare an unknown with **reference compounds run on the same plate** at the same time, rather than with a table.
- Two different compounds can have the same Rf in one solvent; running again in a different solvent helps.

MARK-SCHEME LANGUAGE

- "Rf = distance moved by spot ÷ distance moved by solvent front" (distractors: dividing by the whole paper length; the reciprocal).
- Drawing a TLC plate for a given Rf: "origin / start line above the bottom of the paper"; "straight line for the solvent front, labelled"; spot at the right fraction of the distance. DO NOT AWARD a wavy line or the top of the paper as the front.

Often asked as: calculate Rf from a diagram or measurements (MCQ or 1–2 marks); draw a plate for a given Rf; why the baseline is in pencil; why the solvent must be below the baseline.

COMMON MISTAKE

Measuring to the top of the paper instead of the solvent front, or from the bottom edge instead of the baseline.
Freehand or wavy lines, no solvent front drawn, a spot impossibly close to the front.
Not labelling which spot is which compound.

SPEC 15.23

HPLC and GC

WHAT THIS SPEC POINT ASKS

Know that HPLC and GC are column chromatography methods that separate substances by their different retention times, and that they can be linked to mass spectrometry (e.g. in forensics and drug testing in sport).

15.23 High-performance liquid chromatography (HPLC)

- **Stationary phase**: very small solid particles packed into a narrow steel **column** (small particles = large surface area = better separation).
- **Mobile phase**: a liquid solvent **pumped through at high pressure**.
- A **detector** (often UV absorbance) records each component as it leaves the column. The chromatogram plots **absorbance (y)** against **time (x)**.

15.23 Gas chromatography (GC)

- **Stationary phase**: a liquid with a high boiling temperature coated on a solid support, inside a long coiled column in an **oven**.
- **Mobile phase**: an **inert carrier gas** (helium, nitrogen or argon), which must not react with the sample.
- The sample is injected and **vaporised**. Components that are more volatile, or less attracted to the stationary liquid, come out first.

15.23 Reading a chromatogram

- **Retention time**: the time from injection to the component reaching the detector. Identify a component by comparing its retention time with a **pure reference sample** run under the **same conditions**.
- **Peak area** is proportional to the **amount** of that component.
- Retention time gets **longer** with a stronger attraction to the stationary phase, a slower flow of mobile phase, a longer column or (in GC) a lower temperature.

15.23 Linking to mass spectrometry

- Different compounds can have the **same retention time**, and some unknowns have no reference sample. So the column output is fed into a **mass spectrometer** (**GC–MS**, **HPLC–MS**).
- The chromatography **separates** the components; the mass spectrum (M^+ and fragmentation pattern) **identifies** each one, by comparison with a computer database.
- Uses: **forensics** (drugs, explosives, fire accelerants), **drug testing in sport** (urine and blood samples), food and environmental testing.

MARK-SCHEME LANGUAGE

- "(use the pure compound to) find the retention time". IGNORE retention factor/Rf here.
- Name the technique for volatile liquids: "gas chromatography / GC / GLC". DO NOT AWARD several techniques.
- HPLC chromatogram axes: time on the x-axis, absorbance on the y-axis.
- Carrier gas: "inert".

Often asked as: what retention time means and how to use it; which technique suits a sample; how to reduce the retention time (MCQ: increase the carrier gas flow rate); why MS is added; scaling a sample mass for analysis.

COMMON MISTAKE

Using **Rf** (paper/TLC) when the question is about HPLC or GC: they use **retention time** (only about one in five scored on this).
Thinking GC or HPLC alone identifies an unknown with certainty.

BEYOND THE SPEC, BUT EXAMINED

- **Peak area** ∝ **amount** of each component, used to say which is most abundant (WCH14 Jun 2023).
- Preparing a standard solution for HPLC and scaling up the mass of a natural product (e.g. spinach for oxalic acid) needed for a given concentration (WCH14 Jan 2025).